SHADOW FUTURES

How Markets Choose Winners—and Lose Sight of the Alternatives

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First edition · October 2026

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The other life of a song

In one version of a small online music market, a song called Lockdown finished first. In another, it finished fortieth out of forty-eight. The band was called 52Metro. The recording competing for attention was the same.[1]

First place invites a story. The band understood its audience. The song had something the others lacked. Listen again, knowing it won, and you might hear the qualities that explain its success. Fortieth invites a different story. The song failed to connect. Other bands understood something 52Metro did not. Listen again, knowing it lost, and its shortcomings might become obvious.

Usually, we get to tell only one of those stories. Someone succeeds, and the rest of us work backward. We examine the product, the strategy, the years of effort. The ending helps us decide which details mattered. We rarely get to check our explanation against another history in which the same work met a different fate.

For Lockdown, someone had kept the other chart.

MusicLab was a website where people could discover unfamiliar music. In research published in 2006, Matthew Salganik, Peter Dodds, and Duncan Watts studied choices made by 14,341 participants across two experiments involving forty-eight songs. Some listeners saw no download totals. Others could see what earlier listeners had downloaded. These listeners were randomly assigned to eight separate worlds, each with its own evolving popularity record. Every song began with zero downloads.[2]

A download in one world did nothing to the chart in another. Different groups of listeners encountered the same recordings, but accumulated different histories. In one experiment, popularity appeared as numbers beside songs in a grid. In the other, the socially

influenced listeners saw a list ranked by popularity. Earlier choices could help determine what the next person noticed.[1]

The stronger the social influence, the more unequal and unpredictable success became across worlds. Appeal still mattered. Songs that attracted listeners without popularity information rarely went from the very best to the very worst, or vice versa. But appeal did not dictate the finishing order.[2]

Lockdown had acquired something more revealing than a good review or a bad one: another result. Its first-place finish was real. So was its fortieth-place finish. Any convincing account of its success now had to leave room for both.

The same question turned up a few years later among projects asking for real money.

On Kickstarter, researchers led by Arnout van de Rijt selected 200 new, unfunded projects. They gave an initial donation to 100 chosen at random. The first backer, whose presence later visitors could see, had picked a project by chance. The researchers then watched who else would put money behind it.[3]

Among projects that received no experimental donation, 39 percent subsequently attracted money from other donors. Among those given the initial donation, 70 percent did. That later support excluded the researchers' own contribution. The first donation had helped bring others behind it.[3]

Notice what an observer arriving afterward would see. Some projects had attracted several backers; others had attracted none. It would be reasonable to inspect the successful pitches for clues. Perhaps their plans were clearer, their ambitions more compelling. Without knowing about the random donation, however, the observer could give the pitch credit for an advantage that had been assigned by an experiment.

The researchers also tested larger initial advantages. Their effects showed diminishing returns: more manufactured success did not produce proportionately more subsequent success.[3] An initial boost

mattered, but increasing it did not keep buying the same additional response.

What it does establish is already uncomfortable. People can respond sensibly to a record of success while helping create the next success in that record. The later backers' money is real. The project may put it to excellent use. Yet counting those backers does not tell us how much support the project would have received without its experimental head start.

Return to Lockdown. Add another thousand listeners to the world where it finished first. We might learn much more about that audience's enjoyment. Those additional listeners would not recreate the world where the song finished fortieth. More observations of one history and more chances to compare histories are different things.

That difference is the starting point of this book. Markets do more than distribute money and attention. They help determine which alternatives get tried, which improve through use, and which produce a record anyone can inspect. A market can become very busy while leaving a particular question poorly answered: how much did productive work, rather than the path of earlier rewards, contribute to this success?

I call the cumulative chance of rewards going somewhere other than the current favorite the comparison budget. At each opportunity, take the chance that someone else receives the reward, then add those chances across opportunities. The favorite can change. This counts chances, rather than actual rival victories; it does not make every purchase or download an independent experiment. The question is how much opportunity the allocation process leaves for informative comparison.

MusicLab makes the distinction tangible. Forty-eight song titles can remain on a screen even while attention flows mainly toward a few. Keeping the titles available is different from getting the recordings heard. A long list of alternatives can coexist with a thin record of what they can do.

The comparison budget is the economic object I propose for studying that scarcity. In the models behind this book, it limits how precisely a reward history can reveal the effects of work, quality, and investment. Under specified conditions, opportunities outside the favorite can dwindle fast enough that even an indefinitely growing record cannot reliably settle attribution. A small chance that persists can still accumulate; concentration alone does not establish the result.

Success breeding success is an old idea. Robert Merton described how recognition could bring scientists resources for further work. Armen Alchian explained market selection under uncertainty decades before MusicLab.[4] My question concerns the evidence left behind by that selection: when does an economy stop producing enough comparison to explain the rewards it continues to distribute?

Lockdown's other chart suggests what is missing. I call these feasible alternative histories shadow futures, borrowing words from Steven Johnson's account of Stuart Kauffman's adjacent possible.[5] They share the same underlying economic conditions and rules, but early events differ. Later choices and effort can respond. The recording stays the same; different first listeners leave a different chart for everyone afterward.

Sometimes policy changes which alternatives can become real. In 1956, an American antitrust settlement opened 7,820 existing Bell System patents to royalty-free licensing. Bell Laboratories had helped create technologies including the transistor. Other businesses now had access to thousands of patented inventions without paying royalties.[6]

Decades later, Martin Watzinger and his colleagues studied the consequences. Comparing Bell patents with matched patents outside Bell, they found a lasting increase in subsequent innovation outside telecommunications. They did not find the same increase within telecommunications, where access to customers remained obstructed. Permission to use an invention was not always enough to make a business from it.[6]

The result gives the missing comparison a physical form: equipment that another company might build, offer, and improve. An invention behind a licensing barrier cannot tell us everything about what other inventors could do with it. Opening access changes the rules; it is an intervention, rather than a replay of an unchanged economy. But it can let possibilities previously confined to speculation develop observable results.

There is a question here for antitrust beyond the next price increase. What does a restriction prevent us from discovering? Existing competition policy already takes innovation seriously. The additional inquiry I pursue is whether control over opportunities can impoverish the evidence on which future economic decisions depend. A remedy should earn its cost by improving those decisions. Bell's uneven results show why simply announcing access is insufficient.

That question becomes more urgent when the thing being bought can help decide who gets hired, what work gets done, and which ideas receive attention.

Consider a customer-support worker with an AI assistant suggesting replies during a conversation. The worker can accept, edit, or ignore them. This was the setting studied by Erik Brynjolfsson, Danielle Li, and Lindsey Raymond, using data from 5,172 agents. Their published study found an average increase of 15 percent in issues resolved per hour after access to the assistant, with especially strong improvements among less experienced and lower-skilled workers.[7]

Here is a reason to welcome the technology. A newer worker can draw on assistance that helps close a gap in experience. The study also found evidence of learning. AI can make useful knowledge more available to people who have had less time to acquire it. Any account of the future of work has to make room for gains like these.

The researchers examined periods when the AI system went offline and agents continued handling their customer conversations. Workers who had used the assistant still handled conversations faster than they had before gaining access. The pattern was stronger among

people with more previous exposure to its suggestions. This was evidence consistent with learning: some benefit could remain when the immediate stream of advice stopped. A worker's improved performance could therefore reflect both what the software supplied now and something the worker had acquired through using it.[7]

That interruption gives the comparison a concrete purpose. Watching a successful assisted conversation tells us what worker and tool accomplished together. Watching work continue during an outage helps investigate what the worker can do when suggestions disappear. Neither record alone describes every possible workplace. Together, they let us ask a sharper question about how a productive ability develops. The distinction matters when an employer decides whether software should guide beginners, handle their tasks itself, or change the training through which the next generation learns the job.

But a customer-support result cannot settle what happens in a software project, any more than Lockdown's winning chart can settle its fortunes everywhere. The work, tools, and opportunities to measure them differ. To see how quickly the comparison itself can change, follow what happened to researchers studying programmers.

METR, an organization that evaluates AI systems, ran an experiment in early 2025 in which experienced developers completed tasks with or without AI tools. In that setting, AI access made completion slower. When METR expanded the research later that year, it encountered a different problem: some developers did not want to participate if participation meant doing work without AI.[8]

They were being offered fifty dollars an hour to work on their own projects. Still, some declined. Others held tasks back from random assignment because they did not want those tasks landing in the condition where AI was forbidden. In surveys, between 30 and 50 percent of developers reported withholding some tasks for that reason.[8]

By February 2026, METR said the newer results gave an unreliable measure of the productivity effect. Changed pay and difficulties measuring time also complicated matters. Its researchers thought AI

was likely helping developers more than before, while emphasizing how weakly their data measured the size of that improvement.[8]

Think of a task the programmer keeps out of the study. It may be exactly the task on which AI saves the most time. The developer has a reason to keep using the tool, while the researcher has a reason to want the missing comparison. Both can be acting sensibly. The work gets done, but that opportunity to measure the difference has gone.

METR began redesigning its experiments. When developers change how they work, researchers have to change how they measure it. The question is how to preserve useful comparisons without asking people to abandon the tools they now rely on.

The same practical question runs from the programmer's desk to the data center. Training a model, running it to produce answers, and comparing it with another system are different activities. Running a trained model is called inference. A service can generate an enormous record of answers while leaving unanswered whether a different model, supplier, or organization of work would serve us better. AI can also make those comparisons cheaper. Which possibility prevails depends on how we deploy it.

Then the argument reaches the pay packet. Think again of the customer-support worker resolving more problems with AI assistance. The worker is doing useful work. So are the people who built the tool. The employer supplies an organization in which both can operate. Measuring the productivity improvement helps us understand what changed. It does not, on its own, tell us how to divide the additional income among them.

Now remember the Kickstarter projects. Their later support was influenced by a donation the researchers deliberately assigned. In ordinary business, we rarely have such a clearly marked intervention. We see the revenues after customers, reputation, investment, and opportunity have acted on one another. That makes the familiar claim that someone deserves precisely what the market paid them much harder to establish than the claim that they created something valuable.

Taxation enters here because governments must choose rules even when contribution cannot be divided neatly from position. Incentives matter; promises and legitimate entitlements matter. Uncertainty about a fortune's origins supplies no tax rate. It does require us to explain our tax rules without pretending that every pound or dollar arrives with a reliable account of who caused it.

There is also a separate question about conduct that removes valuable comparisons. When closing a route to customers prevents other people from learning something useful, could part of the cost fall outside the business making that choice? If so, might access rules, public trials, or a carefully justified tax address it? We will need evidence of the loss and the costs of a remedy. The size of the winning business cannot answer those questions for us.

We already know how to treat some comparisons as worth maintaining. At Rothamsted in England, agricultural experiments begun in 1843 have preserved differences in how plots are treated across generations. Soil and plant samples have been kept for more than 180 years. Some plots have received particular fertilizer or manure treatments since the beginning, apart from fallow periods. Methods have also changed to keep the experiments useful.[9]

Those fields are productive in two senses. They grow crops, and their maintained differences help produce knowledge. Maximizing this year's harvest on every plot would answer a different question from the one the continuing experiments make possible. Someone has to value the comparison enough to keep it alive.

That is the journey ahead: from a song to a first donation, from patents to AI, from the opportunities people receive to the rewards we decide they deserve. We will follow both what an economy produces and what it allows us to learn.

Lockdown finished first. Lockdown finished fortieth. The second chart survived because somebody built a place for it. The harder question is what happens to the futures for which we never make room.

The evidence success leaves behind

The advertisement that got the credit

A customer searches for eBay, clicks an advertisement for eBay, and buys something on eBay. For the advertiser, the journey looks reassuringly complete. There was an ad, then a click, then a sale. Each step can be recorded. Money went out to attract the customer; money came back when the customer bought. Here, apparently, is an unusually clean account of a contribution.

Except that the customer was searching for eBay already.

Economists Thomas Blake, Chris Nosko, and Steven Tadelis studied large advertising experiments at the company. When advertising against its own brand terms was suspended, customers largely substituted unpaid search links for paid ones. The brand advertisements had no measurable short-term benefit. Experiments with more general search terms found a different pattern: new and infrequent customers could be influenced, while frequent customers accounted for much spending without a corresponding change in purchases.[10]

One brand-keyword test stopped eBay advertising on Yahoo and Microsoft's search platform while retaining Google advertising as a comparison. Almost all the lost paid-search traffic and attributed sales reappeared through unpaid links. Customers had another route to the shop. Closing the paid route revealed how many sales the advertising ledger had mistakenly credited.[10]

This is a useful place to begin because it requires no quarrel about whether businesspeople work hard or whether customers know what they want. The advertisements existed. Someone designed them,

arranged their placement, and paid for them. Some were useful to some customers. The difficult question was the difference they made to a specified outcome. Counting activity was an excellent way to establish activity. Establishing contribution required something more.

For the shopper who wanted eBay, the alternative was close at hand: another link on a search page. In other settings, the alternative is much harder to recover. It might be another business that never attracted a first customer, another worker who never received an assignment, or another technology that never developed a record of successful use. We can study what happened in great detail while remaining uncertain about what those alternatives would have revealed.

This book concerns that uncertainty, and a way markets themselves can help create it. An early reward can affect the opportunity to earn the next reward. When that happens, the market is also helping select the evidence that later observers will use to explain its success. The economy we inhabit is partly a system for producing things and partly a system for deciding which possibilities become observable.

Go back to the eBay search page. To discover the advertisement's short-term contribution, the researchers needed customers who could arrive without it. A manager looking only at the completed sequence of paid clicks and purchases would face a different task. More accurate recording of that sequence would not recreate the missing comparison. The distinction will matter whenever we hear that enough data should eventually explain everything.

The work and the difference it made

I use contribution uncertainty for uncertainty about the effect of productive inputs on rewards in a specified economic setting. The inputs might include effort, quality, investment, or risk-bearing. The question concerns how an outcome would change if an input changed, while the relevant economic rules continued to apply. A result can be valuable, and the work behind it entirely real, while the size of that effect remains difficult to recover.[11]

The advertising example also reveals why the outcome must be specified. Increasing visits to a website is different from increasing purchases. Attracting a new customer is different from giving an existing customer a more convenient route. A short-term experiment may leave longer-term effects unresolved. Before asking how much something contributed, we need to say what it contributed to, compared with what, and over which period.

Lockdown's two rankings raise a related question with a different design. MusicLab preserved independently developing markets for the same songs. The song's first-place finish and its fortieth-place finish were both outcomes of listeners choosing. They tell us that one observed ranking cannot be treated as a complete account of what that recording was capable of achieving. They do not calculate the value of every rehearsal or assign a percentage of the song's success to talent.[12]

It is easy to move too quickly between those claims. Once we learn that success depends partly on a history, we may want to declare the winner undeserving. Once we notice that a winner made something useful, we may want to declare the reward fully explained. Neither conclusion follows. A recording can give pleasure in several possible histories, and the amount of attention it receives can differ greatly between them.

Consider another familiar difficulty. In George Akerlof's account of the used-car market, the seller may know more about the car's quality than the buyer does. A buyer unable to distinguish good cars from bad ones may offer a price that drives good cars away. The information

problem changes which trades take place. Akerlof showed why uncertainty about quality belongs inside an account of how a market works.[13]

My starting question allows the inputs to be observed. Suppose the buyer can inspect the work, rather than suspecting a hidden defect. That still leaves the effect of those inputs on the outcome to be learned. Knowing how much an advertising campaign cost does not identify the sales it created. Knowing what a musician recorded does not reveal every audience the recording could have found. Better disclosure and better comparison can solve different problems.

The distinction is particularly important when success changes the product itself. Kickstarter's experimental first donations helped some projects attract subsequent backing. In an ordinary business, an early customer might also finance an improvement that makes the next customer happier. The advantage can become productive. By the time we inspect the business, asking whether it is good and asking how it became good are separate inquiries.[14]

I will take the work seriously throughout. An explanation that erases the engineering, practice, organization, and willingness to bear losses explains too little. So does an explanation that treats the finished reward as a precise measure of those contributions. The task is to understand how real productive activity interacts with the opportunities through which it develops and becomes visible.

Who gets a chance

A first success can change the opportunities available afterward. The Kickstarter experiment gave researchers a way to observe that process by assigning early advantages. Later donors could respond sensibly to the record they encountered while helping widen the gap between projects.

Each choice does two things: it allocates resources now and helps determine what the next observer will be able to inspect. An expanding record of success can therefore coexist with fewer opportunities to evaluate alternatives.

The comparison budget gives that inquiry a defined object. At each allocation, given the history so far, identify the candidate with the greatest chance of receiving the reward. Take the chance that someone else receives it instead. Accumulate those chances across the opportunities under study. The favorite can change. This measures the allocation chances left outside the current favorite, rather than the number of times an outsider actually wins.[11]

A list of awards and rejections does not, by itself, tell us those chances. The budget is not a count of winners or the number of firms on a website. To apply the idea empirically, we need evidence about the allocation process: which alternatives are eligible, what changes their prospects, and what is recorded beyond the award. The quantities cannot be read straight from market shares.

MusicLab helps us see why a catalogue is insufficient. A title remaining on a list does not mean listeners encounter it on comparable terms. Conversely, a single business can arrange useful tests of alternatives, as eBay did with advertising. Counting organizations and counting informative opportunities are different inquiries. Institutional design can matter as much as the number of logos on the screen.

In my allocation models, the comparison budget limits the information the reward record can provide about productive input effects. Under further conditions, it can remain finite over the whole future of a market. Then adding transactions need not let an observer reliably settle attribution. This is a conditional result about a specified experiment, including its information and rules. It is not a declaration that a dominant company is permanently inexplicable.[11]

The small word "finite" does substantial work. A tiny but persistent chance of selecting an outsider can keep accumulating. A shrinking chance can also accumulate without limit if it shrinks slowly enough. Even then, comparisons must distinguish the input effects we care about. We will investigate both possibilities before using the result to discuss actual institutions.

Another transaction can demonstrate that an established service continues to work well while revealing little about an untested alternative. Commercial value and information about a particular comparison are separate properties of the same event.

Useful evidence may exist elsewhere: a field experiment, an independent technical evaluation, or measured outcomes beyond the award itself. If those observations distinguish the contribution explanations, they belong in the inquiry. An argument about missing evidence would be worthless if it manufactured scarcity by discarding evidence that exists.

eBay's experiments illustrate this constructive response. The company could change where advertising appeared and observe what followed. Its advertising ledger did not have to remain the last word. Before declaring an attribution problem intractable, we should ask what a feasible intervention could reveal. The comparison budget helps diagnose a specified record; it does not forbid investigators from improving their experiment.

Other possible histories

The seed of this book's name was a phrase in Steven Johnson's account of Stuart A. Kauffman's adjacent possible. Kauffman, author of *At Home in the Universe*, explored self-organization and complexity. The adjacent possible describes possibilities reachable from what already exists. Johnson called it “a kind of shadow future.” Existing materials and discoveries put some next steps within reach; taking them can make further combinations possible.[15]

Think of Bell's transistor. An invention can become a component in devices its inventors never built. Who can work with it matters to which possibilities are explored. Johnson's image made me ask an economic question: what happens to possibilities that depend on opportunities going to someone who never gets them?

I chose *shadow futures* because the phrase holds together two views of the same opportunity. Before a customer places an order, more than one business may have a future that could grow from it.

Afterward, we can follow the business that won: the revenue it receives, the investment it makes, the work it learns to do. The other path recedes from view. It may never leave enough of a record for us to judge what could have developed there.

I use the term for feasible alternative allocation histories with the same starting conditions, underlying productive effects, and rules for how people respond. The first customer can change; so can the later investment financed by that first sale. The rules stay fixed while events unfold differently. Kauffman's adjacent possible concerns the opening of new possibilities. My question concerns what we can learn when only one of the available paths becomes observable.

MusicLab makes the idea visible. Lockdown could finish first or fortieth without becoming a different recording. The researchers kept the other chart. An ordinary business rarely gets an equivalent record of the opportunities it missed ten years ago.

The comparison budget connects those missing paths to the paper's impossibility result. It adds the chances of awards going outside the current favorite. Under the model's rules for how productive work affects those chances, a finite lifetime total can leave different accounts of contribution impossible to distinguish reliably from the reward record. Even watching that record grow indefinitely cannot provide a method that recovers the right contribution across every explanation the model allows.[11]

That is why I chose a phrase about unrealized futures for a result about knowledge. A path we do not observe can contain a comparison we need to explain the one we do. Another order for the winner adds to its history. It need not tell us what the rival would have learned, built, or earned if an earlier order had gone the other way.

This book is itself a product of the adjacent possible. Kauffman's work, Johnson's image, and the economic literature made a further question available to me. Its future will depend on who encounters it and what they can do with it. A professional economist may give these ideas a larger audience, sooner than I can. Credentials, an established readership, and access to the right conversations can open doors that

an argument alone cannot. That is another way opportunity can shape an idea's reception.

If someone else tests the idea more rigorously, explains it better, or helps more people use it, I will be content with that outcome. Credit matters, as does the freedom to challenge a claim. What I do not want is an epistemic monopoly over an idea that could benefit humanity. I want the comparison to remain open, including when an alternative to my account proves better. I cannot ask readers to take shadow futures seriously and then assume that the history giving me the most recognition must be the best one.

An imagined better outcome is not evidence. We need a feasible route and a defensible account of how people respond. Changing an early award within unchanged rules explores another history. Changing the allocation rules is an intervention. The eBay experiments changed an arrangement; they did not replay an industry.

A simulation needs the same discipline. Programmed markets reveal consequences of their assumptions; explaining an actual buyer requires evidence that those assumptions fit.

There is another distinction we will need: predicting what comes next and explaining why the outcome has the shape it does. In Lockdown's winning world, an observer who expects more listeners to choose an already popular song could make useful forecasts. Those forecasts would not reveal the other world's chart. Getting the next choice right does not itself account for the song's lead.

Blackwell and Dubins's work helps make the statistical distinction precise: forecasts can converge under specified conditions. Whether the same record identifies productive contribution is another question. We will examine the circumstances in which the two inquiries diverge.[11]

Merton described how scientific recognition could bring resources for further work. Alchian examined economic selection without perfect foresight. Research on increasing returns, reinforced processes, and statistical learning helps explain how a history of

rewards develops. The comparison-budget bound is my proposed way of connecting those processes to the precision with which contribution can be measured.[16]

An early research grant makes the stakes concrete. Thijs Bol, Mathijs de Vaan, and Arnout van de Rijt studied Dutch scientists whose applications fell just above or below a funding threshold. The near-winners subsequently accumulated more than twice as much further funding over eight years as the near-losers, excluding the initial grant itself. Part of the gap arose because unsuccessful applicants entered later funding competitions less often. The distinction involved who continued to present work for evaluation, as well as how evaluators treated it.[17]

The research proposal that never reaches another panel leaves a different kind of absence from the one a rejected proposal leaves. We can inspect the latter decision. The former requires asking which earlier events affected participation. When opportunity influences the decision to compete again, a record of later funding awards can conceal part of the process producing its own pool of candidates. That is a reason to follow the people outside the winner's history, even when the published results look complete.

From evidence to institutions

At the customer-support desk studied by Brynjolfsson, Li, and Raymond, AI assistance helped workers resolve more issues per hour on average. The gains were particularly strong among less experienced and lower-skilled workers. That is evidence of useful capability reaching people who could benefit from it. It should count in any serious discussion of the technology.[18]

But suppose we want to know how organizations should train future workers. Measuring the gain from an assistant available today does not finish the inquiry. We would need to compare arrangements that affect which tasks beginners attempt, what guidance they receive, and what they learn over time. A service that solves the current customer's problem and an institution that develops tomorrow's

expertise can be evaluated together, provided we actually preserve evidence about both.

METR's developer research exposes one difficulty. As programmers became less willing to do certain work without AI, the research team found that participation and task selection made its follow-up estimates difficult to interpret. The work being studied was changing which comparisons people would supply. The team responded by reconsidering its design. Learning about the technology required attention to the institutions producing evidence about it.[19]

This is one reason the future of economics belongs in a book about AI. More computing can make some experiments cheaper and some records easier to analyze. It can also make one system so convenient that its alternatives are rarely tried. A larger stream of answers may reveal much about current use while leaving important alternatives unresolved. What we learn will depend on questions, access, and design as well as on processing power.

Bell's 1956 settlement provides a historical warning against treating access as a slogan. Opening existing patents to royalty-free licensing was followed by greater innovation outside telecommunications. The study did not find the same result inside the sector, where barriers to reaching customers remained. A policy can loosen one constraint while leaving another in place. An entrant needs a usable route from permission to performance.[20]

For antitrust, that suggests an inquiry into what a restriction prevents other people from learning. The relevant harm could extend beyond a higher price today if lost comparisons impair valuable decisions tomorrow. Existing competition policy already considers innovation and other dimensions of competition. My proposal concerns the evidence produced through access, with the practical costs of preserving or replacing it included in the judgment.

For taxation, two questions must remain distinct. One concerns the confidence with which a reward can be divided into contribution and positional advantage. Another concerns whether an action that removes useful comparisons imposes costs on others. The first limits

a claim about exact attribution. The second could support a corrective policy only with additional evidence about consequences, incentives, and feasible instruments. Neither question supplies a tax schedule by itself.[11]

Policy must compare feasible arrangements. An independent test, better access to existing records, and a change in market rules produce different evidence at different costs. A feasible alternative, a probable improvement, and a worthwhile policy are different judgments. Taking an alternative seriously includes allowing evidence to defeat our hopes for it.

A different account of success

At Rothamsted, continuing agricultural experiments make the value of maintained comparison unusually tangible. Different treatments have been preserved while methods have also been adapted to keep the experiments relevant. The record matters because someone has continued the arrangements that produce it. Useful knowledge has a material history: plots, treatments, samples, labor, and decisions about which differences to keep.[21]

An economy cannot turn every field into an experiment or give every rejected supplier a second chance. Nor should it. The interesting question is where preserving an alternative would teach us enough to improve what comes next. Sometimes the answer will be an independent route into a market. Sometimes it will be an experiment inside a large organization. Sometimes it will be a better measurement, and sometimes the available evidence will support doing nothing.

We will begin with the explanations attached to winners, then examine the comparison budget and shadow futures more closely. The historical chapters ask how opportunities harden into infrastructure and who controls access to a test. The AI chapters follow the investment that makes a model possible, its repeated use, and the workers whose abilities develop alongside it. The final

chapters consider what competition policy and taxation can reasonably do with the argument.

Throughout, a winner will remain a winner. Lockdown attracted listeners in the world where it finished first. The AI-assisted customer-support workers resolved problems. What changes is the inference we draw from an outcome: how far it explains itself, which further comparisons would matter, and which institutional choices can be justified with the knowledge available.

The eBay customer gives us a modest version of that discipline. A complete record of the paid click could have encouraged a confident story about the advertisement. Making another route observable changed the explanation. In more consequential settings, we should be equally interested in the route that never became a record. The first task is to notice when the winner's evidence has become the only evidence we allow ourselves to see.

The winner and the explanation

An ending looking for a beginning

Put Lockdown at the top of the chart and imagine that the other chart has disappeared. The first thing we know is that listeners selected it. The next thing we may want to know is what made them do so. We could listen for a memorable passage, compare the song with its competitors, and form a view about the audience. A good explanation might begin with something that really is in the recording. The trouble starts when we decide that the ranking has told us how much that feature mattered.

The song's fortieth-place finish makes that leap visible. It does not make every judgment about the recording false. It changes the kind of explanation a judgment can support. A quality capable of helping the song in both worlds cannot, without something further, account for the difference between the worlds. We need to understand how that quality met the sequence of attention, choices, and displayed popularity in each one.[22]

Follow a MusicLab listener. The website offered unfamiliar songs by unfamiliar bands. A participant chose a recording, listened, and was asked to rate it on a scale from one star to five. Afterward came an opportunity to download it. Downloading was optional. The experiment recorded several steps between a title appearing on a screen and a song acquiring another mark of popularity.[22]

A rating and a download answered different questions. The rating expressed a response after hearing the recording; in the social-influence worlds, the download entered the popularity record shown to subsequent listeners. Imagine someone who liked a track

but chose not to keep it. That person's favorable judgment would not become another download on the chart. Even this simple market contained more than one way of being successful. The researchers had to choose which outcome to explain.

Now change the screen. In the first experiment, songs appeared in a grid, with positions assigned randomly for each participant. Some listeners also saw download counts. In the second, socially influenced listeners saw a single column ordered by popularity. An early download could now help move a title up the list encountered by the next listener. The recording had stayed fixed while its route to an audience changed.[22]

Think of a title near the bottom. To attract a listener, it has to be noticed among alternatives carrying more evidence of earlier attention. Once heard, it still has to win a download. A final download total combines those stages. A musician interested in reaching listeners would want to know where the obstacle lay: getting the title seen, getting the song played, or getting someone to keep it. The total alone cannot separate them.

Looking first at popular titles can make sense. A listener with ten spare minutes has a practical search problem. Earlier downloads may offer a useful shortcut. But that shortcut helps decide which recordings receive the next hearing. The chart both reports choices and influences the conditions under which further choices occur. It cannot be read as forty-eight recordings repeatedly taking the same unchanged test.

For Lockdown, separate worlds let the researchers investigate what an ordinary chart conceals. Each world began without a download advantage, then acquired its own record. An admirer hearing the first-place song has evidence about an encounter with that recording. The fortieth-place world adds evidence about the process through which encounters accumulate. Enjoying the song answers a real question. Explaining its finishing position requires the wider record that the experiment preserved.

Ordinary economic success comes with a similar invitation. We see the company that reached customers, the worker who developed a reputation, or the project that secured its backing. We can inspect choices those winners made. What is harder to inspect is how another beginning might have changed the circumstances under which the same choices were available or effective. That is why evidence about the beginning deserves attention before we start reading the ending as a verdict.

Giving success away

The Kickstarter researchers did something direct about beginnings. They intervened before selected projects had raised any money. The projects already had proposals, ambitions, and creators. The researchers did not change those proposals. They changed whether an early visitor encountered a project with a first donation or a project still waiting for one. This allowed later support to be compared with a difference in initial success deliberately introduced by the research design.[23]

The distinction between those two kinds of change matters. Improving a pitch and adding its first backer could both increase subsequent funding. Without the experimental assignment, an observer might mistake the second mechanism for the first. A creator could offer a polished explanation of why a successful pitch worked, and much of that explanation might be sensible. It would still need to account for the advantage that came from already having support.

The experiment found that projects given an initial donation were more likely to attract subsequent backing. Later support was counted separately from the researchers' donation. It was not enough to announce that the treated projects finished with more money, since the researchers had put money into them. The revealing result concerned what other people did afterward. The design separated the initial gift from the additional support that followed it.

That gives us a useful discipline for other claims about success. When someone says an investment brought a return, ask which part

of the return has been counted. A first customer supplies revenue directly and may help attract the next customer. Those are different benefits. If we count the initial payment as evidence of the power to attract later payments, we have smuggled the intervention into the outcome. The Kickstarter design avoided that elementary trap.

The team did not stop at crowdfunding. Its experiments also intervened in product-review ratings, Wikipedia awards, and petition signatures. For Wikipedia, the subjects were highly productive editors; a randomly selected subset received a community award. The researchers then observed further awards from other people. Recognition could influence subsequent recognition even among people who were already contributing substantial work.[23]

The editor is a particularly instructive figure. The work exists before the experimental award. A later observer who sees several honors can reasonably think the editor has done useful things. That may be entirely correct. What the number of honors cannot establish unaided is how much each later recognition reflects the work and how much reflects the fact that the editor has already been recognized. Genuine contribution and a feedback effect can occupy the same page.

There is also no need to suppose that every later donor or award-giver merely copies a number. Early support can invite closer inspection. Closer inspection can reveal something worth supporting. An institution may benefit from such shortcuts when nobody can investigate every candidate. The experimental result concerns the consequence of receiving the early advantage. Determining the precise channel, and whether it improves decisions, requires further evidence rather than an assumption that influence is automatically foolish.

The researchers' larger interventions supply a useful warning against a seductive sequel. They compared smaller and larger manufactured advantages and found diminishing marginal returns. In the funding treatment, moving from one experimental donor to four did not create a proportionate multiplication of later success. A system can amplify the first step and respond much less strongly to

the next few. The force of feedback is a question about its rules and setting.[23]

That result prevents us from taking the favorite story about a snowball and making it the entire theory. Some early differences persist. Some run out of force. Some are overcome by other differences. If every unequal result were offered as evidence of irresistible cumulative advantage, nothing could count against the idea. The experiments are useful precisely because they can show both amplification and its limits.

For a project creator, the practical difference might be between improving the proposal and finding a way to secure credible initial backing. Both could deserve effort. An observer trying to learn which bottleneck matters should not treat the final funding total as if it had already performed that diagnosis. The first donor has changed the situation in which the next donor decides. Understanding the record means following that change through time.

The extra vote

A similar problem can begin with something smaller than a donation. In an experiment published in 2013, Lev Muchnik, Sinan Aral, and Sean Taylor randomly gave online comments an initial positive vote, a negative vote, or no vote. Over five months the study covered 101,281 comments on a social news website. The manipulation changed a visible score attached to words that had already been written.[24]

An artificial positive vote increased the likelihood that the next viewer would vote positively by 32 percent relative to the control group. Final mean scores were 25 percent higher. The percentages describe relative differences in those outcomes; they do not mean that a third of readers had been converted from disagreement to agreement. One extra mark on the screen had changed the subsequent path of evaluation.[24]

Here the productive object is unusually easy to hold in view. The comment did not become better written when the researchers clicked the positive button. A later reader could still find a good argument in

it. But the experimental advantage was not an improvement in the argument. If a final score were used as a precise measure of the quality of the words, some of the effect of the initial vote would be assigned to the writing.

The study's site did not order comments by their popularity. That is a useful detail because it stops us from explaining the whole result as the familiar consequence of being moved higher on a ranked page. The displayed score was tied to the comment when people evaluated it. Social information could affect the subsequent rating even without that particular route through a popularity ranking.[24]

It would be a mistake to hear the result as a reason to abolish ratings. A comment score can still help a reader find something useful. The problem is what we ask the score to prove. A device that assists selection need not provide an uncontaminated measure of the selected item's contribution. We can value the convenience of aggregated opinions while investigating how those opinions were produced.

Negative votes behaved differently. Readers tended to correct the artificial negative treatment, leaving final mean ratings without a statistically detectable difference from controls. The upward and downward nudges were not symmetrical.[24]

The topic mattered too. The researchers detected positive herding in politics, culture and society, and business. They did not detect it in economics, information technology, fun, or general news. Those categories had comparable activity levels. The same button and the same visible score did not produce a uniform response across the site.[24]

Compare the comment with Lockdown. In both settings, someone encounters an object together with information about earlier responses. The object may be appealing. The earlier response may also matter. The observer who sees only the accumulated result faces a mixture of those influences. The two experiments make the mixture easier to study by introducing or separating the relevant histories.

What they establish is specific enough to be useful. An early signal can affect subsequent judgment. It need not determine every outcome, work equally in both directions, or leave the underlying work unchanged in a longer economic setting. We have evidence for a mechanism and reasons to investigate its strength. We do not yet have a general permission to read all success as accidental or all rankings as worthless.

Recognition becomes an opportunity

In his account of scientific recognition, Robert Merton used the image of a forty-first chair. The French Academy had forty places, but the history of French thought contained distinguished people who never occupied one. Merton's examples included Descartes, Molière, and Proust. An institution with a fixed number of honors cannot turn the boundary between its members and everyone else into a complete ranking of achievement.[25]

The point is easy to appreciate when the chairs are literal. There are forty available places. An outstanding person can remain outside without anyone needing to conclude that the work was poor. A prize can be prestigious because its recipients are excellent, while failing to include everyone of comparable excellence. The quality of the winners and the completeness of the distinction are separate properties.

Merton went further. Recognition could become a resource for future research. Eminent scientists could receive disproportionate credit for contributions, and credit could help bring facilities for further work. His argument drew in part on Harriet Zuckerman's interviews with Nobel laureates. The honor was both an account of earlier achievement and a possible influence on what the scientist could do next.[25]

That second role changes the economic question. An extra chair would alter an honor roll. Access to a laboratory can alter the work that later appears on the roll. If we inspect achievement only after the opportunities have diverged, we may find a real difference in

productive capacity. The difference can be an outcome of the earlier allocation as well as a reason for the later one.

The Dutch grant study introduced earlier makes this process less dependent on a striking phrase. Bol, de Vaan, and van de Rijt examined applicants close to a funding threshold. Those just above it later accumulated more than twice as much additional grant money over eight years as those just below. The original award was excluded from that total. A narrow early distinction was followed by a much wider difference in further funding.[26]

The design matters as much as the result. Comparing all successful applicants with all rejected applicants would entangle the award with differences that helped determine who won. Focusing on applicants near the threshold makes the comparison more credible under the study's assumptions. It does not establish that every pair of nearby scientists possessed identical talent. It provides a way to investigate an early award's consequences without treating the entire difference in later records as proof of initial superiority.

The competitions had names: Veni for the early-career award and Vidi for the midcareer grant. Among applicants closest to the Veni threshold, about 59 percent of winners later applied for Vidi, compared with 40 percent of nonwinners. The researchers checked whether the application difference simply reflected people leaving academia or the country; it did not. An earlier decision had changed which proposals reached a later panel, before that panel judged anything.[26]

A scientist can stop applying, a supplier can stop bidding, and a musician can stop trying to reach an audience. The alternatives do not wait patiently for evaluation. An apparently thin field may reflect earlier decisions that drove people away. The missing proposal is harder to inspect than one that received a low score.

There are several possible practical responses, and they need different evidence. Helping promising unsuccessful applicants reapply would address participation. Giving a reviewer a clearer account of how close a prior decision was might affect the

interpretation of a failure. Funding everyone would involve a much larger commitment with different costs. The study makes a mechanism visible; choosing a response requires asking which change would improve the institution's work.

We should also distinguish subsequent funding from the underlying scientific achievement we hope funding makes possible. A grant is an input and an award, not a direct measure of every discovery it supports. More money might permit important work, but it is not itself proof that the resulting work contributes more. An institution can create a gap in resources and a gap in recognition without those gaps carrying identical information about scientific value.

The comparison with the online comment is instructive. An artificial vote changes the evaluation of fixed words. A grant can help change the future words, experiments, and projects themselves. Both involve feedback, but the productive consequences need not be the same. Understanding cumulative advantage requires following what the reward does after it is received, rather than treating attention, money, equipment, and achievement as interchangeable quantities.

What failure selects

A story about early advantage should be able to survive an encounter with a successful loser. In a 2019 study, Yang Wang, Benjamin Jones, and Dashun Wang compared junior scientists whose applications narrowly received or missed US National Institutes of Health funding. Early rejection was associated with greater departure from the NIH system. Among those who remained active, however, near-misses subsequently produced work with greater citation impact than narrow winners.[27]

The result creates two stories that are tempting to tell separately. One concerns people pushed out by an early failure. The other concerns people who continued and later did distinguished work. An institution evaluating only the second group could congratulate itself on the power of adversity. Someone looking only at departures could

describe failure as a straightforward destruction of promise. To understand the process, we need to keep track of who is included in each outcome.

For a scientist still submitting proposals, the ability to continue may be part of what distinguishes that person from someone who stops. Later success among survivors does not by itself show that giving everyone the same early setback would improve scientific work. The researchers investigated whether selection alone could explain their findings and argued that it did not fully do so. Their interpretation deserves examination alongside the distinction between the survivors and the original applicant pool.[27]

Imagine reading a shelf of papers by researchers who overcame rejection. The papers are real achievements. The biographies might offer useful accounts of persistence. But the shelf contains no place for papers that were never written by people who lacked the resources, inclination, or opportunity to continue. Admiring the work on the shelf and evaluating the institution's treatment of the original applicants are different tasks. One requires attention to achievements; the other also requires attention to absences.

This is the same discipline we needed for Lockdown's missing chart. An outcome can be impressive without revealing every route by which the observed group came to exist. A scientist's later work may genuinely be better. The additional question is what an observer can infer about a process that also affects who remains visible. The existence of a successful survivor cannot, unaided, make the cost of the missing careers disappear.

Nor should we turn the funding result into a convenient reversal of the Matthew effect. Further funding and citation impact measure different outcomes. A person can receive less money and produce influential work. Another can receive more money for research that is valuable in ways the chosen citation measure does not capture. The question of which reward predicts which achievement must remain specific enough for evidence to answer it.

Subsequent researchers tested both patterns across a wider set of funding programs. A 2025 reviewed preprint by Vincent Traag and colleagues examined fourteen programs from six funders in Europe and North America. It found the relationship between early and later funding more robust than the apparent advantage in later citation impact following rejection. The latter varied with setting and method; selection into reapplication offered an important explanation.[28]

That is useful news for anyone tempted by a universal story. A striking finding about one group can identify a mechanism worth studying without becoming a rule for all careers. Replication can change our confidence about how far it travels. If a comparison is valuable enough to influence policy, it is valuable enough to examine under arrangements that might reveal its limits.

For a funding agency, the practical choice is not whether to believe that winners are talented or that losers are resilient. It is how to allocate resources while learning what those allocations do. Does an early award change later applications? Does it help people do better work? Does a rejection remove promising candidates from future consideration? Different records are needed for the different answers. A single later honor roll cannot contain them all.

An early rejection need not settle a scientist's career. Another funder, laboratory or employer can provide a route back. People respond to setbacks, and institutions can revise earlier judgments. Which opportunities remain available, to whom, and for how long helps determine whether an early disadvantage persists.

The road with the fuel

Long before online charts, Armen Alchian offered an economical way to separate a successful outcome from a wise decision. His 1950 paper imagined thousands of travelers leaving Chicago along randomly chosen roads. Gasoline stations existed on only one road. Travelers who happened to choose it could continue; the others would run out of fuel. The continuing travelers might look unusually far-sighted to someone who saw only their progress.[29]

Alchian was illustrating how an environment could select surviving behavior without requiring every individual to understand the selection in advance. If gasoline supplies moved to another road, the pattern of continued travel would change even without travelers changing their paths. This was an explicitly unreal thought experiment, designed to make a point about selection under uncertainty, rather than an account of actual motorists.[29]

The travelers help clarify what is at stake in a record of winners. Progress along the fueled road is a real outcome. It would be absurd to say that the travelers had not moved merely because their initial route was chosen randomly. But observing that progress would not establish that their original decisions were unusually well informed. The fact to explain and the explanation attached to it remain distinct.

In the grant studies, continued participation has a similarly consequential role without being reducible to Alchian's deliberately simple setup. The scientist who keeps applying remains eligible for later awards. The one who stops no longer appears in the same competition. We need to investigate what changed their choices and opportunities. The mere fact that one is still visible cannot establish the whole account of why the other disappeared.

The thought experiment also reveals a possible abuse of skepticism. An observer could insist that the successful travelers were only lucky and therefore dismiss everything they did afterward. That would explain too little if, in a richer example, continuing the journey allowed them to acquire knowledge and develop skills. An early advantage can create conditions for later productive activity. The existence of the advantage does not make that later activity unreal.

For a Wikipedia editor, recognition can bring another person to a useful contribution. For a business, revenue can finance improvements. These are different mechanisms. Alchian's example exposes an inference from survival; explaining a particular reward system still requires investigating how it selects, rewards, and develops its participants.

Markets can select viable activities under uncertainty. People can imitate, adapt, and learn without possessing a perfect account of the economy. Choosing a business that works is one task; measuring how much each input contributed to its rewards is another. The first can be useful while the second remains unresolved.

The response should be to examine what would discriminate between explanations. In the travelers' example, discovering where the fuel was located changes the interpretation of the journey. In the online experiments, the researcher's record of the assigned advantage changes the interpretation of the later score. In a real procurement process, a useful independent trial might do similar work. Each additional observation earns its value by answering a question the winner's progress could not settle alone.

We can therefore learn from a successful traveler without treating every mile as a certificate of foresight. The same discipline lets us learn from a successful company without turning every sale into a complete causal audit. It asks us to understand the selection process as well as the selected outcome, and to notice which comparisons that process leaves available for the next observer.

The car the buyer can inspect

Akerlof's used-car model makes another kind of missing evidence easy to picture. A seller learns whether the car is good through ownership. A prospective buyer cannot distinguish its quality as well. The offered price reflects that uncertainty, and a good car's owner may refuse to sell at it. The information problem can remove good cars from the set the buyer gets to choose among. Quality and selection interact.[30]

The buyer's uncertainty helps determine which cars actually appear in this market. Inspecting only cars offered at the resulting price would not show the full population of cars their owners might have sold under different information arrangements. The missing good car matters even though no transaction involving it appears in the sales

record. Akerlof's example illustrates how a market's rules of information can help select its observable contents.

The contribution problem I want to isolate can remain even after a different uncertainty has been resolved. Suppose, in an explicitly imagined extension, an inspection tells the buyer the condition of each car accurately. The hidden-defect problem in the example has changed. But that knowledge would not tell a dealership how much its advertising added to sales, how much a salesperson's work affected a customer's choice, or how a prior reputation brought that customer through the door.

This does not make inspections unhelpful. An inspection answers a valuable question about the car. It may enable a trade that otherwise would not occur. The mistake would be asking the inspection to provide an account of every input's effect on the dealership's income. A certificate of condition and an estimate of contribution are different products, even if both are called information.

eBay's paid-search study provides a real counterpart. There was no need to doubt that an advertisement had appeared or that a customer had clicked. The records could establish the event. The experimental suspension showed how much apparent advertising success could be supplied through another route instead. The question was about an effect, and the added evidence came from changing the conditions under which that effect could be observed.[31]

It helps to picture the two records side by side. In one, the customer follows the paid link. In the other, the customer can reach the same destination without that advertisement. We are interested in the difference in purchasing, rather than the mere existence of the first route. A detailed account of the first route is useful for running a website. The second arrangement is useful for investigating what the advertisement contributes.

Broader searches required another test. Someone typing “cell phone” need not have eBay in mind. Using Google's geographic bidding system, the researchers stopped nonbrand advertising for sixty days in 68 US market areas, selected using a random draw

followed by matching on past sales. Sales could then be compared with places where advertising continued. New and infrequent customers responded more than frequent customers. An advertisement could be useful to a shopper without creating the purchase that its click would later appear to explain.[31]

This is where a familiar response to disputes about success reaches its limit. Show the work. Publish the records. Demonstrate that the product functions. Those are useful demands, and sometimes they settle the question actually being asked. But they establish contribution only to the extent that the observations distinguish the relevant alternatives. When the dispute concerns how much a factor changed an outcome, evidence that the factor existed can leave a substantial part unresolved.

In MusicLab, everyone could in principle inspect the same recording. The result did not require one world to receive a secretly defective version. The experimental setting separates the song from some of the history through which listeners encounter it. A recording can therefore be known and its possible market outcomes still varied. This is the observed-input question in a form that does not depend on concealing the product.

For economic arguments about merit, the distinction is consequential. Demonstrating a useful product can defeat the claim that nothing valuable was made. It does not necessarily establish the precise size of the reward attributable to making it. These claims should not be forced into a single contest in which proving the first automatically wins the second. We can respect the achievement and continue investigating the explanation of the income attached to it.

What success makes possible

An early advantage can finance productive capabilities that become valuable in their own right. Experience with customers can improve a product; revenue can pay for research. Later success may then have better immediate reasons than the original allocation did.

The improvement does not identify its own origin. Opportunity and the response to it have acted together. Another recipient might have developed an equally useful capability, a worse one, or something different. Observing the actual improvement cannot choose among those possibilities.

The work performed along the successful path remains real. A useful explanation must account for both its productive effect and the circumstances that made it feasible. Labeling the first advantage luck cannot measure the later work away; documenting that work cannot establish what a different initial allocation would have produced.

Obtaining the missing comparison can be expensive. Switching systems, training staff, or running an experiment consumes resources and may interrupt reliable service. The economic question is which uncertainty matters enough to justify investigating, and whether a smaller test can resolve it.

Bell's patent opening makes the distinction between knowledge and opportunity concrete. The historical study found that access to inventions and access to customers were different constraints. Outsiders could receive permission to use knowledge while remaining unable to turn it into a market record in the obstructed sector.[32]

Recognition can invite attention to existing work; revenue can finance new capability. Both change the record the next evaluator encounters. Treating every later award as a fresh judgment on unchanged starting abilities misses how earlier awards shaped the conditions of judgment.

The grant studies reveal another response: participation itself can change after an early result. When preparing an application is costly, repeated rejection can make another attempt unattractive. The observed field of candidates is partly an outcome of the reward process, rather than a fixed list on which that process merely acts.

By then a directory of remaining suppliers may conceal as much history as it reveals. It tells the buyer who is available now. It does not show all the firms that could have developed under different earlier allocations. This is one reason concentration deserves careful

description. A smaller field could reflect superior performance, productive scale, restricted access, discouraged entry, or several influences together. Counting the survivors cannot settle the account of how they became the survivors.

Turning the chart upside down

Salganik and Watts devised a particularly revealing challenge to the idea that a displayed record simply reports what an audience wants. In another MusicLab experiment, published in 2008, they allowed a popularity order to develop and then reversed the displayed download counts in two new worlds. A song with many downloads was presented as having few; one with few was presented as having many. After this intervention, new downloads were recorded honestly.[33]

The swap could be remarkably concrete. Before inversion, She Said by Parker Theory had 128 downloads and Florence by Post Break Tragedy had nine. In the altered displays, their apparent counts were exchanged. No band needed to improve a recording for the screen to give later listeners a different account of its success. The researchers had changed the history people believed they were entering.[33]

This was a different question from the first MusicLab experiment. Independent histories under unchanged arrangements help reveal how outcomes can vary. Deliberately altering the popularity record tests a response to an intervention. Both designs are useful, but we should keep their jobs distinct. The altered chart does not become an ordinary replay merely because the same recordings remain available.

Some apparent popularity became self-fulfilling. Yet the inversion did not permanently reverse the market as a whole. The songs with the greatest independent appeal tended to recover their popularity. The manipulation also reduced overall downloads. The result supports influence while showing that influence encounters preferences and can affect how much people take from the market, as well as which song receives their attention.[33]

That recovery is an important part of the ending. The best songs were not helpless objects waiting for a number to tell them what they

were worth. Listeners still heard recordings and responded to them. A useful argument about history needs space for those responses to overcome some disadvantages. Otherwise, it would have no way to explain why the inversion did less than a fully self-fulfilling account might predict.

The lower download total raises a separate question about the whole market. A band could benefit from a favorable display while listeners collectively found less music they wanted to keep. An individual's rank and an audience's total enjoyment need not move together. The experiment measures downloads rather than every aspect of welfare, but it makes clear why explaining who wins is different from evaluating the outcome for everyone affected.

For Lockdown's first chart, we now have several warnings against explaining too much from the ending. Initial support can attract further support. A positive vote can influence later evaluation. A grant can change who applies again. A manipulated chart can divert attention, and strong preferences can also resist it. The evidence gives us several mechanisms, rather than a single principle under which every winner is either inevitable or arbitrary.

The next step is to ask what the record itself can teach us once those mechanisms are operating. Suppose another thousand choices arrive. They might test something new, or repeat a route already strongly favored by the existing history. We need an account of the opportunities for informative comparison inside that sequence. Counting transactions is an appealing place to start, but it cannot be the end of the inquiry.

The comparison budget

A movie behind several doors

The film is Good Will Hunting. One image emphasizes Matt Damon and Minnie Driver; another emphasizes Robin Williams. In a 2017 account of artwork personalization, Netflix engineers used these possibilities to illustrate a choice their system had to make. A member drawn to romance might respond to one invitation, a member drawn to comedy to another. The same film could be introduced through different parts of its appeal.[34]

There is a small economic experiment hidden in that image. Showing an invitation supplies a chance for somebody to respond. Showing a different invitation supplies a chance to learn something else. An artwork that remains in a designer's folder can be beautiful, but the folder does not contain the audience's response to seeing it. An additional play following the established image does not automatically answer the question about the image still in the folder.

Netflix's engineers described a practical response: deliberately vary some selections, record how they were selected, and use responses to improve later choices. They explained that learning could carry an immediate cost when a member received an image other than the one currently predicted to work best. Their account concerned how to combine that experimentation with personalization, rather than simply displaying the same apparent winner indefinitely.[34]

A company with a large audience can create comparisons inside its own service. The same scale that helps an established choice dominate can spread the cost of testing another. What matters is how the service assigns those opportunities.

The image also keeps us honest about the question. Comparing two invitations to watch Good Will Hunting would not measure the

contribution of every actor, the director, the screenwriter, or the company carrying the film. A useful experiment can answer a narrow question very well. Its strength depends partly on resisting the urge to attach every neighboring question to the same result.

The engineers also asked what happens when an image changes between visits. A member might remember a film by its picture and then fail to recognize the new invitation. Or the replacement might finally persuade the member to try it. If several pictures preceded a play, deciding which deserved credit became harder. The team treated the frequency of image changes as part of the learning problem.[34]

Getting somebody to press play was not enough, either. A striking picture could attract an audience that soon abandoned the film. The engineers described using the quality of engagement when labeling their observations, to avoid rewarding that kind of clickbait. Now the record needed to connect the member, title, image, and subsequent response. A million presentations would be less revealing if the file lost those connections.[34]

The comparison budget names one feature of the process producing such records: accumulated allocation chance outside the current favorite. Brief probability examples will make that accounting concrete before the documented systems show what an application additionally requires.[35]

What we add up

Start with two alternatives that each have a one-half chance of receiving an allocation. Either can be designated the current favorite when they tie. The chance outside it is one half, which becomes the first addition to the comparison budget. This is a probability before the result is known, rather than a prediction of the realized winner.

At the next allocation, suppose the leading alternative has a three-in-four chance. The chance outside it contributes one quarter. If the other alternative becomes the favorite at a third allocation with the same margin, the addition is again one quarter. Across these three decisions, the additions total one.

That total does not promise exactly one outsider victory. The awards could repeatedly go to the favorite or sometimes go elsewhere. A quarter added before an allocation stays a quarter in the accounting, whichever alternative wins.

The favorite is redefined at every step. The quantity concerns the concentration of the next opportunity, conditional on the history, rather than loyalty to a permanent incumbent.

The favorite is defined by the chance of receiving the allocation, not by an independent verdict that it is the best product. A buyer can favor a familiar supplier because switching is costly. An experimenter can deliberately give a plausible challenger substantial attention. The reason for the choice and the quality of the option both matter economically, but neither should be smuggled into the word "favorite." The budget first describes how opportunities are distributed.

Calling this a budget does not mean that somebody deposited a fixed fund of experimental tokens at the beginning. The quantity accumulates as opportunities arrive. We can ask how much has accumulated so far, or what the remaining sequence could add under a specified process. The economic issue is whether continuing the market continues to supply meaningful possibilities outside its current favorite.

With three alternatives, an eight-in-ten chance for the favorite leaves a combined two-in-ten chance outside it. Adding another name changes nothing if it receives no chance and the existing probabilities remain the same. A longer directory is not itself a larger comparison budget.

Conversely, a company can preserve a variety of informative trials while remaining a single company. Netflix's account of artwork experimentation illustrates the institutional possibility. A legal organization and an alternative being tested are different units of description. Before counting, we must specify which allocations and which alternatives are relevant to the question. The number of corporate registrations cannot perform that work for us.

The definition is intentionally narrower than a general index of competition. It does not put the quality of a service, its price, a worker's income, and a consumer's enjoyment into one score. Those things may matter greatly. The comparison budget describes a feature of the opportunities through which evidence can emerge. To evaluate the consequences of changing it, we need to connect those opportunities with useful information and consequential decisions.

The allocation needs a consistent unit. For a study of image selection, candidate images could be the alternatives and a presentation the allocated opportunity. Whether somebody subsequently plays the film is an additional outcome. For a procurement study, suppliers and contracts could take those roles instead. We cannot count presentations for one question, switch to completed plays for another, and call the resulting total the same budget. The definition becomes useful by fixing the economic question and the process before counting.

In the models underlying this book, that connection takes a precise form. When verified productive inputs enter the allocation rule in the specified way, and their differences remain within the stated bounds, the comparison budget places an upper limit on the information the allocation history provides about their effects. The scarcity of outside chances can therefore limit how precisely contribution can be recovered from the record. A large budget, by itself, provides no matching guarantee.[35]

The news story that was not shown

Yahoo's front page gives us a documented example of an organization designing a record for later comparison. In research presented in 2010, Lihong Li, Wei Chu, John Langford, and Robert Schapire studied personalized news recommendation. Part of the underlying traffic came from a randomized bucket: candidate articles were selected at random for the highlighted position, with user and article information and the resulting click or nonclick recorded.[36]

The unshown article creates the difficulty. A person encounters one highlighted story, and the system sees whether that person clicks it. The same encounter does not reveal whether the person would have clicked every other candidate. The researchers' procedure started from that limited observation, rather than treating a visitor's arrival as a completed test of every story in the pool.

Random selection made a different use of the record possible. A proposed recommendation rule could be evaluated against logged encounters in which its choice matched the article actually selected. Under the paper's assumptions, this replay procedure supplied a reliable way to evaluate the rule using existing experimental traffic. The record was useful because of how the articles had been chosen and which responses had been retained.[36]

It is a beautiful answer to one version of the data problem. The observer does not invent a visitor's reaction to an unseen story. The evaluation uses reactions to stories that were genuinely shown. Its credibility comes from the logging design and the assumptions connecting those encounters to the proposed rule. The large database helps, but the relevant form of variation was built into the process before the later analyst opened it.

Now imagine a different file in which every visitor had been shown the same predicted favorite. The analyst might still learn much about that story and those visitors. What would be absent is the same coverage of choices under the randomized design. Replaying another rule would require further assumptions or further data. A more powerful computer could search the file faster without putting the missing article in front of a visitor who never saw it.

Yahoo recorded what visitors did after an article appeared, including nonclicks, as well as information about each visitor and article. A study of contribution has to use that evidence alongside the selection record. Counting only which story won the position would discard observations the experiment was designed to collect.

A replay based on yesterday's visits cannot, by itself, tell us how a new recommendation rule would change tomorrow's audience.

Readers might return more often; publishers might offer different stories. Studying those changes requires a design or model that accounts for those responses, rather than assuming that the same encounters would keep arriving.

Imagine four groups of viewers, each always assigned its own particular image. Across an equally sized population, the images would have equal shares of presentations. Yet within a group, the rule leaves no chance for another image. Aggregate diversity can hide a complete lack of alternatives within each group.

The converse is possible too. One image could dominate overall presentations while a carefully selected trial supplies useful evidence about a question involving another audience. The minority share alone would not tell us whether that evidence is weak or strong. We would need the selection rule, the groups involved, and the responses observed. Equal totals can conceal different processes; unequal totals can coexist with purposeful learning. Neither pattern in a chart substitutes for inspecting how each opportunity arose.

An allocation record can contain more than the winner. Task characteristics, eligibility terms, selection chances, and measured outcomes may supply additional information. Those observations should be included explicitly, even when they answer questions outside the simple allocation model.

A connection someone gets to make

A recommendation can also change a person's economic network. Karthik Rajkumar and colleagues studied randomized variations in LinkedIn's People You May Know recommendations. Different variants changed the kinds of connections suggested. The research, published in 2022, linked these experiments to subsequent job mobility. The effects of weak connections depended on how tie strength was measured; simply seeking ever weaker relationships was not a universal prescription.[37]

The connection makes the sequence particularly tangible. A recommended person can become a contact. That contact can help

information about an opportunity reach a job seeker. A subsequent job change is then part of a history influenced by an earlier selection. The study's design allowed the researchers to investigate aspects of this process that an ordinary list of members and their later jobs would have left entangled.

Imagine studying a member's career after the connection has been made. The profile could show qualifications, previous employment, and a real new role. The question about the recommendation is whether changing who was suggested affected the path into that role. The person's abilities need not be fictitious for the route to matter. A useful match can involve both a qualified worker and information that reached that worker through a particular contact.

By varying connection recommendations and following later employment outcomes, LinkedIn's researchers could investigate how an introduction changed someone's opportunities. The useful evidence came from deliberately changing which connections people encountered.

What made a tie weak? The researchers used both the number of mutual connections and the intensity of bilateral messaging. These measures describe different relationships: knowing few of the same people and rarely exchanging messages need not coincide. Moderately weak ties did best by the mutual-connections measure; the weakest ties did best by interaction intensity. Even the practical meaning of trying a different connection depended on the measurement.[37]

The authors also examined the risk that treating one member could affect another member's outcomes. Networked experiments face this problem because people are connected. We should not call two routes independent simply because different people or different accounts appear on them. Their relationship is part of what the design must investigate. This caution applies just as readily when several suppliers depend on a common distribution channel.[37]

A network can improve access to opportunities while complicating attribution of their causes. References, connections, and observed

performance need to be distinguished in an account of how the next allocation occurs.

Two long futures

Suppose the chance outside the favorite stays at one in a thousand at every allocation. Each opportunity adds a thousandth to the budget. A thousand opportunities add one; another thousand add another one. The process is heavily concentrated, but its accumulated outside chance keeps growing.[35]

Concentration and a finite lifetime comparison budget are therefore different properties. A tiny chance that persists differs from a chance that vanishes quickly enough for its total to remain bounded. Nor does either calculation promise that awards arrive on schedule or reveal an informative input contrast.

Now consider a different, explicitly hypothetical allocation schedule. Start with a one-in-four chance outside the favorite. At every subsequent opportunity, halve that chance. The first few opportunities contribute most of what the entire future can add. Continuing forever adds successively smaller pieces, and all those pieces together amount to one half. There can be indefinitely many decisions while the accumulated outside chance has a finite ceiling.

The contrast concerns the whole sequence. A single photograph of either process could show a dominant favorite and a neglected rival. The later history depends on whether meaningful chances continue arriving or become scarce fast enough. To distinguish those possibilities empirically, we need evidence about the rules and their evolution. A concentration chart is a record of realized shares, not a measurement of that future sequence.

A third schedule makes the need for care clearer still. Let outside chances decline, but so slowly that the additions continue accumulating without limit. We do not need the arithmetic of that schedule to see the conceptual point. "Shrinking" is not a synonym for "finite in total." Before a claim about disappearing comparison does

serious work, the rate of disappearance must be established in the setting being discussed.

The halving schedule is an accounting illustration, not a fitted model of a named company. An actual supplier's prospects could rise after a useful demonstration or fall after an incompatible product change. Its future chance could depend on who won earlier contracts. The budget follows the relevant conditional chances through those changes. Replacing that evolving process with a single fixed market-share number would remove precisely the history the object is meant to describe.

The schedules expose a distinction ordinary language can blur. A very small chance may feel equivalent to none when deciding whether to make one costly attempt. For what an indefinitely long record can reveal, persistence matters: a long sequence can accumulate what each opportunity adds only in a tiny amount.

The relevant time horizon matters too. A business, worker, or policymaker may need an answer long before an indefinitely accumulating process becomes helpful. An infinite lifetime budget could coexist with painfully little relevant opportunity over five years. The finite-record limit is therefore important in its own right. We should ask what can be learned in time for the decision, rather than using an eventual possibility as a reason to dismiss present ignorance.

Nor can the opposite inference be made automatically. A finite amount of opportunity need not leave every practical question unresolved. A buyer might already have enough evidence to choose a service sensibly. The stronger issue is a promise of reliable, increasingly precise attribution throughout the admissible explanations of the model. That is a more demanding promise than choosing an option that works well with the information currently available.

For Netflix's illustration, the immediate decision might be which image to show a particular member. For the public procurement example, it might be whether another supplier merits a pilot. We should specify that decision before evaluating whether the available

comparison is adequate. A budget is an input to the inquiry. It does not tell us that every uncertainty deserves to be eliminated before anybody may act.

How a lead can close a contest

There is a long mathematical history behind the difference between feedback that preserves comparison and feedback that eventually overwhelms it. One classic model begins with an urn holding colored balls. Draw a ball, return it, and add another of the same color. A color selected early becomes more likely to be selected again. Robin Pemantle's survey describes this family of reinforced processes and the many different behaviors produced by their rules.[38]

The urn isolates a feedback rule: an earlier draw changes the chances of later draws. Changing that rule can change how the sequence develops, even when every version can be described casually as success breeding success.

In the familiar version, adding a ball of the drawn color can change the long-run proportions while other colors continue to appear. The history can become unequal and still keep producing comparisons. To find out whether those chances have a finite lifetime total, we need to examine how the rule changes them after each draw.

A stronger version gives accumulated success a much more powerful effect on the next selection. For a simple example, imagine weights that grow with the square of the accumulated score: doubling a score quadruples its weight. Research on appropriate strong-feedback models establishes a striking outcome. After some finite, random point, a single alternative receives every subsequent allocation. The result depends on the specified reinforcement conditions, not merely on a large lead.[39]

Roberto Oliveira studied the onset of such dominance, building on earlier monopoly results. Thomas Gottfried and Stefan Grosskinsky later distinguished several feedback regimes, including strong and weaker forms of monopoly. Their results explain how different reinforcement rules can produce different forms of dominance.[39]

What the comparison budget adds is an information question about the resulting record. In the strong-feedback model developed here, the remaining chances outside the eventual winner are summable. The budget stays finite. Under the other conditions of the attribution result, indefinitely extending that single allocation record cannot supply a universally reliable recovery of the contribution effect. The prior dominance result and the information interpretation do different jobs.[35]

There is a feature here worth holding onto. An alternative can retain a positive chance at every finite stage, yet receive no further awards along the realized history. A positive chance is not a promise of an eventual event. When those chances get small quickly enough, their continuing existence need not create an endless supply of informative deviations. Nominal eligibility and sustained practical contestability can therefore diverge within a precisely specified model.

Entry, new products, new tasks, or independent tests can reopen comparisons. Whether positive feedback closes them permanently depends on the allocation rule and which changes remain possible. Gottfried and Grosskinsky distinguish reinforcement processes with different long-run behavior; the rule determines which result applies.

The urn and Netflix's artwork system both use selection rules that change with earlier results. The urn makes the feedback easy to inspect. Netflix shows how a business can design those rules to keep learning.

The calls made to choose a policy

An experiment with rice farmers in India gives the selection question a different purpose. Maximilian Kasy and Anja Sautmann worked with Precision Agriculture for Development on ways to enroll farmers in a phone-based extension service. Six strategies combined morning or evening automated calls with different advance text-message arrangements. The recorded success was completion of five questions during the call. It was not a measure of a farmer's eventual harvest.[40]

The experiment proceeded in waves. Results from completed calls informed assignment to later strategies. The aim was to help select an approach for use after the experiment. That purpose matters: maximizing the number of completed calls during the trial is a different objective from collecting the evidence needed to make a better choice for the much larger implementation that may follow it.[40]

The distinction is familiar once we put a calendar beside the experiment. This week's calls can deliver useful service. They can also teach the organization which method to use next month. Giving every remaining call to the current leader may improve the immediate recorded success rate while leaving a close challenger insufficiently tested. A reasonable allocation depends partly on whose outcomes, at which dates, the organization is trying to improve.

Kasy and Sautmann's exploration-sampling method retained substantial testing of plausible alternatives instead of simply concentrating on the strategy most likely to be best. Results from earlier calls helped determine which strategies received later ones.[40]

The case supplies an important counterweight to the idea that experimentation must mean equal shares forever. Once an approach looks clearly unpromising, continuing to assign it the same share may be a poor use of opportunities. A close challenger may deserve more investigation. Relevant comparison depends on which uncertainty remains, how much resolving it would matter, and the consequences of sending another call along each route.

Adaptive design ties opportunities to the uncertainty being investigated. A limited experiment on a relevant task can distinguish plausible explanations; a nominally balanced allocation on incomparable tasks may add little. The assignment process must be related to what the observer wants to learn.

The measure of success should remain visible as well. Completing five questions is something an organization can record promptly. It is useful for choosing among recruitment methods on that outcome. If the later decision concerns agricultural income or the quality of advice, additional evidence is needed. The outcome recorded for fast

adaptation should not acquire authority over every consequence simply because it arrives in time to guide the next wave.

This is especially important when comparisons have different costs. An ineffective image may waste a small amount of a viewer's attention. An unsuitable software deployment can demand weeks of staff time. A policy pilot may affect people whose welfare matters during the experiment as well as afterward. The agricultural study's stated objective helps identify the decision being optimized; other settings may require a different balance.

An institution can therefore preserve comparison while making increasingly selective choices. It need not suspend judgment until every candidate has been exhausted. The discipline is to state why the next allocation is useful, which possible responses would change the decision, and what costs are justified by that learning. Those questions turn a general preference for experimentation into something a buyer or program manager can actually implement.

What a large budget cannot buy

Equal chances at every allocation would make the budget grow, but still need not identify an input effect. If the measured input is identical across the alternatives, reallocating awards changes the recipient without creating the contrast needed to learn that input's effect.[35]

For a less abstract picture, imagine testing two artwork files that display exactly the same image. More exposures can help estimate how that image performs with particular audiences. They cannot reveal the difference between the two visual treatments because the treatments do not differ. A system can create a great deal of nominal variation while leaving the intended comparison untouched. Counting chances is necessary for one bound; it is insufficient for a guarantee.

A change of name need not create an independent trial either. Products may share an underlying service, a bottleneck, or a faulty outcome measure. The budget does not certify independence or

meaningful variation merely because a different alternative receives an award.

Yaneer Bar-Yam's distinction between large quantities of detailed data and information relevant to an intervention is useful here. His framework concerns how to identify what matters for a system-level question, rather than treating detail as automatically sufficient. The comparison budget addresses a narrower allocation setting, but the shared discipline is to ask what the record can distinguish for the question under study.[41]

Yahoo's article evaluation depended on randomized logging and observed responses. The farmer calls depended on the measured completion outcome and adaptive assignment. In each case, the record's usefulness followed from how it was generated, rather than its size alone.

Suppose the buyer can obtain an independent performance test without awarding another full contract. That test belongs in the analysis. If it distinguishes effects the award history leaves uncertain, the observer now has a different experiment from the allocation-only record. Applying an information limit while excluding the new test would manufacture a conclusion about ignorance by removing evidence from consideration.

Likewise, known selection probabilities can be useful information, not merely quantities an analyst may assume into existence. A designed experiment may log them. An ordinary market may require estimates based on observed rules and behavior. If obtaining those estimates supplies additional information about contribution, that information has to be counted. An empirically defensible budget needs a defensible observation model alongside it.

Defining the object and measuring it are therefore separate achievements. The rice-farmer experiment could implement assignment probabilities deliberately; an analyst of an ordinary market cannot assume comparable access to its mechanism. Sometimes only a range of plausible budgets will be justified. That can still organize an investigation, provided the range and its

consequences remain visible. Precision written into a definition should never be mistaken for precision supplied by observations.

The comparison budget cannot be read from a market-share table. A rival's sales tell us what happened, not its chance on the next job. Even a well-measured budget needs evidence about differences in what suppliers do and how well they perform. Frequent trials would tell us little about engineering if they never compared meaningful differences in engineering.

A large company that keeps testing

Microsoft's Bing supplies a particularly direct challenge to the idea that the number of firms tells us how much learning remains. In a 2013 paper, Ron Kohavi and colleagues described more than two hundred experiments running concurrently inside the service. Some users were held out to assess the experimentation system's overall effect. The organization was deliberately maintaining differences in experience rather than relying solely on the record of the version everyone already used.[42]

One example involved adding links within advertisements so people could navigate directly to parts of an advertiser's site. The initial version increased revenue but worsened user metrics and page-loading time. Adjusting the presentation, including how many main advertisements appeared, changed that balance. The investigators were tracking more than a single increase in a commercial total. A result that looked attractive on one outcome still needed interpretation alongside others.[42]

That example gives a practical meaning to a question we have repeatedly asked of the supplier record. Which outcome was measured? If we keep only a revenue increase, we lose part of what the experiment revealed. If we keep only the existence of a large company, we lose the comparisons it arranged. The informative object is the combination of the intervention, assignment, recorded responses, and decision, rather than a favorable statistic standing alone.

It also illustrates the difference between a proposed improvement and an improvement demonstrated on relevant outcomes. The extra links offered a plausible convenience. Testing allowed the organization to discover consequences that the proposal's rationale had not settled. A company can have skilled designers and still benefit from creating occasions on which their expectations meet a comparison. The trial does not diminish their skill; it helps direct it.

The public question remains broader. An experiment inside a company can be valuable for its chosen decision while leaving questions about market access or independent suppliers unresolved. We should credit the first accomplishment without pretending it answers the second. A business may preserve many comparisons concerning its own product and few concerning alternative economic arrangements. Each claim must identify the question and the information set to which it refers.

Sometimes the missing comparison calls for another supplier. Sometimes a company can learn more cheaply by changing its own trial or using records it already keeps. The question is which route will reveal something useful about the decision at hand, and what it will cost.

The price of keeping a door open

Keeping a comparison available has a price. Testing can require moving data, training staff, checking outcomes, and accepting disruption before its result is known. The hoped-for information must justify those costs. Some alternatives will sensibly remain unevaluated because testing them would cost more than the decision could gain.

Netflix's engineers described their artwork setting as favorable partly because exploration costs could be spread across a large membership. The cost still depends on what a person experiences during the test. A setting that makes outside scrutiny difficult can sometimes make internal experimentation economical.[34]

The practical question is which uncertainty could change a consequential decision. A small reliability difference may justify

investigation when it affects years of expensive operation. A difference incapable of outweighing known switching costs may not. There is no economic instruction to maximize the comparison budget everywhere.

Existing records, a shared test environment, or a limited task can sometimes answer a question more cheaply than a full deployment. Whether they do so depends on their design and on the inference required. A lost route has no automatically adequate substitute.

The order matters. Start with the question, identify a comparison capable of answering it, and assess the cost of producing that comparison. Only then decide whether an access arrangement or purchasing rule helps. Beginning with a preferred number of suppliers and assuming that the resulting record will be informative skips the hardest part of the task. The directory can be long while the relevant test remains absent.

Learning can strengthen the case for an incumbent. An artwork experiment can confirm that the established image works better for the relevant audience. An informative test must allow a result to change our view, including by increasing justified confidence in an existing choice.

The comparison budget measures the accumulated allocation chance outside the current favorite. What those chances reveal depends on the inputs, observed outcomes, and response rules. Preserving more comparison is worthwhile when the resulting evidence improves decisions enough to justify its cost.

The next chapter examines the histories left unrealized. Holding underlying conditions and response rules fixed makes it possible to distinguish variation in outcomes from the effect of changing the economy itself. That distinction is essential to explaining a reward.

Shadow futures

One economy, different histories

The same recording can have another history.

In MusicLab, Lockdown finished first in one experimental world and fortieth in another. The separate groups encountered the same recordings but different developing popularity records. Their charts supply observed alternatives to the idea that one realized ranking explains itself.[43]

A theoretical replay asks us to hold the underlying effects, starting conditions, and response rules fixed while allowing early allocations to differ. Later investment and behavior can then change because they respond to a different history. The economy follows the same rules without producing identical decisions.

Holding every later input fixed would describe another exercise. If success normally finances investment, suppressing that response changes the assumed economy. A replay preserves the rule connecting opportunity and behavior, rather than freezing behavior regardless of what happens.

A different early allocation need not permanently reverse the outcome. An initial lead can fade, be overcome, or compound. A model must allow the original rules and their uncertainties to determine which follows. Otherwise its preferred alternative has been written into the conclusion.

A shadow future is a feasible allocation history over a stated horizon under the same underlying input effects, starting conditions, and response rules. Different events can produce different subsequent behavior under unchanged rules. The object describes histories the modeled economy can generate, including those it never realizes.[44]

MusicLab's groups contained different listeners rather than identical people sent down cloned lives. Its observable parallel markets therefore illustrate historical variation without constituting literal independent reruns of every feature of an economy.[43]

MusicLab recorded parallel markets. A theoretical replay describes histories implied by specified rules. The first supplies experimental observations; the second helps define what we seek to explain. Estimating the probability of an imagined history still requires evidence about the process producing it.

An ordinary archive contains the investments and outcomes that occurred. It lacks development that would have been financed by revenue never received. That omission becomes important when we ask the realized record to explain the effect of productive inputs on success.

What changes, and what stays

Changing an input before an allocation is an intervention. Changing the rule that allocates opportunities is another intervention. Both are legitimate economic questions, but neither is the same exercise as replaying the original process under unchanged conditions.

For the engineering question, we might specify an additional month of work on reliability, a particular starting date, and the contracts available over the next two years. We would then compare expected rewards with and without that extra work, allowing later investment and exposure to respond as the model says they would. The result is a contribution measure for that intervention and horizon. It is not simply the difference between whichever two histories happen to interest us.

Consider the customer-support study introduced in the prologue. Brynjolfsson, Li, and Raymond examined the introduction of an AI assistant and measured changes in agents' work, including issues resolved per hour. Their published study found average gains, with larger gains among less experienced and lower-skilled agents. The comparison concerned access to a particular assistance system in a

particular workplace. It did not measure each worker's share of the value created by the entire organization.[45]

A contribution question must specify its target. Does more engineering change reliability, whether customers observe reliability, or the chance of receiving revenue? An intervention on work and one on information about work can have different consequences.

Opportunity can finance investment, and investment can improve performance. Explaining the resulting reward requires tracing that chain. Recognizing the role of an early allocation does not make later productive work disappear.

MusicLab's separate markets help us see variation in success under a common experimental arrangement. They do not tell us the effect of rewriting Lockdown's chorus. To estimate that effect, we would need a design that compared the relevant recordings under suitable conditions. The song's different rankings and the effect of changing the song are related questions, but the first does not supply the answer to the second.

This is why contribution needs an explicit comparison. An additional engineer, a better recording, and a new distribution route are different changes. Their effects may unfold over different periods. A large reward could be sensitive to one and barely sensitive to another. Saying that success reflects contribution leaves these choices unresolved. Naming the change makes the claim something we can investigate.

This prevents explanations from becoming stories chosen after the outcome. Improving an alternative's initial quality, changing its starting position, and changing its early allocations are different operations. Each needs its own place in the analysis.

The fork in the laboratory

There is a striking predecessor involving an animal with no procurement department. Line Chloé Le Goff and Philippe Soulier studied reinforced choice using Argentine ants. In their laboratory setup, an ant entered a fork, chose a branch, and was removed. Another ant followed. The maze began without a pheromone trail. A trial ended after one hundred choices; the researchers recorded fifty trials.[46]

Earlier passages could influence later choices through deposited pheromone. The authors considered a model in which the response to the accumulated trail depended on unknown settings. They studied estimation from repeated trials and the limits of estimation from a single sequence. Their paper connected reinforcement with difficulties recovering parameters, even from an indefinitely extended path.[46]

The ants expose a statistical temptation common to reinforced processes: a longer record of the established route seems to promise an eventual explanation of why it prevailed. Yet continuing that route differs from observing the process begin again.

The fork was carved into white PVC and partly covered by Plexiglas. No food was present during the experiment. Only one ant occupied the apparatus at a time, and it could not return after choosing a branch. The researchers designed these restrictions to make their behavioral assumptions more credible: an arriving ant encountered the trail left by predecessors, rather than other ants at the junction. The apparatus helped define what a choice could tell them.[46]

The resulting paths did not all look settled. In fewer than ten of the fifty trials, the researchers described a branch as appearing to have been selected. In the others, selection was not obvious after one hundred passages. The same protocol had produced visibly different histories. Those finite records gave the researchers something that continuing traffic down one established branch could not: repeated observations of how a path developed from its beginning.[46]

Le Goff and Soulier had already shown that a long history of reinforced choices need not reveal the forces behind those choices. In a market, that raises a further question: can the opportunities left open to rivals tell us how much we can learn about why the winner won? The comparison budget gives those opportunities a precise meaning and connects them to the problem of measuring contribution.

The statistical question is how much opportunity the process retains to distinguish input effects. A shorter record can contain substantial relevant variation while a much longer one follows an established route. The design, assignment chances, and measured outcomes determine the difference.

A million entries are not a million beginnings

Before considering the strongest result, take a familiar example that goes the other way. Imagine an ordinary coin with an unknown, fixed tendency to land heads. One explanation says heads occurs half the time. Another says it occurs three quarters of the time. Any particular finite sequence of heads and tails is possible under both explanations. Shared possibilities do not make the explanations equally persuasive.

A long sequence containing about three heads for every tail will increasingly favor the second explanation. Under the usual assumption that the tosses are independent with an unchanged bias, the proportion of heads supplies information that keeps accumulating. We do not need to watch every imaginable coin sequence. The one we observe can become enough to distinguish the two fixed biases reliably as the number of tosses grows.

An unobserved possible history does not establish a lasting attribution limit. A coin has other possible histories too. The issue is whether observations along the actual process separate the relevant explanations sufficiently as the record grows.

Return to Yahoo's news experiment from the previous chapter. Its randomized article displays and recorded clicks were designed to support a particular evaluation. The logging mechanism mattered

because the researchers did not observe how the same encounter would have responded to every other article. Their analysis used the opportunities the design actually generated, with assumptions about those encounters and the evaluation procedure.[47]

The complete experiment must be specified. Awards alone supply one information set; measured reliability and independent tests can supply another. A laboratory result directly informative about the input effect cannot be excluded when evaluating what the record reveals.

In the model, the information limit concerns a specified family of candidate explanations for how productive inputs affect allocation. A candidate explanation is not an arbitrary story about the company. It supplies probabilities for the recorded outcomes at each stage. Two such explanations can disagree about engineering's importance while using the same purchasing procedure and the same rules for how suppliers respond to earlier contracts.

The comparison budget then asks how much conditional allocation chance accumulates outside the current favorite. The favorite can change as the purchasing history moves forward. As Chapter 2 showed, a small chance can persist long enough to accumulate without limit. The strongest result concerns the more demanding case in which the lifetime total is finite. It also requires restrictions on how different explanations separate as those chances diminish. Those restrictions specify how far the next-award probabilities can differ between candidate explanations.[44]

To apply the result, an investigator needs to establish the feedback rules and the chances they leave for alternatives. Ant traffic, song charts, and market shares describe outcomes. Establishing a lasting learning limit also requires evidence about how the process generates them.

When waiting cannot finish the job

First, the candidate explanations must describe a common predictable experiment. Input and state rules are fixed functions of the information available before each award. If success finances investment, each explanation retains that response rule rather than changing behavior to fit the observed record.

Second, the candidate explanations retain the same possible next recipients after every history. If an award is possible under one explanation and impossible under another, that award could decisively distinguish them. The shared-possibility condition excludes this shortcut.

Think again of the coin. Both fixed biases permit heads and tails at every toss, yet the biases can be distinguished in the long run. Shared possible outcomes therefore do not suffice. Third, the model requires the separation between candidate next-award distributions to be controlled by the remaining chance outside the favorite, in each direction of the comparison. As opportunities diminish, the experiment's ability to separate those explanations diminishes with them.

For the bounded input differences in this allocation model, the required control can be established. A separate engineering measurement could nevertheless reveal an input's effect directly. Such evidence must be counted; the allocation record is only one potential source of information.[44]

Fourth, under each candidate explanation the cumulative lifetime chance outside the favorite must remain finite, apart from events assigned probability zero. A dominant observed share or several years of declining rival chances cannot establish that property of the whole process.

The strong-reinforcement models discussed in Chapter 2 supply one sufficient setting under specified assumptions. Ordinary positive feedback does not. A persistent route to trials can preserve accumulating comparison; chances shrinking fast enough can have a

bounded lifetime total. Similar observed shares can arise from either arrangement.

Under all these conditions, the complete history cannot support a single contribution-estimation method that becomes reliably accurate under every candidate explanation. For a contribution target that differs across the explanations, no universal procedure eventually recovers the right value everywhere merely by continuing to watch this one history. This includes procedures more sophisticated than simply equating sales with merit. Better calculation cannot create the missing separation.[48]

The technical step concerns collections of histories. Under these conditions, a collection assigned probability zero by one explanation has probability zero under the other. Their probabilities need not be equal otherwise. A universally reliable method would have to separate histories enough to return different contribution values under different explanations. The model establishes that continuing this experiment cannot deliver that separation. A method could work under one candidate, but cannot promise accuracy under them all.

Under the theorem's conditions, continuing the specified history cannot deliver an attribution procedure whose error vanishes under all candidate explanations. Operational experience can remain useful for other purposes. The limitation concerns the information produced, rather than an analyst's patience.

The result also reaches an apparently modest alternative: avoid estimating a precise number and simply decide which of two different contribution values is correct. Under the same conditions, there is no sequence of such tests for which both kinds of mistake vanish. One cannot escape the attribution problem merely by replacing a detailed merit calculation with a binary verdict.

This is a conditional statement about a class of economic histories, supported by classical probability tools. It is not a finding that actual markets always exhaust their comparisons. The ambitious part is the connection between a defined allocation quantity and a persistent limit on learning input effects. Keeping the conditions visible makes

that connection stronger, because it shows what evidence would be needed to apply or challenge it.

What the record still reveals

The attribution limit leaves many facts intact. Recorded outages, completion times, and accepted work can establish performance. Contracts can establish payment obligations. The result concerns the recovery of a defined contribution effect, rather than a general absence of knowledge.

Even the allocation record can support one explanation more strongly than another. Awards that repeatedly coincide with an observed productive advantage may increase confidence in its effect. The theorem limits the ability to drive attribution error to zero across the specified alternatives; it does not require all explanations to receive equal support.

MusicLab showed substantial variation across worlds without finding that musical appeal was irrelevant. Social influence changed the relationship between appeal and success. Recognizing alternative histories does not require treating every song as interchangeable.[43]

Execution, predictability, and attribution deserve separate judgments. Demonstrating that work was completed, correctly predicting who receives the next reward, and estimating how much an input changes that reward are different achievements. Strong evidence for the first two can coexist with uncertainty about the third.

Prior information can narrow the range of plausible explanations. A separate engineering test may rule out others. A buyer may also have enough evidence to choose a supplier while remaining uncertain about why the business succeeded. The information needed for that decision can be less demanding than a complete account of contribution.

That uncertainty matters through the decisions it could change. An organization may have enough evidence to renew a reliable service despite unresolved attribution, while an investor assessing a different

technology needs the missing input effect. Complete explanation is not a prerequisite for every practical choice.

Two water supplies, one city

In London during the cholera epidemic of 1854, John Snow investigated a comparison created by two water suppliers. The Lambeth Company had moved its intake upstream in 1852, away from London's sewage. Southwark and Vauxhall drew from a more contaminated stretch. Their customers were intermingled: different houses along the same streets could receive water from different sources.[49]

Snow sought the water supply of houses where fatal attacks had occurred. This was harder than consulting a tidy customer list. Residents did not always know their supplier. He described using receipts and, when necessary, a chemical test that distinguished the waters. The eventual comparison supported his account of water's role in cholera. The practical work included finding out who had received what.

The inferential opportunity came from a change in supply and the intermingling of customers. Snow did not observe the same household through two literal epidemics, one under each water source. He investigated an existing contrast and argued that the groups' surrounding conditions made it informative. That argument about comparability carried the causal claim; the bare existence of two companies would not have done so.

Before concluding that a record cannot identify contribution, investigators should examine variation supplied by other customers, policy changes, and additional measurements. A result about a sparse observation scheme cannot justify ignoring a richer one that exists.

Snow's example gives verification a precise role. Establishing which water a household received defined the comparison; it did not eliminate the need to assess whether the groups were comparable. Measuring an input and identifying its effect similarly require distinct, complementary work.

An economy need not reproduce its entire past to learn a causal effect. It needs evidence that distinguishes the particular alternatives relevant to that effect. Sometimes naturally occurring variation provides it; sometimes a designed trial helps. The finite-budget argument identifies a setting where continuing the specified record cannot finish that job. It directs us toward better evidence, rather than away from causal investigation.

Agreeing about the next allocation

Two explanations can converge on the same prediction for the next allocation while assigning different roles to productive inputs and accumulated advantage. Agreement about what comes next need not settle why the process took its present form.

David Blackwell and Lester Dubins established a classical result in 1962 about conditional forecasts coming together as information accumulates. Under an absolute-continuity condition relating the observers' probability laws, their conditional distributions for the future merge. This is a precise result with assumptions, rather than a general claim that everyone who watches the same facts must eventually agree.[50]

In my finite-budget setting, the required relationship between the fixed candidate history laws is established by the preceding argument. Blackwell and Dubins's result then applies: conditional forecasts of the future allocation sequence merge, even though a universally reliable recovery of the differing contribution values remains impossible. Both pieces matter. Forecast agreement follows from classical work; its coexistence with this attribution limit follows from the specified market model.

Forecasting follows the process already in motion. Contribution asks what an intervention would change within a defensible account of that process. Each can be useful without answering the other's question.

Predictive agreement alone does not establish an attribution problem. Return to the coin with a fixed but unknown bias. Imagine

two observers who begin with different reasonable probability distributions over possible biases. Watching the same independent tosses can bring their forecasts together while the observed frequency also reveals the bias. Their growing predictive agreement is perfectly compatible with successful parameter learning.

In the coin example, the fixed half-heads and three-quarters-heads explanations can be distinguished by a sufficiently long record. The observers' initial mixtures over possible biases and the laws generated by those two fixed biases are different objects.[44]

Imagine a forecasting competition that rewards only predictions of the next supplier. It might identify who best anticipates the purchasing process. It would not automatically identify who best understands the effect of an extra engineering month. Those are separate tests. A forecast score becomes evidence about contribution only when the design establishes the relevant connection, rather than assuming that good prediction must explain every cause of the outcome.

A map with uncertain settings

The distinction between prediction and parameter recovery has another established home: spatial statistics. Hao Zhang's 2004 paper studied models in which observations become more densely packed within a fixed region. Under the conditions he analyzed, not all the relevant covariance parameters could be consistently recovered, even with increasing observation density. Nevertheless, certain predictive conclusions could converge and a particular combined quantity could be estimated.[51]

Imagine measuring soil across a field and using the readings to predict conditions between sampling points. Two models might increasingly agree about those predictions while leaving some of their settings uncertain. The forecasts could become useful before every detail of the underlying variation was known.

The sampling arrangement matters. Zhang distinguished filling gaps within one fixed region from collecting data over an expanding

region. In reinforced allocation, continuing one established history likewise differs from observing additional informative beginnings. More data must be described by how they are generated.

Zhang showed how prediction could improve without identifying every underlying model setting. The allocation question has a related form: whether forecasts of the next reward can become accurate while the effect of changing an input remains unresolved.

A prediction of tomorrow's workload can remain valuable while leaving the effects of more engineering, another supplier, or a different access rule unresolved. The conclusion is about the scope of an inference, rather than the usefulness of prediction itself.

Consider an AI buyer comparing forecasts of its current system's responses with estimates of the benefit of switching systems. Its logs may be excellent for the first problem. They may contain few cases in which another system was tried under comparable conditions. This is a hypothetical purchasing problem, not a conclusion about any named model. Its point is that the relevant comparison follows the decision being considered.

What a simulation can supply

Simulation makes this distinction concrete. Specify starting resources, allocation rules, input effects, and responses to success, then run the modeled process repeatedly. The distribution of histories reveals what those assumptions permit.

A simulation can expose contradictions, show that an imagined outcome is rare under the chosen rules, or compare the implications of persistent and diminishing exploration. Its conclusions remain conditional on the rules and effects entered into it.

The MusicLab researchers could compare actual recorded outcomes across their experimental worlds. A simulated procurement market does not acquire the same evidential status merely because it contains eight worlds, eight hundred, or eight million. Increasing the number of runs can reduce uncertainty about what the simulator

implies. It does not independently validate the simulator's account of how engineering affects contract awards.

Take the attribution problem directly. Suppose two candidate input effects remain compatible with the evidence. Run the simulation under one and it supplies one distribution of shadow futures. Run it under the other and it supplies another. A precise average inside either simulation does not tell us which candidate governs the real procurement process. That is exactly the distinction the attribution question asks us to resolve.

Disagreement between defensible models can identify a useful experiment. Similar recommended actions may instead show that a decision is robust to unresolved attribution. Sharply different recommendations make that uncertainty material; displaying only a preferred simulation conceals it.

Yahoo's offline evaluation illustrates a disciplined use of an existing record. Its researchers specified a logging process and assumptions that allowed that record to evaluate another article-selection policy. The word “replay” there refers to an evaluation method using logged encounters. It does not mean that a complete alternative economic history was observed. What can be learned depends on the design and the question.[47]

Simulation should distinguish historical variation, input interventions, and institutional changes. All three can be investigated, but their interpretations differ. Shadow futures first specify the economy being replayed.

A comparison with a purpose

An unobserved alternative could have been superior, ordinary, or unsuccessful. A missing comparison leaves uncertainty among those possibilities. It supplies no presumption that the unrealized outcome would have been better.

The eBay advertising experiments provide a concrete corrective. As the introduction described, shutting off brand-search advertisements

largely shifted traffic to unpaid search results in that setting, undermining a simple interpretation of paid clicks as additional customers. The same research found a different picture for some other searches and users. The experimental comparison refined the claim rather than making all advertising equally valuable or equally wasteful.[52]

An informative experiment must be capable of disappointing the person who requested it. Results can strengthen an established arrangement or reveal costs that demonstrations concealed. A test designed to vindicate a predetermined winner cannot perform that role.

Estimating the effect of an input also differs from deciding who deserves the resulting gain. Contracts, risk, incentives, and moral claims enter that further argument. Statistical attribution cannot settle them through the word contribution.

A valid contractual claim to payment can survive uncertainty about all the causes of commercial success. A promise and a causal explanation answer different questions.

For competition policy, the relevant question is whether preserving a comparison would improve a consequential decision enough to justify its costs. Users may bear disruption, and some experiments may be uninformative. A missing comparison creates a problem to investigate, rather than a reason to subsidize every alternative.

Beginning again

One way to learn more is to observe independent repetitions of an informative experiment from a common beginning. Under appropriate identification and regularity conditions, repeated finite experiments can support consistent estimation even when extending one reinforced path cannot.[53]

Separate organizations do not automatically supply independent repetitions. Shared staff, reputation, shocks, or development can

connect their outcomes. Independence and a common informative beginning require a design, rather than a change in letterhead.

MusicLab separated its social markets so that popularity information developed within each world. The design created a way to observe differences a single market would not reveal. Applying an analogous design to suppliers would require identifying which influences actually cross between the proposed units. The useful lesson is about constructing a comparison, rather than giving every administrative subdivision the status of another world.

Repetition must still provide the relevant variation. If measured input profiles are identical across alternatives and no other observation separates their effects, beginning again cannot create the missing contrast. A large budget supplies opportunity; experimental design connects it to the target.

Nor does a repeated experiment have to answer every question. A trial could identify short-run reliability while leaving the consequences of years of accumulated experience unresolved. A buyer might deliberately choose that limited question because it is useful and affordable. The appropriate response is to state the horizon and scope, then use the evidence for what it supports. Modest, interpretable knowledge can improve a decision substantially.

Le Goff and Soulier compared what could be learned from one reinforced sequence with what could be learned from repeated trials. Changing the observation scheme can make a previously unresolved question answerable. In an economic setting, a new test or another measured outcome may supply evidence that ordinary transactions leave out.

The histories that get a chance

A productive economy can generate different histories, with real work and achievement along each. Shadow futures give those histories a precise place in an explanation of the one observed. Policy then poses the further question of which changes would improve the attainable outcome.

Existing evidence, additional measurements, new experiments, and redesigned allocation rules can support different decisions. The comparison budget describes a constraint within a specified process; shadow futures define histories left outside its realized record.

The next chapters examine institutions governing actual routes into markets. Bell's 1956 patent decree changed access to existing inventions under a public rule. Its consequences can be studied through historical evidence, without assuming every previously excluded inventor would have succeeded.[54]

The economic task is to identify which comparisons can teach us something valuable, how to preserve them, and when their benefits justify the cost. That task begins before the next winner is known.

The histories that became infrastructure

How far can a judgment travel?

A market can grow while the number of independent places to try again shrinks.

Imagine a carpenter who disappoints a buyer in one town. Another town might offer a different buyer, unfamiliar with the dispute and willing to commission a small job. Now imagine that both towns purchase through one platform, which carries the carpenter's ranking from one to the other. The second town is still there. The second judgment may have disappeared. The carpenter's ability to start again depends on whether another buyer can make a fresh assessment.

Adam Smith began his account of the division of labor with a pin workshop. Specialization raised what its workers could produce together. Its reach still depended on finding buyers. In his chapter on the extent of the market, Smith contrasted the narrow opportunities of scattered settlements with the commerce made possible by water transport. A larger reachable market could support work that a small village could not.[55]

Distance therefore did two things. It limited the customers an artisan could serve, and it could separate the people deciding whether to give that artisan a chance. Removing distance could expand both income and competition. Whether it also joined previously separate judgments is another question. A new road connecting buyers is different from one authority deciding which sellers every connected buyer may see.

The old town was hardly an automatic refuge. Smith also described restrictions imposed by guilds and settlement rules, including

obstacles to moving in search of work. A newcomer could meet a closed corporation rather than an open door. History offers no clean passage from free local markets to oppressive global ones. It offers changing combinations of reach, productive scale, and permission.[55]

Nor did formidable scale begin with software. By 1920, the Supreme Court's United States Steel opinion was recounting a huge industrial combination and disagreement over whether integration reflected the physical requirements of steel production or suppressed competition. Pig iron and finished steel were physical goods; that had not kept their producers small. Furnaces, transport, capital, and organization could already assemble power over a wide territory.[56]

The question is how far a common selection system can reach, and what remains outside it. One intermediary serving ten towns may offer customers more choice than ten isolated villages ever had. It may also carry one exclusion across all ten. Increased scale changes the institutional problem when the route to a fresh evaluation disappears.

In 1972, the physicist Philip Anderson called an essay “More Is Different.” Anderson worked at Bell Laboratories, and his subject was how new levels of physical scale and complexity could require new explanations. His title raises a useful economic question: when does an increase in reach change the kind of independence an economy can sustain?[57]

Consider what is being scaled. A bigger furnace changes the cost of producing steel. A railway changes where steel can be sold. A shared admissions rule changes which producer can reach those buyers at all. These changes can reinforce one another, but they are not interchangeable. The transition to digital markets adds another possibility: an evaluation can travel with the seller and be reproduced across transactions at very little cost. The important change is not that physical limits have vanished. AI's power supplies and computing clusters make those limits conspicuous. It is that physical reach, distribution, and judgment can become organized at different scales,

with one ranking governing opportunities across a much wider network.

Bell gives that question a physical object: a telephone attachment that a customer could buy, yet risk losing service by using.

An invention leaves the laboratory

The transistor existed before the government opened Bell's patents.

In December 1947, John Bardeen and Walter Brattain made a semiconductor device amplify a signal at Bell Laboratories. Their point-contact transistor belonged to a research program drawing on physics, materials, instruments, and the work of several people. In his Nobel lecture, Bardeen emphasized the earlier theoretical work and improvements in materials on which the discovery depended. The invention was an achievement, and it had a history.[58]

On January 24, 1956, a different kind of event changed what others could do with Bell's inventions. An antitrust settlement required extensive licensing of Bell patents, with thousands of existing patents made available without royalties. The settlement also imposed obligations concerning technical information. It left the Bell System integrated. This was an opening of technology inside a continuing institutional structure, years before the later breakup.[59]

Between those events lie two questions that are easy to collapse. How did a remarkable invention come into being? What could happen when other people were able to build on it? Bell's accomplishment supplied the starting material for the second question. Recognizing the importance of that accomplishment does not tell us which arrangements would allow the material to produce the most useful subsequent work.

The semiconductor story makes that distinction tangible. A transistor on a laboratory bench is not yet a reliable supply of components, an affordable radio, or a workable design in someone else's product. Moving between those stages requires further decisions and further learning. Whoever gets to undertake that work

can discover things the original inventors did not yet know. Whoever cannot undertake it leaves fewer results for later observers to examine.

This is one way a history becomes infrastructure. Earlier work leaves designs, machines, procedures, and skills that make some later projects easier. It can also leave permissions and restrictions that determine who may try them. The resulting advantage can be productive. Engineers using an established body of knowledge may work better because somebody has already paid to develop it. A serious account of opportunity must include that benefit.

Bell's records let us follow both routes. A technology could leave the laboratory through a license and a training program. A finished attachment could reach a shop yet remain barred from a customer's telephone. The invention, the right to build on it, and the right to use the result were separate economic opportunities.

Follow those opportunities rather than asking whether Bell was simply good or bad. One route led from a major laboratory to inventions others could use. Another led from an independent device to a customer whose telephone service could be threatened if the device was attached. The differences between those routes help explain why releasing knowledge and opening a market can have different consequences.

The price of a license

Bell had shared transistor technology before the 1956 decree. In April 1952, representatives of forty companies attended a nine-day technology symposium after paying a $25,000 patent-licensing fee. The program included a manufacturing visit. The Computer History Museum's account and collection identify this as part of a broader effort to spread semiconductor techniques. Access did not begin on the day of the settlement.[60]

That chronology prevents a convenient origin story in which the government released a wholly sealed invention and an industry immediately appeared. A licensing fee, a technical gathering, and

compulsory licensing represent different arrangements for passing on knowledge. The relevant question is what each arrangement enabled, for whom, at what cost, and with what further restrictions. “Open” is too broad a description to answer all four.

For an engineer, access can require more than permission to practice a patented invention. There may be a difficult production process to understand, a material whose quality matters, or a component that performs differently outside a demonstration. A legal document can remove a restriction without supplying every skill needed to use the resulting freedom. The distinction is familiar to anyone who has followed a recipe and discovered that instructions and competence are different possessions.

The symposium's combination of licensing and instruction is therefore revealing. Making an invention usable can involve transmitting practical knowledge alongside the rights. The economic object reaching another company is a bundle of opportunities and requirements. A cheaper license helps one part of the problem. It need not remove the other costs of turning an idea into something worth buying.

The government's contemporary account of the 1956 judgment recorded a requirement to furnish technical information at a reasonable charge. It also described annual patent lists and a mechanism for settling royalty disagreements. These were ways of making an access obligation operational. A general statement that firms should be allowed to innovate would have left the mechanics unresolved.[59]

The incumbent has a legitimate interest here too. Producing instructions, answering technical questions, and maintaining a stable interface require work. An access rule can shift those costs onto an organization that does not receive all the resulting benefits. Ignoring that work would make openness look cheaper than it is. Charging for every missing piece without scrutiny could make nominal access unusable. The design has to address both possibilities.

Bell's case thus begins with an economic sequence rather than a single unlocked door: permission, usable knowledge, practical implementation, and a route to a customer. Each stage can fail even when the previous one succeeds. The next case concerns a device that had already reached the stage of manufacture and sale. Its difficulty lay at the telephone itself.

A cup on a telephone

Hush-A-Phone was a cup-like attachment that fitted onto a telephone. It was intended to provide privacy for the speaker and reduce surrounding noise. The company had been selling it since 1921. By the time its dispute reached the federal appeals court in 1956, more than 125,000 had gone into use, according to the court's account.[61]

The telephone companies' tariffs prohibited attachments they had not supplied. They claimed the right to suspend or terminate service for violations. Vendors and users were told that Hush-A-Phone could not be used, and some distributors stopped selling it. The independent maker's ability to build a cup had not secured its customers' ability to use one.

The Federal Communications Commission had upheld the restriction. The appeals court examined the distinction between effects on the conversation and harm to the system. Using the cup for maximum privacy could reduce the clarity and volume heard by the other party. But the court saw no findings supporting the claimed injury to the telephone system generally. It set aside the Commission's order and sent the case back, leaving the antitrust questions open.[61]

This was a dispute about a mundane object, which is precisely why it is useful. We do not need to settle whether Hush-A-Phone was destined to transform communications. Its claim could be considered at the level of what the device did and whom it affected. A privacy aid might involve a trade-off that the two people in a conversation were capable of choosing for themselves.

A subscriber could cup a hand around the mouthpiece to seek privacy. The court asked why an attachment producing a similar effect

should be treated differently. The cup also left the hand free to write. A restriction presented as protection of the telephone system reached down to an ordinary choice about how to hold a private conversation.[61]

A general network cannot function if every user is free to damage everyone else's service. That concern deserves a technical answer. The question is whether a particular restriction follows from a demonstrated harm, a reasonable precaution, or a blanket preference for equipment supplied by the operator. Those possibilities lead to different rules. They should not be made indistinguishable by calling all of them protection of the system.

When a customer cannot use an independently supplied device, the market produces less experience of that use. Fewer people learn whether it is worth the inconvenience, whether a different design works better, or whether another application is possible. A prohibition can affect the record of performance as well as the maker's immediate sales.

The maker had already sold devices; this was no wholly untested invention. Its complaint went to the FCC in December 1948. Hearings followed in 1950, oral argument in 1951, and a final Commission decision only in December 1955. The appeals court noted the four-year delay after argument. Permission was a question about time as well as principle: an independent producer waited while the established system continued to operate. The court's record supports that chronology, not a calculation of the sales or discoveries the delay prevented.[61]

A footnote exposed the unequal starting positions. Telephone companies could offer equipment unless the Commission stopped them; an independent maker faced authorization before competing. The court called this unfair and said delay aggravated it, while leaving the legal question unresolved. A permission system can allocate time itself: one side continues accumulating customers while the other waits. That interpretation concerns a regulator's procedure as well as private control. It is distinct from the judgment on the cup.[61]

What happened after the decree

Decades later, Martin Watzinger, Thomas Fackler, Markus Nagler, and Monika Schnitzer studied invention after Bell's 1956 settlement. They compared follow-on patenting associated with Bell patents with carefully matched non-Bell patents. Their published finding was a lasting increase in innovation outside telecommunications, while the corresponding increase did not appear inside telecommunications, where customer access remained constrained.[62]

The design matters because a simple before-and-after count would be difficult to interpret. Semiconductor and other technologies were developing for many reasons. A comparison group provides a way to ask whether the patents affected by the licensing change developed differently from otherwise comparable patents. The credibility of that inference depends on the matching, the timing, and the absence of other differences that would account for the result.

The study's measure also has a scope. Follow-on patents and their connections to earlier patents are evidence about a particular dimension of invention. They do not put a monetary value on every resulting product, measure every improvement that went unpatented, or reveal what every unrealized company would have achieved. We can take the finding seriously without turning a patent measure into a complete account of social welfare.

The contrast points to a practical question: can a business actually use the knowledge it is allowed to obtain? A patent license may be worth little if the route to customers remains closed. The Hush-A-Phone dispute makes the telephone network's rules about use tangible.

This does not establish that every obstacle to entry should disappear. A buyer may have sound reasons to require reliability, security, and compatibility. A supplier may fail those requirements even with full access to the underlying knowledge. The valuable distinction is between testing whether an alternative satisfies the requirements and making its failure inevitable through a rule that prevents the test.

Bell's study measures follow-on patenting after a licensing intervention. To investigate the opportunities behind that result, we would also need to know which teams gained access, what work they attempted, and how their chances of reaching users changed.

The decree was an intervention in the economy's rules. It was not a replay of Bell's earlier history with a different lucky first award. That distinction helps explain what its evidence can do. It can inform our understanding of a change in access. It cannot show us every other route the same unchanged economy might have taken, or give each later inventor an exact share of credit for the resulting industry.

A list of patents is a record of inventions already made. A route to a customer can produce evidence about applications that have not yet been tried. Opening the first does not automatically open the second. This is why an institution can distribute a great deal of existing knowledge while leaving important opportunities for future comparison under the control of the same organization.

An advantage can become useful

One reason these choices matter is that success can improve the thing that succeeds. W. Brian Arthur's 1989 account of competing technologies made this mechanism explicit. Adoption can bring experience and improvement, attracting further adoption. Under the specified increasing-returns conditions, historical events can help select a durable technological outcome that need not have been the best available development path.[63]

Suppose, in a thought experiment, a widely used software system has trained operators, tested connectors, and a large collection of fixes. A rival's underlying design might be promising, yet the established system can be the practical choice today. Its accumulated advantages are services users actually value. The argument for examining its history does not require pretending those services vanish when we recognize how they developed.

Could another history have developed a better set of capabilities? Changing a price at the end may not recreate the development that

never happened along the way. A later subsidy can alter incentives; it does not automatically supply years of missing experience.

Bell's technology sharing makes the constructive side visible. Accumulated knowledge can become an input for other people's work. A strong incumbent need not keep every improvement inside its own boundary. An institution may preserve the benefits of a developed technology while allowing others to learn from and extend it. Whether a particular rule achieves that depends on its practical terms.

There is a further distinction from the strongest theorem in Chapter 3. Historical dependence, increasing returns, and technological dominance do not by themselves establish a finite lifetime comparison budget. Arthur's adoption model and my information argument answer different questions under their respective assumptions. The shared idea that early events can matter should not conceal those differences.

The machine and the workforce

The keyboard offers a familiar version of the problem, and a warning about telling it too confidently. Paul David's 1985 paper used QWERTY to explain historical dependence through the interaction of machines, trained typists, and conversion costs. S. J. Liebowitz and Stephen Margolis subsequently disputed his empirical account, including the evidence offered for the advantages of switching. The keyboard is a contested case, not a settled demonstration of an inferior standard trapping the economy.[64]

The underlying coordination question remains useful. Imagine an employer choosing equipment that existing workers already know how to use. Choosing it can be sensible even if a different arrangement might have worked well had training developed differently. To establish that switching would improve matters now, we would need evidence about both the alternative's performance and the costs of getting there. Pointing to a different imaginable beginning would not settle the decision.

Transition costs change what a comparison measures. A technology can perform well in a demonstration while slowing work during training. An afternoon's performance and a year's performance may answer different questions. The evaluation horizon must follow the decision, rather than be selected afterward to favor a preferred technology.

Hush-A-Phone sharpens the contrast. Its case did not require proving that every subscriber should use the attachment. A subscriber's reasonable choice about privacy was narrower than a claim that the whole network should be reorganized around the device. The scope of the requested opportunity mattered. Requiring proof of a total technological revolution before allowing a modest use would have set a peculiar standard for entry.

For an investigator of historical advantage, these examples impose a useful discipline. Identify the change being evaluated. Describe the cost of adapting to it. Look for evidence capable of distinguishing a promising alternative from an appealing story. That discipline protects the case for useful comparisons: its credibility should not depend on every famous account of lock-in turning out to be correct.

Finding out what nobody knew

Friedrich Hayek put discovery at the center of his account of competition. In a lecture delivered in 1968, published in the English translation cited here in 2002, he argued that competition reveals facts that would otherwise remain unknown or unused. Its point involves uncertainty about the particular results it will produce. If every relevant answer were already known, the discovery process would have little to discover.[65]

One outside learner was Tokyo Tsushin Kogyo, later renamed Sony. Its corporate history describes engineers studying a technical book and reports Kazuo Iwama sent from a 1954 visit to Western Electric. The team had to develop its own manufacturing equipment. Acquiring rights had not delivered a finished factory.[66]

Discovering a useful product and measuring each input's contribution to its maker's rewards are different tasks. A market can show that something works while leaving us uncertain about how its producer became the one able to offer it.

Licensing had not delivered a saleable radio. Sony's account describes a prototype, the TR-52, whose housing warped as the weather warmed in 1955. Plans to sell it were abandoned; the redesigned TR-55 reached the market later that year. A working transistor had left a practical problem unresolved.[66]

The distinction is visible in the objects: a licensed invention, manufacturing equipment, a failed casing, a redesigned product. An opportunity to develop can generate problems nobody has yet priced into a finished offer. Solving them takes further work.

At Hush-A-Phone's scale, a user could learn whether the privacy benefit justified the effect on a conversation. At a laboratory's scale, a company could learn how to turn a semiconductor technique into a product. These are different forms of discovery. Preserving the opportunity for them does not require a planner who already knows all the answers. It requires rules under which relevant answers can be produced and used.

There is no reason to expect that process to be costless or universally successful. A pilot can teach the buyer that a product is unsuitable. A development project can consume resources without producing an improvement. Competition's capacity to discover includes discovering disappointments. The economic justification for an opportunity concerns the value of what might be learned, assessed before knowing which result will arrive.

Who gets to accumulate a record

Merton's account of cumulative advantage in science connects recognition to the resources and attention that support later work. The advantage can change what a researcher is able to do next. It need not consist solely of an audience mistaking an unchanged piece of work for a better one. This is why a serious explanation of unequal success has to allow capability itself to develop along the way.[67]

Opportunity can produce the evidence used to allocate the next opportunity. Paid work finances development, exposes practical problems, and establishes a record. A weaker later application can therefore be both a consequence of earlier exclusion and a reason for another rejection.

The field experiments discussed in Chapter 1 isolated one part of this sequence. Van de Rijt and colleagues randomly supplied initial advantages and observed subsequent success in several settings. Their Kickstarter experiment found more later third-party support for projects given an initial donation. Larger initial advantages did not produce proportionate increases without limit. The causal evidence concerned specified interventions, rather than a universal law of runaway success.[68]

Lacking a record differs from being unable to acquire a useful one. The first is an observation; the second concerns a development process. Treating them as interchangeable allows an existing allocation history to justify its own continuation.

Bell's licensing change can be understood as altering one input into such processes. A firm with access to a useful invention could attempt work previously made more difficult by licensing terms. The work would then generate its own results. This interpretation places the opportunity between existing knowledge and later achievement. It does not assume that removing one obstacle removes all the others.

Alchian's account of profit-based selection supplies another established perspective. A process can select surviving firms without requiring every firm to foresee or deliberately optimize the

circumstances that favor it. In the Bell setting, the relevant environment includes the rules for using knowledge and reaching customers. Surviving within that environment and identifying the contribution of each input remain different analytical tasks.[69]

Selection and institutional analysis belong together. Success can reflect skillful adaptation to existing requirements. Whether those requirements were useful, whether alternatives could have developed under different arrangements, and what future changes would accomplish are additional questions.

Discovery inside the large organization

Bell's own laboratory supplies the other side. Bardeen described a field-effect proposal that initially failed. Investigating the failure directed research toward semiconductor surfaces; the point-contact transistor emerged from that program. The large organization could pursue an unpromising result until it opened another route. Productive internal discovery belongs beside the outside opportunities in this history.[58]

A patent list would not have performed that investigation. It required materials, instruments, theory, and people able to work together. Breaking those capacities apart might have made the experiment harder. Their value helps explain why opening a boundary and dismantling an organization are different proposals. The telephone attachment could be assessed without pretending the laboratory supplied nothing worth preserving.

The 1952 symposium adds a second dimension. Knowledge developed inside a large organization could also be transmitted beyond it. Internal research and outside experimentation can be complements. The question is how the arrangement connects them: what is shared, which uses are feasible, what costs are covered, and whether someone outside can produce a result that the larger organization would not have produced itself.

The relevant information set remains crucial. Internal experiments that reveal the input effect under investigation must be counted. An

outsider's inability to inspect those experiments raises questions of access and verification; it does not establish that the evidence was never produced. The next chapter examines that distinction.

Likewise, an outside supplier is not automatically a fresh, independent history. It may use the same underlying components, draw on the same training, or depend on the same route to customers. These connections can be efficient. They also affect what a comparison can distinguish. Counting company names without examining the underlying experiment would give the institutional arrangement an informational meaning it has not yet earned.

Bell's record therefore supports a question more specific than whether large firms innovate. Which kinds of learning does a particular arrangement make easier? Which does it discourage or prevent? What could a feasible alternative preserve? Those questions recognize the laboratory's productive organization while leaving room to examine the restrictions around it.

The company's own case

AT&T's annual report for 1955 described an experiment in Americus, Georgia. Equipment using transistors carried five conversations over a single pair of rural telephone wires. The report also described experimental use of a Bell solar battery to provide part of the equipment's power. Here was discovery inside the system, directed at a practical service problem.[70]

The report made the company's argument for integration explicit: the people designing equipment, manufacturing and installing it, and operating the service belonged to a unified organization. In discussing the decree, it welcomed the preservation of those relationships. It predicted that major services would continue and that revenues and earnings would not be appreciably affected. Those were the company's contemporary claims and expectations, rather than a later independent evaluation.[70]

The Americus example gives the argument something concrete to rest on. Coordinating a new component, its power supply, and an

operating service can require several kinds of expertise. The case for an integrated organization should therefore be examined as a claim about how work gets done. Dismissing it merely because the organization benefits from integration would leave the productive mechanism unexplored.

At the same time, successful coordination inside an organization cannot establish that every outside restriction is necessary to achieve it. A test of rural equipment and a rule barring a telephone attachment concern different activities. Evidence that the first worked does not settle the second. Following the actual activity keeps a broad institutional defense from standing in for the narrower comparison a particular decision needs.

Integration and specialization create empirical tradeoffs. An interface may introduce failures an integrated service avoids, or permit a component to improve without replacing the whole system. The consequences depend on the technical boundary and must be tested.

Requiring proof that an entire replacement system works before permitting a component test can make that test prohibitively costly. Such a requirement needs a technical or economic justification, particularly when the component can be evaluated safely on its own.

There is a matching mistake in treating the component's success as proof that the whole system could be replaced cheaply. A successful limited pilot can supply useful evidence about that pilot. It may say little about coordination at a larger scale. Advocates of access need to respect this boundary as much as advocates of integration need to respect the limits of evidence for their own arrangement.

The company's report therefore belongs beside the patent study and the attachment cases. Each speaks to a different question. A claim about productive coordination, evidence about follow-on invention, and a judgment about a particular restriction should be allowed to inform one another. They should not be collapsed into a vote on whether the entire institution deserves praise or blame.

A bounded interface can preserve much of an integrated service while allowing a relevant component comparison. Reliability, coordination costs, and productive gains determine whether that arrangement is worthwhile.

A rule for connecting

In 1968, the FCC addressed another independently supplied device: the Carterfone, invented by Thomas Carter. It connected a private two-way radio system with the telephone network. The telephone handset rested in a cradle; the arrangement allowed a radio user and a telephone caller to speak through the base station. This was a practical connection between systems, rather than a new public telephone network.[71]

The telephone companies treated its use as prohibited under their tariff. The FCC found that the Carterfone met a need and did not adversely affect the telephone system. Its decision rejected a blanket prohibition covering harmless as well as harmful interconnecting devices. The Commission preserved the possibility of reasonable standards and restrictions on devices that actually caused harm.[71]

The cradle makes the economic issue visible. A person could possess the radio, the telephone, and a device capable of joining the conversation, yet face a rule forbidding that combination. The performance of the combination could not be inferred from a telephone's success on its own. A new use required an opportunity at the boundary between established systems.

The FCC also allowed the network to change. Telephone companies could improve their systems and revise reasonable standards. Independent manufacturers would have to comply; an owner might have to rebuild a device or stop using it. Permission to connect did not oblige the network to preserve every old attachment forever. It created a boundary with responsibilities on both sides.[71]

That rule changes the entrant's task. Instead of seeking permission simply because its equipment came from elsewhere, it must satisfy protective requirements. Those requirements can themselves be

disputed and evaluated. The economic promise is a workable route to use, with a test of the claimed harm. Whether a particular implementation delivers that promise remains an empirical question.

Notice how the institutional question has changed since the patent opening. Permission to use an invention addresses one barrier. Permission to attach a device addresses another. A technical standard can address a third, while a customer's willingness to pay remains a fourth. A policy directed at one stage should be judged against what happens along the whole route, including the obstacles that remain afterward.

A right to connect becomes useful when someone can actually make the connection. The device must work with the network, and the terms must allow it to be used. Hush-A-Phone and Carterfone let us follow the gap between legal permission and practical opportunity.

Opening a route, paying for it

An experiment useful for future decisions can still be expensive for present users. Staff time, duplicate infrastructure, migration, and temporary reliability losses belong in the cost of preserving an opportunity.

There are costs on the innovation side as well. A policy that makes existing knowledge easier to use can affect the rewards expected from producing knowledge in the first place. That is a prospective incentive question. It requires attention to how research is financed and how the rule changes behavior. The usefulness of follow-on work does not remove it from the evaluation.

Bell's licensing intervention produced different innovation outcomes inside and outside telecommunications. A proposed policy in another industry needs to identify the knowledge at stake, the decisions it could improve, and the costs of preserving or replacing it.

The benefit is the improvement in decisions supported by additional evidence. An independent evaluation or better logging may

answer the question more cheaply; an adequate result may already exist. Some uncertainty will be too costly to resolve.

Timing can alter those costs. Once infrastructure is organized around a single approach, an alternative can become expensive to test. Preserving a limited route earlier may retain an option difficult to recreate later.

The proposal remains conditional. Early pilots can be badly chosen, poorly measured, or retained after they stop helping. An institution committed to learning should be able to close an unproductive program as well as open a useful one. Preserving comparisons is a means to better decisions. It is not an instruction to maintain every experiment indefinitely or to declare every participant entitled to continued support.

The histories inside the system

Bell's transistor, its technology-sharing program, the patent decree, and the attachment disputes belong in one account because they concern successive stages of economic possibility. Research produced something new. Licensing affected who could build on it. Rules about use affected who could reach an actual customer. At each stage, productive organization and institutional permission helped shape what happened next.

The resulting history contains real achievements. It also leaves questions about routes that were harder to pursue. We can investigate those questions through actual interventions and comparisons, as the patent study did. We can use shadow futures to state what remains unobserved. We should keep the evidence for an intervention distinct from the imagined histories that help formulate the attribution problem.

The cup and the cradle bring us back to the carpenter and the two towns. A boundary can let an independent invention reach an established network. It can also make the network owner the judge of what everyone connected to it may try. Digital markets extend that

question: how many apparent destinations still lead to the same decision about who deserves a chance?

New technologies need feasible routes to development, delivery, and evaluation. Accomplishments deserve to be understood within the arrangements that produced them, including the rules governing who can attempt the next one. The histories that become infrastructure matter because later possibilities must pass through them.

Who gets to be tested

Before the first query

The choice can be made before you type.

Open a browser on a new device and enter a question into its address bar. Unless you have changed the settings, a search engine has already been selected to receive it. You can have a strong preference for that engine, no preference, or no clear idea how the selection happened. In each case, the question goes somewhere, a result comes back, and a little more experience accumulates around the service that received it.

In the Google search case, that apparently ordinary sequence became the subject of an extraordinary record. Browser developers, device manufacturers, and wireless carriers had agreements governing default search placement. Google's payments for distribution exceeded $26 billion in 2021, according to the district court's findings. Default access was valuable enough to command an enormous price even when a user paid nothing to enter a query.[72]

The court did not explain Google's success by dismissing its product. Its August 2024 opinion credited skilled engineers, continuing innovation, and high search quality. It also found that Google had unlawfully maintained monopolies in general search services and general search text advertising through exclusive distribution agreements. Product accomplishment and unlawful exclusion appeared in the same account. Neither made the other disappear.[72]

Distribution also shapes what a search engine gets to learn. A query is an opportunity to serve someone. It can also generate information about what people want and how well the service responds. A distribution arrangement can therefore influence both today's

allocation of attention and the evidence available for tomorrow's improvement. The advantage does not have to remain a matter of habit. It can help make the favored service better.

That possibility should make the question more interesting, not easier. If an engine learns through use, a rival may need use to develop. Yet a user can reasonably prefer the engine that has already developed. The established provider's present quality and the rival's difficulty acquiring experience can reinforce each other. We cannot resolve that sequence by announcing that quality matters, or that exposure matters, and refusing to consider the other.

The phone also brings us back to Bell. A route to the customer can govern which alternatives produce a record. For the telephone attachment, the route involved permission to use a device. For a search engine, it can involve the place where a question first arrives. The technologies and legal cases differ. The economic question travels: who gets a meaningful opportunity to be used, improved, and evaluated?

An answer requires more than counting search engines. We need to follow the question through the system. Which service receives it? Which service generates the results? Who observes the response? Who can use that response to improve? Different arrangements can give different answers, even when the screen offers the same number of brand names.

Firefox tries another default

The public record contains a useful change of route. From 2014 through 2017, Firefox's default search engine was Yahoo. Previously it had been Google. According to the court's account, Google's share of Firefox queries fell abruptly from roughly eighty to ninety percent to roughly sixty to seventy percent. Yahoo's share rose from around ten percent to around thirty. In 2017, when the default returned to Google, the query shares moved back.[73]

The users were not compelled to stay with either provider. Some evidently took a route different from the default. That did not make

the default irrelevant. Its importance lay in how the arrangement changed the distribution of queries among people who retained the ability to choose. A friction does not have to be an absolute prohibition to affect who receives an opportunity.

This was an observed change in an actual browser, rather than a clean randomized experiment. Search quality, advertising, and commercial terms also belonged to the history. The court described Yahoo increasing advertising to meet its payment commitment, worsening the experience and contributing to Mozilla's decision to change back. The switch cannot be turned into a timeless estimate of what every default does in every setting.[73]

What it does give us is a concrete contrast. A statement that Google received most Firefox queries has a different meaning once we know that changing the preselected engine changed the shares. The outcome reflects an arrangement as well as individual responses within it. A responsible explanation should investigate both.

There is another lesson in the reversal. An opportunity to reach users can expose weaknesses in the service that receives it. Yahoo's place in the story does not establish that giving a rival distribution necessarily produces a superior product. The additional use can reveal a problem, finance an improvement, or change the commercial pressure on the provider. Preserving an informative opportunity does not dictate which of those results should emerge.

A successful experiment can strengthen the case for an incumbent. Conversely, a change of supplier can follow a poorly designed trial. The identity of the eventual winner cannot establish whether useful learning occurred. That requires examining the test, its outcomes, and the decision it informed.

Firefox's actual history also helps distinguish a changed rule from a replay. Changing the default changed the arrangement through which queries were distributed. It was an intervention. A shadow future under the old arrangement would hold the relevant rules steady and follow another feasible sequence. Both comparisons can help us think,

but the evidence from a changed default answers a question about that change.

When a query teaches a search engine

The district court described several ways that user activity helped search. Queries reveal what people ask. Clicks, returns to the results page, and other interactions can help assess relevance. Query data can help decide what belongs in an index and which pages need to be revisited. More use can improve the work of finding and presenting information.[74]

An unusual query makes that advantage tangible. An engine cannot learn from a question it never receives. Nor does one encounter necessarily reveal which result was useful. A click, an immediate return, and a later reformulation can tell different stories. The court described rankings informed by accumulated interactions. Scale supplies more than a larger inbox: it can supply repeated occasions to learn which answers people find useful.[74]

This is productive information. It would be a mistake to describe every additional query as empty repetition. The relevant distinction is what a query helps us learn. More encounters can improve the established engine's response to encountered tasks while leaving another question less well answered: what would a different service, or a different development route, achieve with a meaningful chance to operate?

A search result also reflects several decisions. The provider has chosen which pages to collect, how to rank them, and how to present the answer. A user's response concerns what was shown under those choices. It does not directly report the response to every result that might have been shown by another system. The record can be rich and still have an uneven relationship to the comparisons we want.

That unevenness gives the word “data” too much work to do. A catalogue of pages, a sequence of user interactions, and results produced by another company's engine are different resources. Each can help a competitor in a different way. Handing over one does not

automatically hand over the capability to generate the others. The remedy proceedings eventually had to address these distinctions in practical detail.

Our comparison budget asks a narrower allocation question: how much modeled chance accumulates outside the current favorite? It does not count queries, bytes, or people. Applying the model would require a defined allocation unit, a contribution target, justified probabilities, and an account of all informative observations. The court's search record is not an estimate of that budget, and a large search share does not establish that its lifetime total is finite.

The favorite here is the alternative most likely to receive the next allocation under the model. It can change with history. A service's share of past searches is an outcome, not that conditional probability; turning one into the other would require separately justified evidence and assumptions.

The historical mechanism and the model can nevertheless direct attention to a common practical problem. Opportunities to serve users affect what providers can learn and become. A challenger can be told to improve before receiving use, while useful improvement partly depends on receiving use. An institution concerned with competition has to examine how that loop works, alongside the benefits that existing scale supplies.

The comparison inside Google

The court record includes a particularly useful complication for a book about missing comparisons. Google evaluated its search quality against other engines using human-rated, side-by-side studies. It also compared its speed with Bing's. A 2017 comparison found a latency disadvantage and helped prompt Project Folly, an effort to reduce delays. The court described the project as successful.[75]

This is evidence of comparison inside the dominant organization. The provider could identify a weakness, examine a rival's performance, and improve. We should not discard that activity because the organization was large, profitable, or later found liable for

unlawful conduct. Those are different facts about the same institution. An account of what it knows has to include the tests it actually conducts.

The example sharpens the question for outsiders. A test of search speed can answer a useful question about search speed. It need not reveal how another provider would develop after several years of different distribution. Nor does it automatically establish the contribution of every input to Google's accumulated rewards. The comparison should receive credit for what it investigates, while leaving other questions open.

For a user waiting for results, a faster response is a real improvement. For a potential entrant, a public demonstration of a faster prototype may still fall short of obtaining a route to users. Both perspectives matter. A system can support productive internal learning while making some kinds of external development difficult. That combination is more informative than a choice between declaring the incumbent all-knowing and declaring that it learns nothing.

Project Folly offers a more useful test than a count of experiments. Google's comparison identified a concrete disadvantage, and work followed to reduce it. An outsider appraising that evidence should ask whether latency is the uncertainty affecting the decision. If so, the test matters. If the question concerns another provider's development under different distribution, measuring response time cannot answer it alone.

We should also distinguish a test's existence from access to its results. A private organization can possess useful evidence that a buyer, researcher, or public authority cannot examine. That can create problems of verification, incentives, or dependence. It does not justify pretending the evidence was never generated. The next step is to specify who needs to know what, and why.

A name for control over knowledge

Zahra Meghani called this kind of control an “epistemic monopoly.” Her 2021 paper examined regulatory agencies' dependence on product developers for the knowledge needed to evaluate technological products. She argued for access to credible sources beyond the sponsor and examined how broad patent rights could let a developer control conditions of independent research.[76]

For a particular question about contribution, epistemic monopoly means control over the opportunities needed to compare alternatives. An actor that controls who gets tested and who can inspect the results can help determine which explanations of the rewards will ever receive a hearing.

The distinction helps us ask something precise about the search case. Distribution contracts can affect who receives queries. Data terms can affect who learns from an existing record. A test's confidentiality can affect who can check the evidence. These controls can overlap, but they are not the same control. Each needs to be connected to the particular uncertainty under investigation.

Verifying invoices, hours, and completed work establishes inputs and execution. Identifying their effects requires suitable comparisons. Access to an existing informative experiment is a different problem from arranging an experiment that has never occurred. Those obstacles call for different responses.

This is why an information argument should begin with a question rather than a label. Which contribution are we trying to understand? Which observations could distinguish the relevant explanations? Who controls the route through which those observations would arise? The term becomes useful when it organizes that investigation. It becomes an accusation without content when it merely replaces “large technology company” with a more alarming phrase.

The product on the bench

Meghani's discussion has a physical counterpart to the search record: a product that outside scientists need to study. Her paper examined a dispute over the regulation of dicamba herbicides. The underlying court opinion, issued in June 2020, described Monsanto's field studies in support of an initial 2016 registration. The company had tested its reformulated herbicide in Georgia and Texas. At that time, it had not made the herbicide available for independent studies. The court's decision to vacate the registration rested on its assessment of risk.[77]

There were results, then, and there were experiments. The problem was not an empty file. A developer had produced evidence about its own product, while another route to producing evidence was unavailable. That is different from saying the developer's measurements must have been false. It asks what could be learned by investigators able to choose and carry out their own tests.

The court later vacated the challenged 2018 registrations on grounds concerning the agency's treatment of risks. It did not base that decision simply on a rule that every product lacked satisfactory data unless independently tested. The historical detail relevant here is the organization of inquiry before the earlier registration, not a general judgment about every herbicide or a statement of current regulatory status.[77]

A scientist with permission to read a company's white paper and a scientist with permission to test the product have different opportunities. The first can examine what the reported study says. The second can investigate another condition, reproduce a result, or find that the first study left a consequential question unanswered. The work may confirm the developer's conclusion. Independence matters because it permits a relevant inquiry, including one the developer did not select.

The example also shows why disclosure and access should be connected. Without enough information about methods, an outside scientist may struggle to reproduce a study. Without the product, a

detailed method may remain something that cannot be attempted. An effective arrangement has to follow the practical requirements of the investigation. The existence of published pages does not establish that every important question can be checked.

Disclosure is useful when the existing record answers the question. Releasing another copy of a performance report cannot replace a failure condition that was never tested. The first task is to identify whether the missing element is access, interpretation, or another observation.

This is not an argument that every outside investigator is better than every company scientist. Expertise, instruments, and experience matter. An independent test can be incompetent, redundant, or badly designed. The institutional objective is to enable credible investigation of consequential questions and to assess its results, rather than to treat independence as a substitute for scientific quality.

More signs, one route

The historical concern is the reach of a judgment. A rejection by one independent buyer can leave other buyers free to form their own assessments. A shared ranking can transmit that rejection across a much larger market. Demand can expand geographically while the routes to independent evaluation contract.

Shared services can also reduce prices, improve reliability, and connect entrants to distant customers. The question is whether those gains preserve a practical route to another judgment. Multiple websites and separate owners do not establish that independence by themselves.

Search syndication gives this distinction a real institutional form. A service can obtain search results from another provider and present them through its own product. This can make entry feasible without duplicating an entire search infrastructure. It also means that several user-facing services need not represent several independently developed engines. Their relationship has to be described before their number is treated as evidence of independent discovery.[78]

The reverse possibility matters too. A single operator might maintain genuinely informative trials of different approaches. Project Folly is a reminder that a large provider can learn through comparison, although its particular tests do not answer every question. Counting organizations could miss that activity just as counting storefronts could exaggerate the independence of their underlying systems.

None of this makes organizational structure irrelevant. Separate firms can bring different incentives, technical judgments, and routes to users. Common ownership can make some questions less attractive to investigate. The point is to establish which differences produce the evidence we need, rather than assuming that every legal boundary creates another informative history or that every internal boundary destroys one.

For a regulator or buyer, the useful map therefore follows functions. Who develops the service? Who distributes it? Who records outcomes? Who can change the experiment? Who can examine the result? The same company name can appear in several boxes. Different names can lead back to the same underlying process. The map tells us more than a count of badges on a screen.

Three different doors

The September 2025 remedies opinion and the December final judgment addressed several routes at once. The court did not require Google to sell Chrome. It restricted specified distribution conditions, required particular forms of data access, and provided for search syndication. These were distinct interventions with different objects. A headline announcing that Google must share would not tell a rival what it could actually obtain.[78]

One door concerned distribution. The judgment restricted tying the licensing of Google applications to the placement of specified Google services and restricted arrangements preventing partners from offering rivals. It also imposed one-year limits on certain agreements. These provisions concerned the route through which a

user encounters a service. They did not themselves give the rival a finished search engine.

A second door concerned data. The judgment specified information from Google's web index and particular user-interaction datasets used in ranking models. An index is a structured record of what the service has found on the web. User interactions can inform relevance and training. Giving a rival access to these resources could help it develop a service, but the judgment did not simply hand over all Google's algorithms, ranking signals, or trained models.[78]

The web-index provisions named concrete fields: document identifiers, their corresponding web addresses, when an address was first seen and last crawled, a spam score, and the device type. They also included information identifying documents Google considered duplicates. These are useful ingredients for another provider's work. They are not an instruction to make the same ranking decision. Nor did the access order transfer the intellectual property rights of third parties whose material appeared in the index. An entrant could still need machinery for collecting pages, interpreting them, and delivering results. The list makes the distinction between knowledge access and a finished capability inspectable: we can ask which specified resource removes which obstacle, rather than treat the whole search engine as a single indivisible secret.[78]

The third door concerned an operating service: syndication. A qualified competitor could obtain specified search results and features through an interface, subject to the judgment's terms. This allows a provider to use capabilities it has not reproduced for itself. It can lower the burden of serving users while the provider develops. It also raises a question about how the arrangement supports eventual independent capability.

The judgment limited reliance on that route. First-year Google syndication was capped at forty percent of a qualified competitor's annual US queries, with tapering over five years intended to encourage investment in independent search capabilities. The court left the pace and measurement of tapering for further determination.

Access supplied a bridge; the terms sought to keep the bridge from becoming the competitor's permanent foundation. Whether that would work required evidence from implementation.[78]

Market access, access to knowledge, and permission to use a technical service change different opportunities. A measure that opens one route can leave the others closed. Evaluation needs to follow the resulting ability to develop and test alternatives.

This is where a comparison perspective becomes more demanding than a general endorsement of openness. Which door produces the evidence relevant to the decision? If the buyer wants to compare interfaces, access to the same underlying service might suffice. If it wants to compare independent methods of generating results, that arrangement may preserve the very dependence being investigated. A remedy must be connected to the question it is expected to help answer.

The court's findings of legal violations supplied the basis for intervention. Choosing a remedy raises a further practical question: will a rival gain a workable way to produce evidence that could change a buyer's choice?

The cost of opening them

Search payments also financed something outside search. The court's 2024 opinion reported that Google's payment to Mozilla exceeded $400 million in 2021 and represented about eighty percent of Mozilla's operating budget. The financial connection supporting Google's default position also helped support a separate browser. A change directed at one market could therefore affect the resources available in another.[79]

Mozilla's chief executive, Laura Chambers, welcomed the court's rejection of a proposed ban on search payments in September 2025. She argued that a ban would weaken independent browsers. She also welcomed greater flexibility to offer other search and AI services and emphasized the need for careful implementation of data remedies. These were the arguments of an interested participant, with its own

experience and incentives, rather than an independent welfare verdict.[80]

The position is economically intelligible. A browser is the software through which a person accesses the web; a search engine supplies answers to searches. More competition in one does not automatically finance competition in the other. A payment can help sustain a browser while also making a particular search relationship financially attractive. The arrangement has to be examined across both activities.

The court considered the risk that a broad payment ban would harm distributors or leave them using Google without receiving revenue. It declined to impose the requested blanket ban. That conclusion did not mean every distribution restriction was acceptable. It meant the remedy had to be assessed in the system where its effects would unfold.[81]

For a comparison policy, the lesson is that one route can support another. Removing a financial tie might open a search opportunity while weakening a browser through which alternatives reach users. Keeping the tie might preserve that browser while maintaining incentives unfavorable to rival search providers. Describing either move as an unqualified increase in competition would skip the interaction that makes the choice difficult.

A remedy can also remove useful experimentation financed by an incumbent's existing business. Any appraisal must count evidence lost as well as gained, alongside changes in production, financing, and access.

This is not a demand that every alternative pay for itself immediately. A comparison can have benefits beyond its first transaction. It is a demand to identify those benefits and their costs. Hayek's discovery argument gives us a reason to value processes whose particular results are unknown in advance. It does not give us a reason to assume that every rearrangement of those processes increases the value of what gets discovered.[82]

Who can examine the test

Data-sharing disputes make the same problem concrete in a different way. A search record can contain sensitive information even when a name has been removed. In the remedies opinion, the court treated privacy as a valid concern and discussed techniques for reducing disclosure risks. The final judgment required safeguards, licensing terms, and information about the contents and preparation of the datasets.[83]

These requirements belong inside the informational analysis. Data made safer to share can become less useful for some purposes. A record whose preparation nobody understands can be difficult to interpret. A technically useful record might still be inappropriate for unrestricted circulation. The objective is a defensible opportunity to learn, with the people represented in the record included in the accounting.

The judgment's definition of a qualified competitor also involved more than asking for a copy. It included approved data-security standards, privacy and security audits, and a sufficient plan to invest and compete in or with the specified markets. Access was an administered relationship with obligations. It was not a public download button available to anyone who described themselves as a challenger.[78]

Such conditions can make a remedy safer and more purposeful. They can also affect who can use it. A small organization may lack the capacity to meet requirements that a large rival can satisfy. That possibility does not prove the requirements are wrong. It tells the designer to examine whether they are necessary and proportionate, and whether a less costly arrangement could preserve their function.

Mozilla's response raised a related concern: data remedies should avoid simply transferring power from one major technology company to another. That was a stakeholder's warning about design. It deserves consideration alongside the potential benefit of making useful resources available. The name of the recipient cannot substitute for

examining what the recipient will be able to test and how others will benefit.[80]

For a public evaluator, access to a test and access to its results are likewise separate. It may be possible to arrange an independent evaluation without releasing every underlying record publicly. It may be necessary to disclose methods so that a result can be interpreted. A confidentiality condition might protect legitimate information or prevent meaningful scrutiny. The right response follows the specific obstacle and purpose.

A published pass score may conceal performance on rare but costly failures. Detailed results could resolve that uncertainty without exposing every line of code. If the relevant condition was never tested, however, fuller disclosure cannot produce the missing result.

That distinction separates two forms of control that often travel together. One governs what happens and therefore what evidence is generated. The other governs who can inspect the evidence afterward. An institution can open its files while restricting new inquiry, or permit inquiry while keeping useful results difficult to check. A serious account of epistemic power must follow both.

A screen that can be tested

Mozilla's 2023 study randomly varied browser choice screens during simulated device setup in Germany, Spain, and Poland. Information, list length, and timing were experimental choices. One comparison held the five-browser screen constant while changing when it appeared.[84]

When the screen followed a click on the preset browser's icon, roughly nineteen percent selected that browser, compared with eleven percent during setup. The preset was Samsung Internet or Edge. The preceding click was a proposed explanation, not a separately established mechanism.[84]

Chrome was an alternative here. Timing could favor a position occupied by another brand.

These simulated choices do not establish sustained browser use or independent search capability. A remedy's sequence can nevertheless affect its immediate outcome.

The study changed a rule and observed responses. It therefore supplies evidence about an intervention. A shadow future would preserve the old rules while outcomes unfolded differently. The distinction keeps us from asking one experiment to settle both how a screen works and how a whole competitive history would have developed.

There is a constructive implication for the institution imposing a remedy. It can ask how to learn whether the remedy works. Which outcomes would show that a meaningful opportunity had opened? Which would indicate that users were merely clicking through another screen? What evidence would distinguish a temporary shift from a sustained improvement? Those questions should influence the record collected as the remedy operates.

They also create a route for revising an intervention that disappoints. A choice screen might need different information, a different timing, or a different role within a wider package. A trial can increase confidence in an incumbent or identify a problem with a challenger. An institution that values evidence should be able to learn both kinds of lesson without treating them as failures of principle.

The work after the judgment

The subsequent Google record brings implementation back into view. In their September 17, 2026 status report, the plaintiffs described the work of the technical committee, the development of processes, and efforts to implement the final judgment. The report discussed template licenses, requests for information, and exchanges over proposed terms. An obligation to provide access had become a sequence of organizational and technical tasks.[85]

This was the plaintiffs' account, rather than a court finding that every obligation had been fulfilled. It describes the practical route to access: who qualifies, which records can be used, how privacy is

protected, what documentation is supplied, and which terms remain disputed. A rival's opportunity depends on how those details are settled.

A deadline in a status report is a plan, not a completed exchange. The proposed dates for templates and responses did not show that every later step had occurred or that a rival had received useful data. To judge the remedy, we would need to follow those steps through to their effects.

The legal dispute continued beyond the district court's decision. Google publicly announced an appeal in January 2026 and argued that the ruling discounted user preference, product quality, and competition. It also objected to the data and syndication requirements. In July, the plaintiffs' appellate brief defended relief and challenged the rejection of a payment ban. These were opposing parties' arguments, rather than an appellate resolution.[86]

That continuing disagreement matters for interpretation. We should not describe a contested remedy as a completed experiment whose results are already known. We can examine the theory of what it is intended to change, the terms actually ordered, and the evidence produced as it operates. Later results could support, complicate, or undermine expectations on either side.

A formal promise of access becomes economically meaningful through specifications, a budget, security arrangements, usable records, and a way to interpret outcomes. Those implementation requirements help determine whether comparison actually occurs.

The same attention should be given to a claim that a remedy has failed. A route could be open but unattractive because the alternative is poor. It could be attractive but burdened by unnecessary conditions. It could be operating while its learning benefits take time to appear. The investigator needs evidence capable of distinguishing these accounts, rather than choosing one from the identity of the winner.

Before a rival can grow

An alternative can disappear before it leaves enough evidence to judge. A database of completed transactions will contain no entry for a product whose development stopped before its first sale. Sharing the incumbent's records cannot recreate the engineers' unencountered problems, the customers' unmade requests, or the improvements that might have followed. Existing knowledge and opportunities to produce new knowledge require different remedies.

Matthew's Gospel tells a brutal story of preemption. Herod hears of a child described as a future king and seeks to eliminate the threat before it can grow. In Matthew's account, he orders the killing of Bethlehem's young boys, while the child he seeks has been taken to Egypt. The threatened successor is to be removed before he can become a rival. The economic question is what happens when an incumbent can close a challenger's route before there is a performance to compare.[87]

That question already has an economic literature. Colleen Cunningham, Florian Ederer, and Song Ma studied pharmaceutical acquisitions and subsequent project development. Acquired projects overlapping the buyer's existing drugs were less likely to continue development; the pattern was stronger where competitive pressure was weak. Their research investigates a mechanism through which buying a potential competitor can prevent competition from arriving. These are empirical findings with identification assumptions, not a randomized test of every acquisition.[88]

An acquisition can also finance development that would otherwise stop. Counting today's competing products can miss the consequences of either decision: continued funding may let a project mature, while closure may remove a chance to learn what the entrant could have built.

The Google record must stand on its own findings about distribution and competition. The pharmaceutical study supplies no finding about Google's motives. But it helps state the timing problem: demanding that every prospective rival first produce a mature

superior service can be circular when the challenged restriction concerns the route through which that service could mature.

This proposed information rationale requires a causal account of the lost trial, a valuable decision it would inform, and a remedy worth its costs. Existing antitrust already considers innovation and exclusion. Shadow futures sharpen a question about the record we will inherit; they do not by themselves establish illegality.

Whose future is missing?

Return to the address bar. The question you type reaches an engine, and the engine returns a result. That useful act is real. So is the work that made it possible. The institutional question concerns the other routes by which useful acts might be attempted and evaluated, and the arrangements that govern who gets to learn from them.

For a buyer or public authority, the task is to make that question practical. Identify a decision worth improving. Find the comparison that could improve it. Ask who can produce and check the evidence, what doing so costs, and what existing knowledge can already resolve. Sometimes another route to users matters. Sometimes better access to a record or a well-designed internal trial supplies what is needed.

Rewards also carry personal consequences. What does a documented achievement establish about what its makers deserve? Before turning to AI's factories and workplaces, we need to separate evidence about contribution from the moral and practical reasons for rewarding it.

Who deserves the winnings

The song does not work harder in the world where it wins.

Lockdown, by 52Metro, is the same recording in the MusicLab market where it finishes first and the market where it finishes fortieth. The musicians have not composed a different chorus for the more successful world. Listeners arrive in different sequences and encounter different popularity records. The reward being distributed in this experiment is attention and downloads, rather than a documented royalty payment. But the experiment places an awkward question beside any argument that moves too quickly from success to deserving it.[89]

If we had seen only the winning market, what exactly would first place have entitled us to say?

That people downloaded the song, certainly. That it appealed to at least some listeners, also. That its makers deserved our admiration might be an entirely reasonable personal judgment. That the distance between first and fortieth measured an equally enormous distance in creative effort would require evidence the ranking does not supply.

This matters because we use the language of deserving to do several jobs. We use it to praise an achievement, enforce a promise, encourage a future undertaking, and justify the distribution of money. A sentence that sounds plausible in one of those jobs can become misleading when carried into another.

Consider a prize for building a machine. Its winner may deserve to be paid because the machine passed the announced test. A government might offer the prize because it wants more people to attempt the problem. An engineer might deserve particular scientific credit for the crucial design. Someone else may have supplied the financing that made construction possible. Those are four intelligible claims. They need not produce the same division of the proceeds.

The temptation is to find one number that settles all four. Profit is a convenient candidate. So is a salary, a patent award, or the size of a company. Each is already expressed in the language of measurement. Why not read the moral verdict directly from the account?

Because an account records what was paid under particular arrangements. Explaining what would have happened under different arrangements is a further task. Deciding which arrangements we ought to choose adds another. We can value the first record while asking the other questions.

In the previous chapter, the question was who gets an opportunity to demonstrate what they can do. Here it is what follows once someone succeeds. The distinction becomes easier to see beside a light bulb and a vehicle that can drive itself. In both cases there was a real achievement. In neither does acknowledging the achievement finish the economics of its reward.

The inventor and the bill

In his Nobel lecture, Shuji Nakamura describes buying a reactor to grow semiconductor material. The commercial machine had been designed for a different material from the gallium nitride he wanted. Some attempts produced no growth; others produced a black layer where the material should have been transparent. He spent a year and a half modifying the equipment. Buying the reactor was the beginning of the work.[90]

His two-flow design introduced another stream of gas that gently pushed the reacting gases toward the surface where the material grew. He reported uniform, high-quality growth over a two-inch surface. This was a physical change producing an observable result. The question of what the inventor deserved began with an achievement we can describe, rather than with the amount he was eventually paid.[90]

Nakamura's account also credits Isamu Akasaki and Hiroshi Amano, whose earlier advances helped establish gallium nitride as a workable route. His lecture explains how blue LEDs can produce white light by combining their output with a phosphor. The familiar

bulb therefore leads backward into a history of materials, equipment, and distinct technical advances. It does not lead to a single undifferentiated stroke of good fortune.

Someone who concluded from MusicLab that technical accomplishment is imaginary would have learned the wrong lesson. The reactor's performance supplies observations beyond a record of who received the next contract. Laboratory measurements, changed designs, and reproducible physical results can answer questions that a popularity ranking cannot. They belong in the evidence.

Now put a second document beside the lecture. In January 2005, Nichia announced a settlement with Nakamura, its former employee. The company's statement said it would pay approximately 609 million yen in remuneration and a further 235 million in interest damages. The agreement covered a broad collection of inventions, patent applications, and know-how from his employment, rather than just one blue-LED patent.[91]

The lecture and the settlement concern overlapping history, but they answer different questions. One explains technical developments through an inventor's account and cited research. The other records an agreement resolving an employment-invention dispute. Neither can simply be substituted for the other.

Interest makes the distinction particularly visible. Part of the payment concerned the passage of time. Adding interest changes what is owed without changing the semiconductor material, the reactor design, or the original experiment. It would be strange to conclude that the inventor's technical contribution grew by precisely the amount of interest accruing on a disputed payment.

The same point applies less visibly to the rest of an award. A legal settlement may reflect enforceable rights, litigation costs, compromise, and the scope of claims being released. Those considerations matter to the people involved. Settling their dispute does not require the payment to become a universal measure of the invention's social value.

Nichia's statement emphasized the risks carried by the employer and defended the settlement from its own perspective. That is an interested party's account, not an independent allocation of all credit. Yet the underlying economic question deserves attention: what should a company receive for financing uncertain work? Praising an inventor does not supply an answer by itself.

Imagine, as a separate illustration, that a laboratory needs both a costly machine and someone capable of making it useful. Without the machine, the experiment cannot happen. Without the capable operator, it fails. Removing either may destroy the entire result. If we describe each indispensable input as causing all the output, we have not discovered a way to pay out the same proceeds twice. We have discovered that contribution needs a specified comparison.

Replace the operator with whom? Replace the machine at what cost? Allow how long for another laboratory to discover the method? Ask about the next batch of devices, the next decade of production, or the eventual availability of efficient lighting? These are different questions. An answer about one does not automatically answer the others.

For the hypothetical laboratory, the relevant replacement might be an available machine from another supplier. For a historical invention, it might be another research team pursuing a different route. The word contribution can sound exact while quietly moving between those alternatives. A responsible calculation must say what changes and what stays in place.

This is why shadow futures need discipline. Replaying history under the same underlying conditions and response rules lets us examine how outcomes vary. Removing an inventor, changing a financing arrangement, or replacing a technology is a separate intervention. We cannot establish the effect of removing a person merely by imagining a less fortunate version of that person's career.[92]

Technical evidence can show that a reactor modification solved a production problem. Dividing the invention's full commercial and

social returns among inventors, employers, suppliers and predecessors is a separate question.

The distinction protects the inventor as well as the company. If the original employment payment were low, that fact alone would not show a small contribution. If a later settlement were large, its size alone would not measure the contribution either. Actual payment cannot serve as the evidence for deserved payment without making every existing arrangement certify itself.

With Lockdown, the two rankings expose a gap between a fixed recording and its reception. With a blue LED, measurements can narrow important technical gaps while leaving broader questions of credit and reward. There is no single level of uncertainty that applies to every claim about an achievement. We need to know which claim is being made.

A prize waiting in the desert

On March 13, 2004, fifteen driverless vehicles set off on DARPA's 142-mile Grand Challenge. The best managed about seven and a half miles. Nobody collected the million-dollar prize. In October 2005, five vehicles finished a new, 132-mile course. Stanford's Stanley won the two-million-dollar prize in six hours and fifty-three minutes. The prize account changed from no payment to a large one; the history of work was longer than that entry.[93]

A prize waiting at the finish line has a useful property: its purpose can be understood before anyone reaches it. It invites people to attempt a problem. The reward can help justify expenses, attract collaborators, and concentrate effort on a testable objective. Those reasons do not depend on pretending that the eventual winner will be the sole author of autonomous driving.

Stanley's finishing time established something specific about performance on that course under those conditions. It did not measure the total contribution of every member of the team, every competing team, or every earlier researcher whose work entered the

vehicles. A competition can deliver a clear winner while leaving that larger accounting unfinished.

There is no contradiction here. The prize organizer needs an administrable rule. The competitors need to know what qualifies. Paying the winner according to the announced rule can be justified by the value of a credible promise. The rule need not claim to reproduce a complete moral history of the technology.

Return to the first race. Nobody collected the million dollars. If the only evidence of worthwhile activity were a payment to a winner, the official prize account would show nothing to praise. But that is a property of the payment rule. It tells us that the specified threshold was not reached. It does not establish that no participant learned anything useful.

Equally, failure should not receive an automatic certificate of social usefulness. A vehicle that stops in the desert may reveal an important limitation or merely repeat a mistake that was already understood. Whether the attempt generated valuable knowledge depends on what it tested, what was recorded, and who could use the result. Failure is an outcome to investigate, not an achievement by definition.

DARPA announced the second challenge the very next day after the first ended. The decision preceded Stanley's victory by roughly eighteen months. It created another occasion on which teams could turn development into a public performance. We can inspect that institutional choice without assuming the organizer already knew which team would finish, or that every further contest would justify its costs.[93]

DARPA's documented sequence does not tell us what the optimal prize should have been. Doubling the headline amount between the two events was not a randomized test of prize size. The routes, technologies, participants, and time available also differed. The sequence demonstrates that an institution can announce a reward, observe failure, and arrange a further test without having solved every question of individual merit.

That is already a substantial lesson. Discussions of incentives often become arguments about whether a particular rich person is admirable. The more useful prospective question is what an offered reward induces people to attempt. The effect may differ across people, projects, and stages of development. Admiration is an unreliable substitute for investigating it.

Suppose a new contest promised a reward too small to cover even a credible attempt's costs. Increasing the prize might bring a valuable team into the competition. Suppose instead that all capable teams were already participating and a larger prize changed only the amount transferred to the winner. The two increases would have different consequences. These are illustrative possibilities, not findings about DARPA's actual entrants.

In 2007, the task changed again. DARPA's Urban Challenge required vehicles to negotiate a staged city, deal with other moving traffic, and obey traffic rules. Six of eleven finalists completed it; Carnegie Mellon's Tartan Racing team placed first. A desert finish had not settled performance in traffic. Changing the test could expose a different set of capabilities and produce another winner. The institution kept asking questions beyond who had won its previous prize.[93]

The comparison budget adds the modeled chances of opportunities going outside the current favorite. A cash prize is not itself that budget. A million dollars can reward a contest with many informative attempts, or a demonstration whose outcome was effectively settled in advance.

The races left more than a winner's name. How far did a vehicle travel? Did it complete the course within the limit? Could it handle traffic under the new rules? Those observations distinguish failed attempts and successful ones more finely than the payment account does. Any application of the information argument must count that evidence. Discarding it would create an ignorance the actual test did not impose.

We can therefore defend a prize without treating its winner as the owner of all the knowledge it helped generate. We can credit unsuccessful teams for identifiable useful work without deciding that every attempt should earn the same payment. The practical task is to arrange rewards and tests so that worthwhile effort and usable learning can happen together.

The finish line resolves who receives a promised prize. It leaves us with the next design decision.

Paying for an attempt

William Kerr, Ramana Nanda, and Matthew Rhodes-Kropf describe entrepreneurship as experimentation. Their account begins before a startup has a market price that can supposedly reveal its worth. Financing an initial test can uncover whether a larger investment makes sense. The ability to stop after an unfavorable result can make an undertaking feasible when an immediate commitment to its entire cost would not be.[94]

That observation gives capital a role more interesting than waiting for luck. Someone must choose a test, pay for it, interpret the result, and decide whether to continue. A good arrangement can save resources by learning early that a promising idea will not work. The person who closes the project may have made a useful decision even though no triumphant company remains to display it.

Staged finance distributes the cost of discovery. Equity, customer-funded development, and grants can each support an initial test before a larger commitment becomes sensible. Their consequences depend on what is learned and how the parties share its costs and benefits.

Kerr and his coauthors also examine the difficulty of prediction within venture capital. In data from one successful firm, the initial partner scores did not clearly distinguish its later spectacular successes from weaker outcomes. This concerns ventures the firm had already chosen to finance. It does not establish that investors lack skill

at selecting or developing investments, or that every rejected idea would have been equally good.

The successive races show why those abilities differ. A team unable to finish in 2004 might return with a better vehicle; a desert winner still faced a new task in a city. A backer could help make that development possible without knowing the final ranking. Requiring foreknowledge as a condition of financing would defeat much of the purpose of financing an experiment.

The incentive argument is prospective: expected returns can change which uncertain projects get funded. Its strength depends on how people respond, which attempts disappear under different terms, and whether alternative financing would support them. The word incentive begins that inquiry.

Contractual entitlement has another basis. Investors and workers can act under agreed terms, complete the promised work, and acquire a claim to payment even when every causal contribution cannot be reconstructed. Reliable promises help people coordinate before outcomes are known.

That defense has limits of its own. A contract can settle what its signatories promised without settling every obligation to people outside it. If a hypothetical contract authorizes conduct that harms others, quoting the payment clause does not answer their complaint. Nor does a contract establish that its bargaining terms were the only possible terms under which the useful project could have proceeded.

The same financial share can have different economic explanations. Compensation for bearing risk differs from returns obtained through exclusive control of an indispensable route to customers. Investigating alternatives and behavior is necessary to distinguish them.

The experimental view of entrepreneurship also gives us a reason to look beyond the conspicuous winner. Useful tests may occur inside a large company, at a small entrant, or through a publicly supported project. The label on the organization does not reveal whether it asks an informative question. We learned that in the previous chapter

when private experiments at Google generated improvements while separate questions arose about access to distribution.[95]

A defensible experiment specifies the uncertainty, the measured outcome, and the decision that follows. Hidden, uninterpretable, or irrelevant results are difficult to justify as public learning. A useful negative result can be valuable without producing a successful company.

This is a prospective defense of experimentation. It permits ambition and large rewards where they are justified by their effects. It also permits scrutiny of the arrangements surrounding them. We can ask for a future worth financing without first declaring that every past fortune was exactly proportional to its owner's contribution.

Different explanations for the same pay rise

A large pay rise does not identify its cause. It can accompany greater output, a different reporting arrangement, or a redistribution of an existing surplus. Those possibilities matter because changing incentives can affect them differently.

Thomas Piketty, Emmanuel Saez, and Stefanie Stantcheva make a related distinction in their analysis of top labor incomes. They separate responses to taxes into changes in productive activity, changes in tax avoidance, and changes in compensation bargaining. The same movement in reported earnings can carry different implications depending on which behavior produced it. Their tax analysis does not require an exact moral audit of every person's pay.[96]

A productive response changes useful output. An avoidance response can change the location or form of reported income without a corresponding change in production. Treating the same earnings movement as the same economic event would confuse those mechanisms.

Bargaining supplies another mechanism. In the authors' framework, bargaining can redistribute income without enlarging total output. An actual negotiation may also produce improvements,

so applying that account requires evidence about the particular activity.

Historical position adds a separate attribution question. Productive work and an early advantage can operate together: opportunity supports development, and development improves later performance. Neither mechanism has to be fictitious for an exact division of the resulting income to remain uncertain.

Being helped by an early advantage does not establish misconduct. Historical position can matter alongside honest productive effort, without either accounting for the entire reward.

Likewise, lower earnings do not establish inferior effort or prove superior virtue. They are outcomes to explain through work, opportunity, bargaining, and institutions before turning them into judgments about character.

Insurance against income loss, strong rewards for useful investment, and access to future evaluation pursue different purposes. Their effects and costs need evidence; none follows mathematically from the existence of alternative histories.

Casey Rothschild and Florian Scheuer provide another useful warning against an easy verdict. Their taxation model lets income combine productive work and rent-seeking that delivers private returns exceeding social returns. The authority need not observe each component separately. Changing taxes also changes the relative attraction of activities, so the effect depends on where the external costs fall and how people respond. A measure of rent-seeking alone does not determine the right corrective income tax.[97]

Resources devoted to useful engineering can respond differently to tax policy from resources spent pursuing privileges. A broad income tax can affect both activities, even when a narrower correction would distinguish them.

Other participants may change their behavior too. Altered returns can make productive work more attractive, or encourage new attempts to capture privileges. A description of waste alone cannot determine which response dominates.

Chapter 3 explains when an expanding allocation record can remain insufficient for universally reliable contribution recovery. A revenue chart cannot establish those conditions. Public finance can still analyze policy without settling every part of an income's history.

The practical implication is to state what information a proposal actually needs. An income-insurance rule might need earnings and eligibility records. A policy encouraging experiments might need evidence about costs and responses. An exact retrospective division of a fortune into contribution and position asks for something more demanding. These projects should not be treated as though they must all pass the same impossible examination.

The risk that gets counted

Risk falls on more than investors. Researchers commit time they cannot recover; employees depend on continued work; competing projects may lose support. Counting only capital exposure can settle a moral argument by excluding other consequences before examining them.

We should nevertheless distinguish them. Losing an investment and losing a job are not interchangeable experiences, and an investor may have claims arising from an explicit financing agreement. The point is to ask who bears which consequence. A complete argument about deserving cannot count the winner's uncertain capital while treating everyone else's uncertainty as background scenery.

The DARPA contest supplies a clean separation. A prize offers an upside for reaching a defined objective. It need not fully insure every entrant against failure. If we were designing another contest, we might consider grants for qualified attempts, shared testing facilities, or different prizes at intermediate milestones. Those are proposals to compare, with costs and possible distortions. They are not conclusions established by Stanley's victory.

An initial grant could let a capable but poorly financed team enter. It could also support a weak team whose attempt teaches little. A large final prize could encourage demanding engineering. It could also

concentrate effort on the scored task while neglecting an unmeasured concern. Choosing the arrangement requires an account of the behavior we want and the evidence we can obtain.

The Kickstarter experiment from Chapter 1 provides a useful check on our intuition. Van de Rijt and his colleagues gave an initial donation to a randomly selected half of two hundred new, unfunded projects. Treated projects became more likely to receive subsequent support from other donors. That is evidence about the effect of an initial advantage on later support, rather than evidence that recipients had suddenly become more deserving.[98]

Now imagine using such a result to design a program supporting new projects. We would want to know which projects the program brings forward, what they produce, and whether the additional donations finance useful work. A finding about subsequent support does not answer all those questions. It gives us a reason to investigate a mechanism that a simple comparison of successful and unsuccessful fundraisers could miss.

It also separates two ways of defending the first payment. One defense says the recipient had already earned it through superior work. Another says the payment could help a worthwhile undertaking become feasible. The second can be persuasive even when we cannot establish the first. Conversely, evidence that a donation attracts other donors does not establish that public money ought to supply every project's first donation.

The value of a proposed experiment is prospective. It should be judged using the information reasonably available when it was funded, including what failure could teach. Later commercial success alone cannot answer whether the original test was worth attempting.

Security can affect participation in experimentation. People unable to meet ordinary expenses after failure may decline a promising attempt. Whether a particular insurance arrangement makes useful experimentation more feasible depends on its financing and behavioral effects.

Claims about risk-bearing deserve equally specific evidence: what was financed, which losses were possible, which alternatives existed, and what behavior the terms induced. The size of a final fortune does not settle those questions.

Nakamura's account separates the purchases from the learning. The reactor existed before it could grow the material he needed. Its owner financed an uncertain undertaking; modifying it required work that the original purchase did not supply. Nichia's statement emphasizes employer risk, while the lecture describes technical changes. Putting the two documents together makes each claim more concrete without making their eventual settlement a scientific measurement.

There is also a difference between bearing uncertainty and creating it for someone else. In an illustrative procurement system, a supplier might invest in reliability and accept the chance that buyers will prefer a rival. Or it might arrange to make rivals' performance impossible for buyers to assess. Both activities can involve expense. Expense by itself does not establish equal social usefulness.

This distinction will matter in the taxation chapter. A possible charge for destroying valuable comparison would need evidence of an external cost and an effective instrument. It would address conduct and consequences. It could not be justified simply by observing that the successful supplier is wealthy, or by treating every obstacle encountered by a rival as socially wasteful.

Decent treatment and an entitlement to a contract remain different matters. An opportunity can be humane to preserve yet too unsafe, uninformative, or costly to purchase. Respect for workers includes taking the needs of those relying on the service seriously.

The risk argument becomes more constructive once it stops serving as a badge attached only to winners. We can investigate arrangements that enable useful attempts, allocate losses tolerably, and preserve reasons to work well. We can abandon arrangements that perform badly. The choice is between particular institutions with particular

consequences, rather than between reverence for a fortune and indifference to enterprise.

What I mean by deserving

I favor substantial rewards for useful work and for making valuable experiments possible. I also favor a level of security that lets people survive failure, and practical routes by which capable alternatives can obtain a hearing.

For Lockdown, they mean I can admire the recording without believing that its position in one market determines the worth of its makers. I can respect a payment promised under an agreement without insisting that the agreement measured every contribution perfectly. I can want a system that encourages more music while admitting that its winners will not form an exact ranking of effort.

Income support can protect households without generating evidence about their work. An experiment can generate evidence without protecting their incomes. Identifying the intended purpose prevents one benefit from being treated as if it automatically supplies the other.

A substantial reward can compensate investment without necessarily granting permanent control over the routes future competitors need to establish themselves. The role, duration, and alternatives to exclusivity deserve their own analysis.

This does not mean we should revise every contract after discovering an inconvenient shadow future. Arbitrarily changing promises after people have relied on them can damage the possibility of future cooperation. A better starting point is to design terms prospectively, explain which obligations survive, and evaluate changes openly. Uncertainty about contribution calls for responsible rules, not permission to improvise against whomever happens to win.

Credit needs similar care. Robert Merton's account of the Matthew effect showed how recognition can accumulate around already eminent scientists. Recognizing that mechanism does not require

denying the accomplishments of eminent people. It invites us to examine how acknowledgment is assigned and how accumulated standing affects access to further work. His analysis is a predecessor to these questions, not a discovery made here.[99]

In the LED story, accurate acknowledgment can name particular advances and their authors. It need not wait until economists settle the division of every dollar earned from lighting. When more detailed evidence distinguishes contributions, use it. When it cannot sustain an exact division, praise can remain specific rather than expanding into a claim that all rewards have been morally calibrated.

A similar restraint helps with criticism. A disappointing reward can be evidence of bad rules, weak bargaining power, an unfortunate sequence, poor performance, or several of these together. The possibility of a shadow future does not tell us which explanation fits a particular person. It asks us to investigate before translating low earnings into a verdict on character.

An objection remains. If we loosen the connection between income and merit, do we weaken the motivation to do difficult things? That is a serious prospective question. We should examine how people respond to particular changes in rewards, rather than answer it by insisting on a claim about past attribution that the evidence cannot support. A useful incentive can be defended as useful.

The driverless-car prize makes that defense ordinary. We do not need to believe that the winner authored the entire technology in order to promise payment for completing a demanding course. We need a credible reason to think the arrangement is worth its costs. The burden becomes harder when the proposed reward includes control over future testing, because we must assess what learning that control permits or prevents.

Another objection points toward equality. If the rankings could have differed, why not give everyone the same reward? Because the premise does not establish equal contributions, equal costs, equal needs, or identical responses to incentives. Lockdown's divergent

rankings unsettle an easy inference from outcome to worth. They do not hand us a complete theory of distributive justice.

We still have to choose. I would rather make that choice with the uncertainty visible than hide it beneath an accounting entry. That stance leaves room for enterprise, contractual reliability, redistribution, and disagreement about their proper balance. It rules out the comforting shortcut that the realized economy must already contain the evidence needed to certify all its rewards.

From the prize to the power cable

The next wave of this argument will arrive with a very large electricity bill. An AI supplier can do valuable engineering and commit substantial capital while its eventual rewards also depend on which systems get deployed first, which users encounter them, and which alternatives can afford a credible trial. These possibilities belong in the same inquiry.

The distinction between a prize and a contribution matters before a computing facility is built. Investors need some expectation of returns. Workers need compensation. Customers need a service that performs. Society may benefit from capabilities that no single payment fully captures. None of those needs makes construction free, and none makes the size of the eventual fortune a complete explanation of what caused it.

Keep the light bulb in view. Whatever we decide about the division of credit, semiconductor material still has to be produced and a useful device still has to work. Keep Stanley in view as well. Whatever we decide about the prize, a vehicle still has to complete the course. An economics attentive to possible histories must remain attentive to physical accomplishment.

That is the task ahead. Follow the actual inputs, follow what is tested, and distinguish the reward promised from the contribution we hope to learn. For artificial intelligence, the argument about deserving begins among chips, cooling systems, and power cables, long before it reaches a debate about who should keep the winnings.

CHAPTER 7

The intelligence factory

In the International Energy Agency's 2026 account of artificial intelligence, one of the consequential technologies is a connection to the electricity grid.

The report describes delays that have little to do with a chatbot's ability to write a sentence. Transformers can take years to arrive. In many jurisdictions, a grid connection can involve a wait of five to ten years. A company may have money, a plan, and customers eager to try its service while still lacking a route by which enough electricity can reach the machines.[100]

This is an unusual setting for a discussion of something described as intelligence. We expect intelligence to concern ideas. Here it also concerns a delivery schedule for electrical equipment. A better algorithm cannot, by itself, carry power down a transmission line that has not been built.

Nor can the phrase data center tell us what a facility will accomplish. A building can house many kinds of computing. The useful economic questions concern the equipment inside it, the work assigned to that equipment, and the arrangements through which its output reaches people. The same physical capacity could support different services and different opportunities to test them.

In its April report, the IEA estimated that all data centers together consumed about 485 terawatt-hours of electricity in 2025. A terawatt-hour is a billion kilowatt-hours, the unit familiar from electricity bills. Its central projection put consumption at roughly 950 terawatt-hours in 2030. That is an outlook, not an observation from the future, and both totals cover more than AI alone. The seemingly weightless answer occupies a material economy.

Someone must obtain the chips, install them, keep them working, and finance the interval before a useful AI product earns anything.

There are strong reasons for some of that work to occur in large organizations. There are also reasons to ask whether the resulting arrangements let other approaches demonstrate what they can do. Understanding both requires following the power cable into the building.

The machine needs a building

At Oak Ridge National Laboratory, preparations for the Frontier supercomputer involved a cooling plant, elaborate piping, and foundations capable of carrying cabinets weighing four tons apiece. The laboratory's published construction account records a failed joint in a large pipe during a pressure test. Water poured out; the team identified the failed weld and repaired it. Before the computer could do its work, the building had to survive the plumbing.[101]

The failed weld belongs in the account of the machine. Buying processors had not completed the job; the supporting systems had to pass their own tests. The pressurized pipe offered a plain measure: a joint had to hold before the planned operation could proceed. Acknowledging that work does not require calculating the crew's share of every future scientific discovery the computer will support.

Frontier is a scientific supercomputer with that proper name. When people call an AI model a frontier model, they mean something different: a system near the leading edge of relevant capabilities at a particular time. The distinction is worth making before the language conceals the machinery. A leading model is produced and operated on physical systems; it is not the building itself.

Begin with the chips. An accelerator is a processor designed to carry out suitable computations efficiently, often by performing many operations in parallel. Graphics processing units, or GPUs, became central tools for this work. A collection of them does not automatically behave like one enormous, dependable computer. They need data, memory, connections, and software that assigns and coordinates the work.

Imagine a team whose members must repeatedly exchange intermediate results. Adding people may help, but only if they can communicate quickly enough to keep the work moving. A training cluster presents an engineering version of that problem. A useful processor waiting for data or another processor is an expensive object accomplishing less than its specifications suggest.

Cooling creates another coordination problem. The electricity used in computation becomes heat that must leave the equipment. At Oak Ridge, an account of the cooling system describes circulating water warm enough to avoid the refrigeration required by older arrangements. The design uses the fact that water can carry heat away even when it would feel warm to us.[102]

That small reversal of intuition matters economically. Cooling is a design problem with alternatives, rather than a fixed surcharge attached to every computation. Improvements can change how much useful work a facility gets from a given expenditure. They can require knowledge that is embodied in installation, controls, and operating practice as well as in the chip.

At LUMI, the European supercomputer in Kajaani, Finland, the operator describes another destination for the heat: the city's district-heating network. Its account says excess heat warms hundreds of households annually. The physical by-product of computing becomes an input into another service.[103]

The important feature is the connection between activities. A facility in one place may be near a usable heat network; another may have no comparable outlet. A household's demand for heating can vary with the season while the computer's workload follows a different schedule. Evaluating the arrangement requires following those actual uses and alternatives.

The LUMI example therefore makes location more than a choice about cheap land. There must be a practical way to move useful heat to people who need it. We should ask what heating it replaces and what equipment and operating costs the transfer requires. The

operator's description establishes a particular use; the wider economic judgment needs a wider account.

That is relevant to a comparison of AI factories even when they offer similar services. Their private accounts can omit different benefits and costs falling outside their walls. A meter reading alone does not identify those effects. Neither does a claim that all computation is virtual. The heat has to go somewhere, and in Kajaani some of it goes into homes.

Counting processors can miss those differences. Counting electricity can miss what the equipment accomplishes. Revenue records what a company receives. Each measure answers a question; none alone explains the whole productive system.

The Oak Ridge construction record makes the missing work visible. Someone had to discover the defect during testing and correct it. That is a contribution we can describe concretely. We do not need an exact division of the supercomputer's lifetime scientific value to acknowledge that preventing a failed pipe from interrupting operation is useful.

The same lesson applies to AI facilities. Engineering competence can be real and consequential even when we cannot infer its entire economic contribution from the winning company's market position. Respect for physical accomplishment and caution about retrospective attribution belong together.

Keeping the chips working

Meta's technical report on the Llama 3 family provides a view inside large-scale training. For its largest language model, the researchers describe a fifty-four-day period containing 419 unexpected interruptions, in addition to planned ones. Their account also reports more than 90 percent effective training time. A long list of things went wrong, yet the operation continued doing useful work for most of the elapsed time.[104]

The two facts belong together. Reporting the interruptions alone would make the exercise sound like a collapse. Reporting the useful training time alone would hide the organizational and technical work needed to achieve it. Automation, diagnosis, and recovery turned a collection of fallible components into a functioning undertaking.

A single failed GPU could require the entire training job to restart. A processor that still worked but ran slowly could hold back thousands of others. Meta's report describes tools for finding those stragglers. The coordination problem was physical: many expensive processors depended on the pace and reliability of the components around them. Their number alone did not establish the useful work they could perform.[104]

The Llama team reported significant manual intervention was needed only three times during that period; automation handled the other issues. That is an operational claim with a specified boundary. An entrant renting well-run computing might purchase the benefit of such procedures without developing them afresh. The institutional question is whether the service lets the entrant investigate a different product, rather than whether every entrant owns its own maintenance operation.[104]

Scale can help in several different ways. The work of developing a useful system can be spread across many uses, provided the resulting service remains useful to those users. An operating team can support more than one project. A costly improvement in diagnosis may pay off repeatedly, rather than needing to be rediscovered for every customer.

A second possibility concerns idle capacity. Imagine a shared facility serving customers whose compatible workloads arrive at different times. Combining their demand could keep equipment occupied more consistently than separate facilities would. This is a conditional illustration of an advantage from pooling, not evidence that every large operator uses its machines efficiently or that every workload can be scheduled interchangeably.

Centralizing production need not make downstream services identical. Shared infrastructure can support different research

methods and applications. The comparison question follows what can vary above that common layer.

Renting access to well-run computing can reduce the need to duplicate physical infrastructure before attempting a product. The provider's expertise can make more projects feasible. Access terms determine which of those projects can actually proceed.

Now imagine requiring each supplier to reproduce the entire cooling plant, network, maintenance system, and staff before it may offer a service. That would create organizational independence at a considerable resource cost. We should not assume the resulting duplication produces enough useful information to justify itself. A desire for comparison is not a reason to ignore engineering economics.

An exclusive capacity arrangement can restrict downstream experiments while preserving useful operating expertise. The productive value of a facility and the informational consequences of its allocation rules therefore require separate assessment.

We would want to examine the reason for the reservation. A long commitment might finance construction that would otherwise be impossible. A priority rule might protect work that cannot easily tolerate interruption. Or a restriction might exclude an alternative despite feasible capacity being available. These possibilities require evidence about the particular arrangement; the existence of a large cluster does not choose among them.

Large shared computing is more than a hypothetical possibility. For 2026, the Department of Energy's INCITE program allocated access to seventy-five computational science projects at Argonne and Oak Ridge. Its published account describes scientific peer review alongside assessments of whether proposed work is technically ready for the machines. The organization operating the facility and the teams using it need not be identical.[105]

The listed projects include William Collins's work on large ensembles of weather extremes using machine learning and research on foundation models for materials. These are allocations for

proposed work, not evidence that every promised result was achieved. They make the institutional possibility concrete: a costly shared machine can support distinct questions chosen by different teams.

INCITE's policy document distinguishes technical readiness from scientific impact. Readiness alone is insufficient for an award. A proposal that needs preparation can be considered for preliminary access through another program. The distinction recognizes a practical problem: a team may need experience with the resource before it can demonstrate readiness to use a large allocation well.

Shared access replaces one expensive hurdle with another, smaller one: a team can apply for time instead of financing an entire facility. Applications still take work, reviewers can make mistakes, and capacity is scarce. The practical question is which promising projects can now be attempted.

Llama's interruption record also counters an easy version of the information argument. The company plainly learned about its own operation. A failing component, a useful recovery method, or a bottleneck in communication can generate actionable evidence inside one organization. Calling the organization large cannot make that learning disappear.

The more demanding question is which question the learning answers. An improved recovery procedure can establish how to keep this cluster operating. It does not, without further work, establish what a differently financed rival would have built or which product a buyer would prefer after a fair trial. A factory can become better understood while its place in the wider economy remains harder to explain.

Three different jobs

The equipment performs different jobs at different stages. In pre-training, a language model's numerical settings are adjusted using large collections of examples. One common task is predicting text from preceding text. The point is to acquire patterns that support later performance, rather than to write a useful answer for a particular customer at that moment.

Post-training then changes the system further. It can use demonstrations and preferences to improve instruction following, usefulness, or particular capabilities. Meta's Llama report distinguishes these stages explicitly. A system that has learned patterns in text is not automatically the assistant its developer wants to offer.[106]

Inference is the use of a trained model to produce outputs. A person asks a question; computation produces a response. The work may include additional steps, such as retrieving information or using tools. That use has a cost each time it occurs. Training a model does not eliminate the equipment, electricity, and operations required to serve it.

For an ordinary analogy, designing a machine differs from operating it to make the next item. The analogy has limits because a trained model is a collection of numerical settings that can be copied. But it helps prevent a common accounting confusion: the expense of developing a capability and the expense of using it are different parts of the undertaking.

DeepSeek's report on its V3 model separates pre-training, extending the context the model can handle, and post-training in its cost table. That division tells us that even a reported training figure can combine several activities. It does not represent a timeless price for producing intelligence in every setting.[107]

Entry can mean training a model, adapting one, or building a service using an existing model. These undertakings require different

resources and permissions. A claim about the cost of one cannot stand for the cost of all three.

A supplier that cannot finance a large general training run could still develop a valuable application. A supplier with a powerful model could still build a poor application. The difference might involve reliability, the customer's information, the surrounding software, or the way a worker uses the output. Counting large training runs does not count every opportunity for useful innovation.

Access to inference does not grant control of model development. A product requiring changes to the underlying model may need other permissions, technical capabilities, or a different design. The ability to submit a prompt does not supply every input needed to compete.

The word frontier also needs a task attached to it. A system can lead on one evaluation and be unsuitable for another use. Its position can change as methods and competitors improve. Treating frontier as a permanent title bestowed on a few companies obscures the very discovery process we want to understand.

In its 2026 report, Stanford's AI Index attributes more than 90 percent of the notable models in its 2025 dataset to industry. The dataset is curated for notability; it is not a census of every model or application. It gives evidence of industry's prominent role in this part of development, while leaving open the reasons for that pattern and the consequences of particular access arrangements.[108]

An assessment of entry must identify the layer being entered and the experiments possible there. A training strategy can be blocked while another useful application remains feasible, or application-level access can coexist with dependence on a single development frontier.

Two budgets

In 2022, Jordan Hoffmann and colleagues at DeepMind investigated how to spend a fixed amount of training computation. Their Chinchilla model used fewer parameters and more training data than the larger Gopher model, with the same total training compute. The researchers reported stronger performance across a range of evaluations. The study challenged the idea that making the model larger was the best use of the available resource.[109]

The concrete choice makes the word budget useful. A research team has access to computing for a period. It can spend that resource on different combinations of model size, training duration, and data. Experimenting with those choices can reveal a better recipe. The factory supplies a constraint; it does not dictate a single design.

The Chinchilla work also involved multiple smaller training experiments. This is a case of comparison produced inside a large organization, directed at a particular technical problem. It is evidence against an argument that scale must always suppress experimentation. Resources can make demanding comparisons possible.

Now distinguish the computing budget from the comparison budget. The first concerns resources used to perform computation. The second, in my allocation model, adds the chances that successive opportunities go to someone outside the current favorite. They are different objects. A very large computing budget could support useful rival tests, or mainly serve a product already selected for deployment.

Spending on competing training recipes produces different evidence from spending on repeated deployment of a fixed recipe. Both can be productive. Only observations relevant to a defined comparison belong in an account of what the spending establishes about that comparison.

Neither description can be inferred from the electricity meter. The meter tells us about energy use. To understand the evidence, we need the designs, measurements, and decisions attached to it. A low

electricity bill can accompany an informative test; an enormous bill can accompany repetition. The economic value of either depends on what is being accomplished.

Chinchilla also changed how its settings were adjusted, how text was divided into units, and the proportions drawn from different parts of the training dataset. The paper records those differences. Its result therefore compares two training packages; it does not isolate model size as the sole cause. The productive lesson is the researchers' ability to investigate a prevailing recipe. Future data, hardware, and uses could make another recipe preferable.[109]

Measured performance across competing services belongs in the evidence. The allocation-only limit cannot be applied by retaining the award record while discarding informative trials conducted before it.

Where deployment mainly reproduces the record of an established service, the questions concern how alternatives obtain use, which outcomes are checked, and whether tasks are comparable. A market share cannot answer those questions.

A million generated answers can reveal performance on encountered tasks without supplying the missing comparison. Applying Chapter 3's result requires specifying the observation process and counting every additional informative outcome and independent test.

Putting the two budgets beside one another helps us ask a constructive question. Can we use some of the computing resource to produce a more informative comparison at tolerable cost? Chinchilla shows the importance of technical experiments. A buyer's pilot might address a different uncertainty about practical performance. Neither requires treating useful scale as a mistake.

The five-million-dollar number

DeepSeek's V3 report contains a number likely to attract anyone interested in the economics of the intelligence factory: approximately 5.6 million dollars. It is the report's calculated cost for specified training work, using an assumed rental price of two dollars per H800 GPU hour. The authors explicitly exclude earlier research and experiments on architectures, algorithms, and data from that total.[107]

The qualification is part of the number's meaning. It does not turn an achievement into a failure. Producing an effective training run with fewer resources can be important. But a reader asking what it costs to create a capable AI organization needs a different account from a reader asking about the computing attributed to one completed training pipeline.

The distinction resembles the difference between paying for a successful reactor run and financing the attempts needed to develop the reactor. The earlier LED chapter showed how repeated modifications could make later production possible. An efficient final operation can rest on costly discovery. Measuring the operation accurately does not make those earlier costs vanish.

Published methods can reduce the cost of reproducing a successful approach. That is one benefit of knowledge transmission, and one reason reproduction cost differs from discovery cost.

Incomplete methods, unavailable data, and different operating conditions can still require fresh experiments. An expenditure figure in a research report does not, by itself, establish the cost at which another organization can reproduce the capability.

The same caution applies in the opposite direction. A company may spend heavily on research that does not improve its final product. We should not assume that every dollar in a large research program was necessary merely because the company later succeeded. To assess necessity, we need comparisons or other evidence about alternatives, rather than either accepting or dismissing the entire bill.

Kerr, Nanda, and Rhodes-Kropf's account of entrepreneurship supplies the established economic starting point. Early tests can inform later commitments, and the option to stop can matter. Bringing that account into AI asks us to examine the sequence of experiments and financing decisions. It does not give the final winner exclusive ownership of the concept of experimentation.[110]

DeepSeek's table is useful because it defines its boundary. Our job is to preserve it. A quoted training cost should tell us which work is included, how resources were valued, and which comparison the figure permits. Without that discipline, a number intended to explain production becomes a rhetorical device for whatever conclusion we already favored.

What can be copied

There is an asymmetry inside the factory. A trained model's numerical settings can be distributed without building an additional power station for each copy. Running the copied model still requires suitable equipment. Reproducing all the research that led to it is a further undertaking. Sharing the artifact, sharing access to computing, and sharing the means of discovery are different actions.

Kenneth Arrow's 1962 analysis of invention explains an older version of this difficulty. Knowledge can be costly and uncertain to produce while being relatively cheap to use or transmit once available. The social benefits of using it can exceed the rewards the producer can capture. Arrangements that create incentives to invent can also restrict its subsequent use.[111]

This is an established economic problem, not a new property discovered when AI appeared. What changes are the particular artifacts, costs, and control points. A model's settings might be available while its training data are not. A technical paper might describe a method while the equipment needed for a competitive implementation remains expensive. Different forms of access answer different constraints.

A common released model can support different services, customer problems, and error-reduction methods where its terms permit. Shared inputs can lower the cost of useful experiments without making their results redundant.

Suppose instead that our question concerns the training process itself: would another dataset or architecture have produced a better model? Two applications using unchanged copies of one model do not perform that comparison merely by having different company names. They may answer important application questions while leaving the training question unresolved.

That is why independence needs an object. Independence of ownership, computing provider, model ancestry, task assignment, or evaluation can each matter differently. Demanding independence in every respect could be prohibitively expensive. Assuming it exists because a storefront has a new name could be equally misleading.

Arrow's appropriation problem also gives an incumbent a serious argument. If immediate unrestricted access prevents it from recovering the costs of valuable research, some research might not occur. We need to examine that possibility. An argument for preserving comparison becomes incomplete when it treats the creation of the thing being compared as somebody else's concern.

The incentive argument still requires a comparison of feasible arrangements. Terms permitting returns and downstream experiments can differ from terms closing a customer route without enabling additional research. Both can be described as protecting investment while having different consequences.

We should compare feasible arrangements rather than compare an actual factory with an imaginary world in which discovery, equipment, and coordination cost nothing. Public support, contractual access, released artifacts, and competing private facilities can have different strengths and costs. Which is useful depends on the problem we want to solve.

The point is neither to award all knowledge to its first producer nor to require every producer to give everything away. It is to investigate

how the terms of producing and using knowledge affect later useful attempts. The factory creates possibilities. Its institutional arrangements determine who can act on them.

From computing access to a useful comparison

Computing access is one stage in producing evidence. A system must also reach relevant tasks, return interpretable outcomes, and permit those outcomes to affect a decision. These requirements connect infrastructure to discovery.

Task selection, comparable difficulty, valid outcome measures, privacy, and operational risk determine whether a service comparison is useful. The evaluation design has to justify its cost.

Distribution can close the route even when computing is available. A default that receives nearly every request can accumulate a record while alternatives obtain little ordinary-use evidence. Availability of an input does not guarantee access to an informative use.

This resembles the distinction between Bell's patent access and customer access, and between search technology and the distribution arrangements examined in Chapter 5. Possessing a useful input is a stage in the process. A practical route to being tried is another. We should not announce that the second problem has been solved merely because the first input is for sale.

Performance can nevertheless justify a strong default. An investigation must include measured quality, actual switching, available comparisons, and the costs of alternatives. Access-dependent evidence is something to interpret carefully, rather than to dismiss.

Limited, well-measured comparisons can sometimes retain the convenience of a default while supplying useful evidence. Their value depends on the uncertainty resolved, implementation costs, and the decisions they can change.

The comparison budget prescribes no universal spending ratio. Small persistent chances can accumulate indefinitely, and large

nominal opportunities can fail to identify a relevant effect. Evaluation concerns the process and its information.

Discovery also needs a route to action. A demonstration has less practical value if its result cannot influence subsequent adoption. Switching terms and interoperability help determine whether learning can change an economic arrangement.

At the factory level, this inquiry asks about available capacity and access terms. At the service level, it asks about routing and usable performance evidence. At the level of research, it asks whether another training strategy can be attempted. A single statement that AI is concentrated cannot answer all three.

Nor does having many firms guarantee that all three are open. Several services could depend on the same model and the same route to users. They could still innovate meaningfully in their own domains. The useful question is what uncertainty their differences can resolve, and what important uncertainty remains outside their experiments.

Taking counterfactual economies seriously requires specifying feasible alternatives, what their tests could establish, and the full cost of producing that evidence. The resulting comparison can justify preserving a route without treating every proposed competitor as entitled to infrastructure.

Who pays for the connection

The grid now returns to the story. On October 14, 2026, the IEA published another report on integrating data centers into electricity systems. It described connection queues containing duplicate or speculative proposals. A large queue can therefore present planners with both an urgent problem and an uncertain measure of how much demand will materialize.[112]

Imagine, solely as an illustration, one developer exploring sites on several grids while intending to build only one facility. Adding every request as if it represented a separate, certain future load would exaggerate the developer's eventual use. Ignoring the requests entirely

could leave the winning location unprepared. The planner needs evidence about commitments and timing, rather than faith in a single headline total.

Here is another economic record that does not explain itself. A proposed project, financed construction, installed computing equipment, and sustained customer demand are different stages. Moving from one to the next can change the risks facing everyone involved. Treating a proposal as completed capacity confuses a possible future with an observed one.

The IEA's October report also describes the importance of allocating connection costs and risks. If new infrastructure is built for a large customer that consumes less than expected or leaves early, somebody must absorb the cost. Contractual commitments and tariffs can determine how much falls on other users of the electricity system. These are institutional choices alongside the engineering.

The IEA's comparison gives AEP Ohio's large-data-center arrangements a concrete form: a ramp-up period, minimum capacity payments, a long commitment, and exit fees. The customer reserving infrastructure cannot assume its obligation ends whenever its appetite for computing falls. That shifts some risk of unused capacity toward the customer requesting it. The terms are a way to allocate risk, not proof that the resulting investment is optimal or that households bear no remaining costs.[112]

Operating commitments matter as well as construction commitments. In another illustrative arrangement, a customer could accept occasional interruptions in exchange for earlier or cheaper access. A computing job able to pause safely might fit that bargain; a service promising immediate responses might find it costly. The relevant unit is the particular workload and obligation, rather than an undifferentiated category called AI demand.

That choice connects the contract back to the engineers. Recovery procedures, stored progress, and scheduling can change which interruptions are tolerable. A physical improvement might make a more flexible arrangement feasible; an inflexible promise to

customers might rule it out. The institution and the machine have to be understood together.

We need evidence to distinguish those cases. A large electricity demand is not inherently a social loss. Nor does calling the demand innovative establish that every associated cost ought to be paid by others. The public question concerns useful output, displaced uses, environmental effects, and the allocation of risk as well as the private business plan.

The information question can also enter, but it must be specified separately. If a valuable pilot uses scarce capacity, it competes with other uses. The value of learning is part of the comparison, alongside electricity, equipment, labor, and the service forgone. Preserving evidence should not become a phrase that makes physical resources free.

Long commitments can support investment while restricting access to scarce capacity. The institutional question is whether a feasible arrangement retains the financing benefit and preserves useful independent experiments. Their costs can exceed their benefits; that possibility belongs in the appraisal.

This is the tension that a price or concentration statistic alone cannot resolve. A low price today can coexist with a poor route to testing tomorrow's alternative. A large producer can supply the very capacity that makes new experiments affordable. The economic task is to trace the mechanisms rather than assign one moral meaning to size.

Leaving the factory

The physical story has carried us from a grid connection to a model ready for use. Along the way, there were pipes to test, interruptions to overcome, training choices to investigate, and costs to define. These are productive activities. The argument about missing comparisons gains nothing by pretending otherwise.

But the completed training run is a beginning for the economic story. A model can be served repeatedly, incorporated into different products, evaluated on changing tasks, and routed toward some users rather than others. Those arrangements affect which observations accumulate and which alternatives develop a usable record.

Follow the next request. Someone chooses a system, the machines produce an answer, and a person or another program does something with it. Does that sequence teach us what we need to know? Does it improve the next decision? Can it reveal when a different system would work better?

The factory has made an answer possible. The next chapter asks what we learn when there are a billion of them.

A billion answers

Inference turns a trained model into a stream of outputs. Each interaction can also create evidence, but what it reveals depends on which requests arrive, how the model is selected, and whether the answer is checked.

The records can include the request, selected model, answer, time taken, and a user's acceptance. A check against an independently known outcome adds another observation. These records answer different questions because they arise from different measurement choices.

The trained model need not change its numerical settings during this exchange. A later development process might use collected examples to update it. The service could also retrieve a newer document or apply a different instruction without retraining the underlying model. We should keep those changes distinct when asking what another successful answer teaches us: the learning can occur in the model, in the surrounding software, or in the people evaluating the result.

Now repeat the exchange a billion times across many workplaces and tasks. The records can become immensely useful. They can expose recurring mistakes, reveal delays, and help developers improve a service. But the number of completed requests does not, by itself, tell us which questions the resulting evidence can answer.

One question concerns this particular order: did the service find the correct deadline? Another concerns the alternative model: would it have done better on comparable work? A third concerns the development process: which investment produced the capability? A fourth concerns an economic arrangement: what would have happened if another supplier had received the early opportunities and resources to develop?

We can learn a great deal about the first question without settling all the others. We can also design the service to collect evidence about more than one. The choice matters before the billionth answer arrives. It matters when we decide what happens to the first request.

Two answers to one question

Chatbot Arena offers a concrete departure from an ordinary exchange. In the platform described by Wei-Lin Chiang and colleagues in their 2024 paper, a user enters a prompt and receives responses from two anonymously selected models. The user compares them and votes; the identities are revealed afterward. The interaction creates a paired preference observation, rather than simply another conversation with a named favorite.[113]

The two columns on the screen do useful work. They bring alternatives into the same encounter, expose them to the same prompt, and delay the information that a famous brand might provide. We can argue about the design while recognizing what it adds. A record of which service people already use supplies a different kind of evidence from a vote after they have seen two responses.

Preference and correctness remain different. A polished wrong answer can be more convincing than a correct but awkwardly expressed one. Assessing accuracy may require evidence beyond the preference judgment.

Arena's paper itself distinguishes benchmarks based on human preference from those based on a known correct answer. That distinction helps us resist treating one ranking as a certificate for every use. A buyer interested in pleasant conversation and a buyer extracting deadlines may need different measurements. Neither has to dismiss the other's question to insist on measuring its own.

Prompts and users arriving at a public platform can differ from those in a specialized application. Long documents, damaged scans, or technical vocabulary can change performance. A general benchmark is evidence to assess for transportability, rather than an entire decision basis.

Unequal task assignments complicate performance comparisons. A service handling clear documents can have a higher acceptance rate than one handling damaged scans even if its underlying capability is no better. Evaluation must examine the assignment process as well as the score.

A paired test can help with that particular problem. We could send suitable examples from each category to both services and assess their outputs using a consistent procedure. The cost includes additional computation and evaluation. The benefit is evidence directed toward a decision that a single unbalanced stream would have made harder.

The Arena design also reminds us that comparison has to be produced. Someone builds the interface, selects model pairs, collects responses, and maintains the analysis. Those activities are part of the institution. The platform's methodological choices affect what a vote means, which differences it can reveal, and how confidently a ranking can be interpreted.

Suppose a purchasing department then uses a ranking to select its default supplier. The ranking can influence subsequent exposure. That possibility does not invalidate the original tests. It creates a new question about how the tests and the resulting deployment interact. A good measured result can become an input into a process that later generates a different kind of evidence.

Service and evaluation can be designed together. Delivering an accurate answer at an acceptable cost remains the productive aim; a suitable comparison can reveal when another system better serves it.

When the judge likes the packaging

Yann Dubois and colleagues tested how an automatic judge responds to presentation in their length-controlled AlpacaEval study. They prompted models to give concise or verbose answers. For one GPT-4 version, the unadjusted win rate ranged from about twenty-three to sixty-four percent as the instruction changed. The underlying model had not received a new training run; the prompt and resulting answers differed. Their statistical length adjustment reduced the swing. The experiment exposes sensitivity to presentation, while leaving us to investigate which changes also improve the answer.[114]

Each response faced a reference answer to the same instruction. The judge's preference supplied the score. It was a measure of which answer the evaluator favored, rather than an independent check against a separately established correct solution.[114]

Arena records users' comparisons of two answers. AlpacaEval uses the judgment of a particular automatic evaluator. An adjustment to that evaluator's scores tells us about its measurements; human voting needs evidence of its own.

Think of the deadline in the purchase order. One service returns the date and identifies the relevant passage. Another returns the same date inside a long explanation of the document. A buyer might prefer either presentation for a legitimate reason. An evaluation intended to measure the extraction task should nevertheless understand how much of its ranking depends on the surrounding prose.

Now imagine that the longer response contains an error elsewhere. The extra detail has changed more than the length of the answer; it has created another opportunity to mislead the worker. Alternatively, the longer explanation might expose an ambiguity that the short answer concealed. Length by itself cannot tell us which happened. The measurement must remain connected to the purpose.

The length-controlled study also makes a counterfactual question explicit: how would the estimated preference compare if the responses had the same length? Its statistical adjustment supplies an

estimate under a model. The authors are not claiming to have observed every possible rewritten answer or eliminated every source of evaluation bias. A defined counterfactual can be useful without becoming an omniscient reconstruction.

For a developer, the practical stakes concern which improvements an evaluation rewards. If a score rises after a formatting change, the developer should investigate whether users actually complete their work better. If better reasoning raises the score, that is worth distinguishing too. The same final ranking can emerge through changes with different consequences for the customer.

Who defines the test helps determine who receives the next opportunity. An accessible benchmark can support discovery, but its convenience is insufficient: its design must expose differences relevant to the intended use.

Choosing before paying

The next design decision concerns which models to call. Sending every request to the most expensive available system may waste resources on tasks a cheaper system handles adequately. Calling every model for every request can produce comparisons, but also an unnecessary bill. The problem invites a selector: software that routes the request according to the work it appears to require.

Isaac Ong and colleagues' RouteLLM research investigates such a selector. Their framework learns from preference data and routes between a stronger, more expensive model and a cheaper alternative. They evaluate the resulting tradeoffs on specified benchmarks and compare their methods with random routing. Their reported improvements concern those models, data, and evaluations, rather than a guarantee for every future workplace.[115]

The purchase order makes the motivation easy to understand. A short document with a clearly labeled date may require less capability than an ambiguous revision spread across several pages. A well-designed router could reserve the more costly service for the

cases where it adds enough value. That is a productive improvement even if the user never sees the selection happening.

Notice what the router must learn. It needs evidence about relative performance on kinds of requests. A general reputation for being stronger is not the same as an advantage worth paying for on every task. A useful selection rule takes variation seriously instead of reading an overall ranking as an instruction to make the same choice forever.

The RouteLLM paper also reports limits to transferring a router across tasks. Data resembling the eventual evaluation can improve performance. Correspondence among training examples, evaluation cases, and actual use therefore matters to the interpretation of a score.

Lingjiao Chen, Matei Zaharia, and James Zou explore a related route in FrugalGPT. Their system can call a model, assess the response, and decide whether to stop or call another. The answer already obtained becomes an input into the next decision. The study compares cost and performance on defined tasks, including classification and reading comprehension.[116]

In the workplace illustration, the system might first ask a cheaper model to extract the deadline. If its assessment of that response is unfavorable, it could ask another model. This differs from selecting once before seeing an answer. It also differs from obtaining an independent human check of correctness. The assessment mechanism itself needs evaluation.

FrugalGPT's paper includes cases where additional calls are made even though the models' answers agree. Its discussion treats this as room for improvement: the selector's uncertainty can cause extra expense. The example makes an important point about automation. Delegating the decision to a model does not remove the decision problem; it creates a system whose judgment about when to continue has consequences.

For a buyer, an economical cascade might lower the cost of trying alternative services. It might also make comparisons less visible if the final interface reports only one answer and one brand. We should

inspect what the system actually records. A useful comparison can occur behind a simple interface, while an interface displaying several logos can conceal dependence on the same underlying service.

Routing and cascades can make it cheaper to compare services as well as to use them. A buyer who can afford only one expensive answer may be able to check several cheaper ones. Whether that happens depends on what the selection system tests and records.

One distinction is essential before we try to measure a comparison budget. A router's estimate that one model will produce the preferred answer differs from the chance that the request will actually be assigned to that model. RouteLLM uses a predicted preference and a threshold to make its selection. The prediction and the assignment rule do different jobs.

A predicted seventy percent chance of producing the preferred answer is not a seventy percent allocation probability. A selector could send every request to the predicted winner. Adding the remaining thirty percent figures would then mismeasure the process that actually assigns work.

A different rule could deliberately randomize some assignments. Its allocation probabilities would describe that design, while its predicted performance scores would describe the selector's expectations. Both can be recorded. Their relationship matters to the analysis, but they are not interchangeable numbers simply because both can be expressed as percentages.

Checking answers reveals performance on the tasks a system encounters; independent evaluation can supply information about alternatives. Those measurements belong alongside the assignment record when we assess what can be learned.

The work of checking also has a production process. Someone may need to open an amendment, resolve an ambiguous field, or determine which document version governs the task. AI could assist with that work, but an automated score remains something to validate against the relevant standard. A service can lower the cost of

producing an answer while leaving costly uncertainty about whether the answer is useful.

The relevant chain runs from the observations training a selector through assignment rules, checked responses, and decisions using those checks. Following it distinguishes efficient production from useful comparison and unsupported precision.

The objectives still need to be named. A router optimized for the best current cost/performance tradeoff need not be optimized to discover a new supplier's potential. It may already have enough evidence for its operational decision, or it may need occasional new observations. We cannot determine which merely by noting that the routing is automated.

A record worth reusing

Long before today's language models, a news page posed a related problem. The Yahoo Today Module could highlight an article for a visitor and observe whether the visitor clicked. It could not observe that same visit's response to every article left unshown. Lihong Li, Wei Chu, John Langford, and Xuanhui Wang investigated how previously collected data could help evaluate another recommendation rule.[117]

Their work used records gathered through randomized article selection. Under their stated conditions, an evaluator can retain suitable events where a proposed rule would have selected the article that was actually shown. The retained record then includes a real response to that choice. The researchers compared offline evaluations with online results in the Yahoo application.

This is an instructive answer to the claim that one history must leave every alternative forever beyond reach. A historical record can contain deliberately created variation. The variation can support questions that the institution operating the system did not answer by simply repeating its current favorite. Collecting the right record changes the possibilities for later analysis.

Randomized request logs with checked outcomes can support evaluation of another selection rule under the relevant coverage and identification assumptions. A literal parallel economy is not required for every counterfactual question.

That last qualification matters. If a service was never available for a particular kind of order, the record may contain no relevant responses from it. Knowing that the probability was zero does not supply the missing performance. An estimator needs suitable observations or additional defensible assumptions, rather than an increasingly elaborate way of describing their absence.

The Yahoo authors call their method replay. Their use of the word concerns evaluating a recommendation policy with suitable logged events. My shadow futures concern alternative histories generated under specified economic conditions and rules. These are distinct uses. A procedure that reuses article records is not a time machine for recreating the entire industry with a different first winner.

Evaluation of current systems, variation across alternative development histories, and the effects of changing a development policy require different comparisons. Randomized request logs can help with the first without establishing how capabilities would have evolved after different early opportunities.

The same record can nevertheless be valuable. If it supports a better procurement decision, it need not explain every earlier fortune to justify its collection. The point of the comparison budget is to clarify a particular information constraint, not to demand a complete causal explanation of the economy before allowing any useful inference.

This gives us a practical reason to preserve details that a success count omits. Which alternatives were available? Why was this one selected? What was measured afterward? A usable assignment record and meaningful outcome can make a small pilot worth more, for a specific decision, than a much larger archive lacking those features.

The pilot also has costs. Someone receives the less suitable answer when the comparison goes badly. Someone checks the outcome. Some

data may be inappropriate to retain or share. The Yahoo research illustrates an opportunity created by deliberate observation; it does not imply that the opportunity is free or that every problem meets its assumptions.

The same ancestor

Our hypothetical buyer opens three services and receives three answers to the order question. Three storefronts are visible. What has been compared?

Perhaps each service has independently trained a different system. Perhaps each has adapted a shared model. Perhaps all three pass the request to the same supplier, with different surrounding instructions. The answers may still differ. The number of interfaces does not identify which part of the production process varied.

Meta's Llama 3 report supplies a concrete example of a relationship between models. It describes using its flagship model to improve smaller models during post-training. Different members of a model family can therefore be connected through the development process. Treating their outputs as evidence from wholly unrelated research histories would miss that connection.[118]

Shared ancestry does not make their performance identical. A smaller model can behave differently, require different resources, and fit a particular application better. Comparing those actual systems can teach us something useful. The mistake would be to suppose the comparison automatically tests every assumption inherited from their common origin.

Imagine that our three services extract the same wrong date because the underlying document representation loses the same footnote. Asking all three may produce reassuring agreement while leaving that error untouched. Now imagine that one service uses a better document parser and finds the footnote. A common language model has not prevented a meaningful improvement elsewhere in the service.

These are illustrative cases, but they identify an empirical task. Follow the chain far enough to understand which components are common, which vary, and where the outcome is measured. Independence is valuable relative to a question. It should not be a ceremonial word attached to a company name.

The same caution applies to repeated answers from one system. New requests and new checks can provide fresh information about that system's performance. Repeating a model call is not automatically repeating the same observation. A thousand carefully checked orders can expose failure patterns that one order could not reveal.

But generating a thousand responses without a new check does not create a thousand independent confirmations of the underlying claim. If the uncertainty concerns the actual delivery date, the order and its amendments remain relevant evidence. A longer conversation among programs might help interpret them. It cannot make an unread amendment irrelevant by agreeing without it.

Yaneer Bar-Yam's “From Big Data to Important Information” distinguishes detailed records from information useful for understanding an intervention.[119] For the delivery order, the decisive detail may be an amendment that thousands of generated answers have never checked.

The workplace can act on that warning. It can measure whether the extracted date is correct, whether the answer helped a worker complete the task, and whether a different service changes the result. Each measurement has a purpose. We gain clarity by naming those purposes before admiring the size of the resulting database.

Predicting the next choice

Forecasts of the next allocation can converge while explanations of productive contribution remain different. Chapter 3 states the conditions under which that distinction persists.

David Blackwell and Lester Dubins showed in 1962 how conditional forecasts can merge under an appropriate relationship between probability laws. Hao Zhang later established an example in spatial statistics where some underlying parameters cannot be consistently recovered even though important prediction can remain effective.[120121]

Forecasting which supplier wins an allocation is a different task from checking whether an AI answer is true. The answer needs evidence of its own.

Acceptance, correctness, and causal contribution are separate targets. Predicting that a user accepts an answer does not establish that it is accurate, or identify which investment caused the capability. A common interface does not make the underlying inference identical.

For the buyer, the next action could be straightforward. If errors matter, check a suitable sample against the original orders. If supplier choice matters, preserve a meaningful comparison. If a historical attribution claim matters, identify the intervention, horizon, and evidence that could support it. The inquiry becomes more useful as it becomes more specific.

Sixteen agents and one compiler

A swarm sounds like a crowd. On a computer, it can mean several copies of a model pursuing assigned work, using tools, keeping records, and returning results. The economic question begins with what those activities accomplish. Sixteen running programs can perform sixteen useful investigations, or spend the afternoon rediscovering one mistake.

In February 2026, Nicholas Carlini at Anthropic reported setting sixteen Claude instances to build a C compiler, which translates code into instructions a computer can run. Compiling the Linux operating system, they stalled on the same bug and interfered with one another's fixes. He redesigned the tests: GCC, an established compiler, handled most files while the new compiler handled varying subsets. Failures became easier to isolate. The reported achievement

still had limits, including inefficient generated code and dependence on other tools. This was an attributed capability experiment, not a measured replacement for a human workforce.[122]

The revised tests helped separate failures. The checking arrangement belonged to the productive machinery as surely as the model did. A candidate program needed reliable tests that could expose its mistakes.

There are different ways to organize that machinery. Anthropic's June 2025 research system used a lead agent to distribute searches among others. Carlini's compiler account explicitly described a setup without a lead agent assigning tasks. In either arrangement, ask who directs the work and supplies the feedback.[122]

Agent systems can organize candidate generation, execution, and checking, as the compiler account demonstrates. Their productive value depends on the full arrangement: human-defined objectives, tools, tests, and judgments about errors, as well as the cost of model calls.

Retained code, test cases, and usable project records can support later comparison when a model changes. The rights to those assets therefore affect the cost of developing and evaluating another approach.

Failed attempts can matter as much as final outputs. Preserving their records may prevent repeated mistakes, while requiring storage, documentation, and protection of confidential material.

Control of tools and records can make switching expensive even when software generation becomes cheap. This is an institutional question about portability and ownership, distinct from the capability demonstrated by an agent experiment.

Ownership alone also cannot establish validity. Carlini's account illustrates useful checking within one firm using related models. A legally independent project can still use a defective test. The analysis follows the route from a proposed design to feedback capable of exposing its errors.

A thousand agents can reproduce one mistaken interpretation; a single check against a relevant document can reveal it. Abundant software labor can multiply attempts faster than it produces trustworthy judgments of their results.

A town throws a party

In a 2023 study, Joon Sung Park and colleagues populated Smallville with twenty-five software agents. One received the intention to organize a Valentine's Day party. The authors report invitations spreading and five invited agents arriving at the cafe. Another resident mistakenly identified her neighbor, Adam Smith, as the author of The Wealth of Nations. Knowledge inherited from a language model had intruded into the neighborhood.[123]

These characters supply a tempting contrast with a compiler. A simulated town can be rerun, with rules changed and histories recorded. But what would establish that its response to a tax change predicts the response of people? Believable conversation and a program's successful execution do not supply that validation. The study investigated believability and agent architecture, not the accuracy of economic policy forecasts.

Simulated demand can reveal implications of assumptions and help design an empirical test. Its preferences may also reflect the instructions and training inherited from a common model. Multiplying simulated participants reduces some sampling noise without correcting a mistaken demand mechanism.

A shadow replay keeps underlying parameters, starting conditions, and response rules fixed while the random history changes. Altering procurement or training policy is a separate intervention. For either operation, the claim about an actual economy requires evidence linking the model to that economy. The party makes generated interaction visible. The compiler's checks point toward the additional work of connecting a proposed answer to a result that can prove it wrong.

When a generated answer can be tested

There is a more encouraging route from generation to evidence when a proposed answer can be checked directly. Google DeepMind's 2025 AlphaEvolve report describes a system that proposes computer programs, runs evaluations, and uses the results to guide further proposals. The language model helps generate candidates; executable tests supply feedback about what those candidates do.[124]

One application concerned scheduling jobs in Google's data centers. A machine can have spare computing capacity but be unable to take another job because another resource, such as memory, is already exhausted. The report describes searching for a better rule for assigning work to available machines. This is the factory's coordination problem turned into an object of automated investigation.

The researchers used a simulator based on historical workloads, tested on unseen recent data, and then measured operation after deployment. They reported recovering an average of 0.7 percent of fleet-wide computing resources that would otherwise have been stranded. That measures available computing capacity, rather than an equivalent reduction in electricity use.

The sequence matters as much as the headline. A proposal entered a test with a defined outcome. A promising simulated result faced another check. Deployment supplied further observations. The program was not considered better simply because another language model produced a fluent explanation of why it ought to work.

Compare that with the simulated party. Smallville can show what its agents do under its rules. A scheduling simulation can also be wrong about a real system. In the AlphaEvolve account, subsequent measurements provided an additional connection to actual operation. We should examine the quality of that connection while recognizing why it adds something that another unvalidated simulation would not.

For software generation, useful evidence comes from executed programs and checked outcomes. Candidate code can help organize a search, but its correctness needs testing beyond the model's ability to produce it.

The test set still needs attention. A program might exploit a quirk of the examples, perform well on clear documents, and fail on revisions. A good development process would investigate such limits. Automating the proposals makes the quality of evaluation more consequential because many candidates can now be optimized against whatever test we provide.

This possibility strengthens the constructive part of the comparison argument. An AI system can help us propose and assess alternatives at a cost that changes which experiments are worth attempting. It can do so inside a large firm or in a smaller undertaking. The organizational label does not tell us whether the tests are informative.

The information also has a boundary. Evidence that a scheduling rule uses machines better can justify adopting that rule. It does not, by itself, measure the contribution of every earlier investor or determine whether another ownership arrangement would have produced a superior computing industry. A successful local experiment deserves its success without being asked to answer every historical question.

Generated proposals can help us test ambitious ideas when we can define an outcome that establishes improvement. The question is how to connect each proposal to evidence that can distinguish success from a plausible story. That connection lets us use AI's capabilities while examining whether the resulting system accomplishes its purpose.

What the billion requests leave behind

Record design determines what inference can follow. A final answer and acceptance signal supply different evidence from a record also containing task characteristics, assignment rules, checked outcomes, and appropriate alternative-system tests. Equal request volumes need not produce equally informative records.

Acceptance can reflect verification, difficulty checking, or time pressure. Comparing outcomes with independently checked evidence helps distinguish those explanations and identify specific failures. The feedback signal must be interpreted before it is treated as correctness.

Production can be concentrated while a limited experiment preserves a relevant comparison. Assignments justified as learning must expose informative differences and be capable of affecting a decision, rather than merely adding a second name to a record.

The favorite can vary by task. Aggregate request totals can conceal a lack of within-task comparison, while a dominant service can host useful internal experiments. Logged assignment rules and checked outcomes matter together.

The comparison budget accumulates modeled allocation chances outside the current favorite, conditional on the relevant history and inputs. It counts neither actual calls nor agents. Small persistent chances can accumulate indefinitely; shrinking chances need not have a finite sum. Additional informative outcomes belong in the observed experiment. A busy agent team supplies no shortcut around these distinctions.

Stanford's 2026 AI Index distinguishes training from inference in discussing energy use. Repeated service remains a physical activity after a model has been developed. That reminder belongs beside the compiler and the order: extra attempts use resources, and their value depends on what they accomplish.[125]

AI can lower the cost of generating and testing alternatives. It can also operate within institutions that concentrate control of records,

distribution, and evaluation. Its useful output and its effect on independent discovery both belong in the economic assessment.

The constructive choice is to identify a valuable uncertainty and the observation that could reduce it. A paired vote, checked document, randomized assignment, reusable log, or independent deployment can help under suitable conditions. Testing today's extractor also differs from testing whether its use teaches workers to recognize tomorrow's difficult cases. The next question follows the person who has to do that work.

The person at the other end

A date in a report depends on more than the model that supplied it. The computing, service selection, evaluation, and person checking the answer all help determine whether the date can be trusted and used.

Someone supplied examples used in development. Someone expressed a preference in an evaluation, prepared a document, checked an answer, or corrected an error. Arena's paired votes make one such contribution visible. In a workplace, the corresponding judgments can be less conspicuous, folded into the ordinary activity of getting a task done.

If an AI service improves the worker's performance, that is a real possibility worth measuring. If it changes which tasks the worker gets to attempt, how expertise develops, or what an employer can observe about ability, those are further questions. An answer that helps finish today's task can also change the conditions under which tomorrow's worker learns.

The next chapter asks what happens to the worker. Its subject is the work behind the answers, the work changed by them, and the first opportunities through which people develop a record of their own.

CHAPTER 9

The first rung

On September 10, 2020, an engineer ran a load test and an AI assistant stopped making suggestions. The interruption appears in an unusual place: an economics paper about customer support. Workers continued answering customers while the software that had been helping them was unavailable. A breakdown had created an opportunity to ask what remained after assistance disappeared.[126]

Usually, a company wants to know whether its new tool works. Does it resolve more complaints? Does it shorten the queue? Does the customer leave satisfied? An outage raises another question. Has the worker acquired something that can survive the disappearance of the machine?

That question belongs near the center of the argument about AI and employment. A worker and a computer can become a more productive team without the worker becoming more capable alone. They can also become a more productive team because the worker learns from the computer. The difference is easy to miss while both are working.

The customer-support study supplies a hopeful case. Erik Brynjolfsson, Danielle Li, and Lindsey Raymond examined the introduction of a generative assistant among 5,172 agents. In the published study, access raised issues resolved per hour by an average of 15 percent. The gains were concentrated among workers with less experience and lower initial performance. Experienced, highly skilled workers gained much less; some quality measures deteriorated slightly for the strongest workers.[127]

These were live text conversations about business software. An agent had to understand the customer's problem, find an appropriate response, and communicate it. The assistant suggested language and information. The human could use, change, or ignore its suggestions.

The measured improvement belonged to that arrangement: a person handling customers with help available.

But the outage evidence adds something. Previously assisted workers retained improvements in handling speed when suggestions became unavailable. The pattern was consistent with learning, and it became stronger with more prior exposure. It was imperfect evidence: outages were not a clean randomized training experiment, and the conversations handled during them could differ. Still, the machine's temporary absence helped the researchers investigate a benefit that its continuous presence might conceal.[128]

The first rung of a career need not be protected from AI. In this workplace, AI appeared capable of helping people climb it.

That possibility should make us more demanding about what we measure. If the company had recorded only assisted output, it would have missed part of the question. If an observer had looked only at the reduced gap between beginners and experts, the observer might have concluded that expertise was becoming unimportant. Yet an assistant can transmit useful practices precisely because someone developed those practices in the first place.

Consider the customer's next unusual problem. A quick answer is valuable if it is right. Someone must recognize when a familiar response does not fit, when the customer has supplied a misleading clue, or when an apparent solution creates another problem. A system that helps more people develop that judgment has a different economic future from one that merely makes their ordinary replies sound equally accomplished.

The study cannot settle that future. It did not observe compensation, and it did not establish the long-run response of total employment. Better service might reduce the labor needed for a fixed volume of contacts. It might also make support useful enough that customers seek more of it. Employers could change targets, hours, staffing, or pay. A productivity estimate does not select one of these responses for us.[129]

The missing wage record matters for a second reason. Suppose a beginner's improved reply draws on patterns learned from excellent workers. The resulting output does not arrive with a receipt dividing its value among the beginner, those earlier workers, the model developer, and the employer that organized the service. The customer has received help. The division of credit remains a further question.

A narrowing performance gap can mean useful knowledge travels farther. Experts might still be crucial for diagnosing unusual failures, developing practices, or improving the system that distributes the old ones. The strongest support workers' smaller gains concern that deployment. They do not establish that a business can dispense with the expertise on which future assistance may depend. A firm needs to examine that continuing work before turning the average gain into a staffing plan.

We should carry both observations forward. A machine can create real value and improve human learning. The reward history of the resulting workplace can still leave important contributions unresolved. Neither observation cancels the other.

The answer and the ability

A mathematics classroom offers a sharper separation between successful assistance and acquired skill. Hamsa Bastani and her collaborators studied nearly a thousand high-school students in Turkey using two versions of a GPT-based tutor. One resembled a standard chat assistant. The other had instructions designed to support learning, including guidance to offer hints instead of supplying the complete solution.[130]

The students practiced, then took an assessment without the tool. During practice, both assisted groups performed better than the group using ordinary resources. Once assistance was removed, the students assigned the standard assistant performed worse than the control group. The tutor designed around learning largely avoided that harm. Its striking practice gains did not translate into a clearly established advantage on the subsequent unassisted assessment.

The researchers had measured two different achievements. One was getting through the problem with help. The other was being able to solve a related problem afterward. A completed worksheet could look like evidence of either. The assessment made the distinction visible.

The experiment measured short-run learning with particular tools and tasks. How assistance was organized changed what students could do when it was withdrawn. Following their later working lives would require a different study.

The tutor's design required human work. Teachers supplied problem-specific information, correct solutions, and guidance about common mistakes. The assistant was instructed to ask for the student's attempt and respond to the difficulty. Behind the apparently effortless exchange stood a choice about what the student should still have to do.

The experiment preserved an occasion to find out. Students encountered a teacher's review, practiced under assigned arrangements, then completed a closed-book, closed-laptop assessment graded with a common rubric. Observing later performance prevented the study from ending when an interface first made the task easier. Workplace assessments face the same question: when should the next observation occur?

Buying an answer and supporting learning are different uses of assistance. A tool can complete a task or help a user develop the capacity to handle later tasks. Immediate deliverables can look similar while the consequences for later ability differ.

Neither approach is always preferable. A customer facing an urgent deadline may reasonably want the repair. An employee already familiar with the problem may gain nothing from another lesson. Instruction takes time, and a tool that insists on teaching every experienced user could become a nuisance. The relevant choice concerns the purpose of this task for this person at this stage of development.

The classroom researchers also examined students' messages. With the standard assistant, students often simply asked for the answer. With the learning-oriented tutor, they exchanged more messages and increasingly asked for help or attempted solutions. The authors saw this contrast as consistent with students using the standard assistant to obtain solutions instead of working through problems. It helps us see a possible mechanism. The message patterns do not separately identify the contribution of each safeguard, since the tutor changed several things together.[130]

Deliverable quality, transfer to an unfamiliar task, and the ability to detect an incorrect recommendation require different assessments. Passing one does not establish the others.

There is no reason to make every assessment an exercise in working without computers. The relevant future skill may be choosing, checking, and combining powerful tools. An accountant need not abandon a spreadsheet to demonstrate professional judgment. A programmer may reasonably use an assistant in the evaluation of programming skill. The test should preserve the ability the employer actually needs to know about.

Assessment should follow the future ability that matters, including recovery when a familiar assistant fails, checking sources, and reviewing output from another system. Preserving a test that answers no useful question can become an expensive ritual.

The first rung is therefore more than a collection of simple tasks. It is an arrangement that lets a beginner attempt something, receive feedback, improve, and demonstrate the improvement. AI can change every part of that arrangement. It can make practice cheaper, feedback quicker, and difficult work accessible earlier. It can also complete the activity from which the beginner was supposed to learn.

The disappearing control group

The experienced programmers in METR's early-2025 experiment present another surprise. Sixteen developers worked on 246 tasks in open-source projects they knew well. Tasks were specified before assignment to conditions allowing or disallowing AI. With the tools available in that study, allowing AI made completion take 19 percent longer.[131]

That result had a precise setting. These were experienced contributors in mature projects, facing real tasks and established standards. The tools were those used during the early-2025 study, primarily a coding environment with then-current models. Nineteen percent longer describes task completion time under the experiment's measurement. It does not mean that every programmer became 19 percent less productive, or that later tools must have the same effect.

The developers' expectations pointed in the other direction. They had expected assistance to save time and afterward still believed it had done so. The discrepancy makes a practical point. A tool can feel helpful during individual moments while adding time across the complete job. Waiting, supplying context, examining suggestions, and correcting mistakes all belong in the cost of producing an acceptable result.

METR collected screen recordings and examined a subset in detail. The records made activities such as prompting, waiting for generation, reviewing output, and writing code separately visible. That is a useful antidote to judging a coding assistant by how rapidly text appears. The desired product is an acceptable change in a working project. A suggestion that arrives quickly can still create work before it belongs there. The recordings do not establish a single cause of every slowdown, and the researchers considered multiple possible explanations. They do show how a better measurement can move the discussion from an impression about assistance to the actual sequence of actions required to finish a task.

Set this study beside the customer-support deployment. One finds substantial average gains, especially for novices. The other finds a

slowdown among experienced developers in familiar repositories. The interesting question becomes which workers, tasks, tools, and working arrangements produce which result. A single number for the effect of AI would conceal the decisions an employer actually has to make.

In its February 24, 2026 account of a follow-up effort, METR reported difficulty obtaining a trustworthy comparison. Some developers disliked the prospect of being assigned to work without AI. Some tasks were kept out of the experiment. The available observations were changing along with adoption.[132]

One participant's published comment included the simple explanation, “I really like using AI!” That is an ordinary preference. It is also a research problem. If a developer avoids entering the experiment because a particularly suitable task might land in the no-AI condition, the measured comparison no longer represents the full set of tasks the researcher hoped to study.

Random assignment cannot recover a task that never enters the pool. It can make the allowed and disallowed conditions comparable among eligible, submitted tasks. The researcher must still ask why these tasks were submitted and what happened to the rest. The boundary around the experiment is part of the evidence.

Study pay also fell from $150 to $50 an hour, potentially altering participation. Developers could run agents concurrently, complicating the meaning of time spent on a task. Unfinished work and differences in task selection created further difficulties. These were reasons to reconsider the design, rather than declare that the original slowdown had cleanly reversed into a precisely measured speedup.

The attraction of the tool could be evidence that it had become more useful. It could also reflect convenience, enjoyment, expectations, or the kinds of work people now chose to attempt. METR believed the newer tools likely helped more, while emphasizing how weakly the follow-up identified the magnitude. The preference to use a product and a credible estimate of its productivity effect answer different questions.

Quality, completion time, and workers' experience should be measured separately. A slower method can be preferred for other reasons. Calling every benefit productivity obscures the choice, while disregarding preference discards something workers value.

This problem is familiar outside software. A worker who has reorganized a day around a tool may reasonably resist surrendering it for research. An employer may dislike paying for deliberately slower work. A researcher cannot assume that preserving a clean comparison costs nothing merely because the comparison would be informative.

Adoption can change what a later comparison means. Users develop habits, forget old procedures, and change which tasks they attempt. Returning to an earlier tool then changes more than a suggestion's availability. An evaluation must specify which stage of that adjustment it measures.

Staged introduction, comparison of assisted training methods, and assessment on new tasks can still supply evidence. Each answers a different question. More observations of the adopted workflow cannot automatically recover every earlier comparison.

Here the distinction between activity and informative opportunity becomes tangible. A company might accumulate millions of AI-assisted edits while losing the practical ability to recruit a comparable group for a particular no-AI trial. The edits could be useful evidence about errors, quality, and day-to-day performance. Their number would not, by itself, answer the missing question about the alternative arrangement.

Adoption can make a tool more useful while making a particular experiment harder to run. METR had to redesign its study because the way developers worked had changed.

From tasks to jobs

Bank tellers offer a historical warning against equating an automated task with an eliminated job. James Bessen reported that the number needed in an average urban branch fell from twenty to thirteen between 1988 and 2004, while urban branch numbers grew by 43 percent. David Autor drew on this account to explain how lower operating costs and expanding demand can change employment. The remaining teller work also shifted toward customer relationships and other services.[133]

Autor also identified deregulation as part of the branch expansion. The figures do not isolate the cash machine's causal effect. They show why counting automated withdrawals cannot by itself tell us how many people a bank will employ. The business can change the number of outlets, the services it offers, and the work it assigns to people.

Autor's argument concerns relationships among tasks. Automation can increase the value of complementary work, but the employment response depends on demand for that work. Saving time can expand production, improve quality, or reduce labor hours. Identifying an automated step is the start of an employment analysis.

There is harder evidence on the other side. Daron Acemoglu and Pascual Restrepo studied exposure to industrial robots across American labor markets over 1990–2007. They found adverse effects on employment and wages in more exposed places. A machine's productivity benefits did not ensure that the affected local workers would gain.[134]

Their setting involved industrial robots performing work such as assembly, welding, and painting. Their analysis used differences in local exposure based on industry composition and robot adoption, rather than randomly distributing robots among towns. The causal interpretation requires assumptions and checks about other forces affecting those places. It is serious evidence about a historical adjustment, with a defined design and a defined scope.

Notice what this research can accomplish without rerunning the whole American economy. Places entered the period with different industrial structures, making their exposure to robot adoption different. The researchers used that variation and investigated competing explanations. Their estimates remain conditional on the design, but the absence of a literal parallel America does not make historical evidence useless. Shadow futures sharpen the demand to specify which alternative is being studied and where the information comes from. They should not become an excuse to dismiss a carefully constructed comparison merely because it occurs within one history. Here the relevant contrast is supplied by differences across labor markets, with the design's assumptions carrying part of the inferential burden.

The teller account and the robot study should make us impatient with easy reassurance as well as easy catastrophe. Automation can reduce the labor needed for particular work. Lower costs and new activities can create demand elsewhere. The balance, location, timing, and distribution of those responses matter. An aggregate gain can coexist with a damaging local loss.

The first rung adds a further adjustment. Firms need workers today, but they may also need people capable of doing more demanding work later. If the tasks that once introduced beginners to a profession become cheaper to automate, the employer can redesign training, recruit experienced staff, hire fewer entrants, or develop another route into expertise. Those choices affect the supply of skilled labor that future firms will encounter.

Preserving obsolete manual work solely because it once trained beginners can be costly. The constructive question is whether an alternative route develops the needed expertise, and who has an incentive to finance it.

A simulation might provide repeated practice without risking a customer's records. A supervised assignment could expose a beginner to a more demanding diagnosis sooner. The classroom tutor offers a real example of deliberately changing assistance around learning. The

customer-support deployment offers evidence consistent with skills spreading through assistance. These possibilities deserve trials alongside the fear that the entry route will disappear.

It is possible, however, for an employer to discover that it can buy experienced judgment elsewhere more cheaply than it can cultivate it. If many employers make that choice, the next generation's opportunities may change. That is a hypothesis about incentives and collective adjustment. It requires evidence about hiring, training, mobility, and demand. It cannot be established by pointing to a machine that writes a good report.

Who gets the difficult assignment?

Responsibility itself can reinforce opportunity. A successful assignment produces a record that supports the next selection. Employers can rationally rely on that record while earlier allocation decisions partly determine whose ability becomes observable.

Absence of a record differs from a record of failure. Neither guarantees future success, but the evidence available in the two cases is different. Evaluating skill requires distinguishing a poor result from a missing occasion to demonstrate it.

AI could widen access to supervised practice by reducing the cost of checking routine steps. It could also concentrate scarce responsibility among fewer already-tested workers. Which effect occurs depends on hiring, training, assignment, and demand.

Platform work provides documented evidence of selection through a record. The International Labour Organization's 2021 report describes ratings affecting task allocation and access to work, with negative ratings contributing to deactivation.[135]

Geographic mobility offers less independence when the same ranking system governs work across locations. If practical alternatives are also unavailable or reuse the same record, moving does not create a new evaluator. This is a hypothesis about dependence, rather than a description of every platform worker.

That last condition matters. The ILO report also records surveyed workers using another platform after permanent deactivation from one. Platforms can open distant demand, and workers can have alternatives. The concern is the situation in which those alternatives become impractical or rely on the same record. We should investigate that dependence rather than assume every worker already faces it.[135]

Europe's 2024 Platform Work Directive addresses one part of the problem. It provides for portability of personal data generated through platform work in automated monitoring and decision systems, including ratings and reviews. Its national transposition deadline is December 2, 2026. At the time of writing, the deadline remains ahead; the existence of the directive is not evidence of uniform implementation or successful remedies.[136]

A portable reputation can help a good worker leave without starting from nothing. It can also make an unfair rating harder to escape. Carry the same flawed score everywhere and portability spreads the original judgment. My proposed safeguard therefore pairs voluntary export with correction, an explanation of context, appeal, and opportunities for independent assessment. A fresh evaluation is different from a right to conceal fraud.

Independent assessment, correction of disputed ratings, and context-sensitive records can make reputation contestable. Each costs something, and an existing score may often be useful. The relevant freedom is a workable way to challenge the evidence and obtain another meaningful observation.

A supervisor's assessment, a practice task, or a second employer's trial can supply evidence missing from the original rating. A reputation system should leave room for those observations to change its verdict.

The employer still needs deadlines met and people paid. It need not solve perfect merit before organizing work. The first rung can disappear because a task was automated. It can also disappear because one institution's early verdict travels farther than the worker's practical ability to find another judge.

Buying a future workforce

Employers can automate routine work, redesign entry routes around assistance, or investigate the consequences before changing staffing widely. These choices trade immediate output against the cost and future value of developing judgment.

Automation may be economical where routine work is reliable, demand for human review is limited, and experienced judgment remains available. A thinner entry route does not make the production saving imaginary.

Assisted training may instead support growing demand and develop future reviewers. The customer-support findings make that possibility worth investigating; they do not establish a universal response across occupations.

Evaluation itself consumes supervision, time, and operational resources. Protecting customers and giving participants a fair basis for taking part belong in its cost. A relevant comparison must justify that expense.

The horizon must match the staffing question. A week of faster output cannot establish whether reducing supervised entry positions creates a shortage of reviewers two years later. Skill development and progression require longer observation.

The measurements should fit that decision. Immediate output and error rates matter. So do the ability to handle unfamiliar cases, the supervisor time required, the quality of later reviews, and whether participants progress into roles the business needs. Retention and the movement of trained workers to other employers may affect the private return to training. Measuring everything indiscriminately is less useful than observing what distinguishes the feasible choices.

Intermediate measures can help explain that development: unfamiliar problems handled, errors caught before delivery, and repeated supervisor corrections. They can reveal an impressive early performance that fails to transfer, or a costly beginning that develops valuable judgment.

A useful study defines what evidence would support expansion, revision, or stopping. The classroom experiment demonstrates why polished output and subsequent capability should remain separate outcomes.

METR's experience also warns that selection can change once users develop strong preferences about tools. A credible comparison may require compensation for extra time, different randomization, or a narrower question. A control-group label alone establishes no comparability.

Employees should understand what the test is meant to establish. A trial intended to compare training methods can become distorted if participants reasonably believe that one condition marks them as candidates for dismissal. A firm has to account for the behavior its evaluation creates. That is part of making the evidence useful, as well as part of treating people fairly.

Training can benefit later employers as workers move. That creates a possible gap between private and wider returns, which requires evidence about mobility, costs, and skill formation before it can justify a subsidy.

The same reasoning cuts against preserving a program simply because it trains people. Training can teach obsolete practices, select poorly, or consume supervision needed for more valuable work. An alternative employer, professional course, or well-designed AI tutor might supply a better route. The comparison has to include plausible replacements, rather than treat the existing arrangement as the only imaginable source of skill.

There are limits to what one firm can settle. It cannot observe the entire labor market's response to simultaneous adoption. It may not know how another employer's staffing choices affect the future pool of reviewers. This is a reason for research across firms, careful labor-market evidence, and institutions that can evaluate shared problems. It is not a reason to substitute a confident story for missing observations.

An employer choosing the fastest tool today may be entirely sensible. A workforce, an industry, or a government may have a longer horizon and different interests. Making those interests explicit is more productive than asking whether AI is simply good or bad for labor. The important choices concern who gains, who bears the cost, what can still be learned, and which arrangements remain feasible.

Who owns the workforce?

The rise of agent work also changes which assets are accumulated. Model calls and checking are operating expenses. Equipment, retained code, reliable tests, and durable productive capabilities may support future production. A large stream of generated work need not leave its purchaser with independently usable assets.

Workers can benefit through wages, ownership claims, and cheaper goods or services. Those are different channels, and their distribution cannot be inferred from a count of agent sessions or completed tasks.

Investment can expand the capacity to attempt new projects or strengthen exclusive control of routes through which others must be tested. Both may involve productive expenditure. Their consequences for future comparison depend on what the resulting assets permit.

A feedback between automated production and reinvestment could help entrants develop or help well-financed incumbents advance faster. Demand, financing, competition, resource constraints, and checking costs determine its strength. A successful agent experiment does not demonstrate an autonomously expanding firm.

Accumulation also faces depreciation, imitation, failures, and rivals. Productive integration can improve services while making independence harder. The relevant assessment includes both those benefits and the feasibility of another development route.

Anton Korinek and Donghyun Suh examine automation and capital accumulation together in their March 2024 scenario model. Investment can complement work that remains with people. Its pace

relative to automation and the assumed range of human tasks can produce very different wage paths. Their model also separates reproducible capital, including compute, from fixed scarce factors. It gives no unconditional forecast that wages or monopoly profits must move in one direction.[137]

Software cannot instantly create an electricity connection or manufacture servers. Scarcity can constrain expansion, but temporary shortage alone does not establish permanent gatekeeping. The stronger claim needs an identifiable obstruction and evidence about feasible alternatives.

Employee ownership, jointly financed equipment, and public computing access distribute claims and authority differently. Their ability to support independent development depends on financing, operation, and evaluation as well as legal ownership.

Broader ownership does not automatically create independent discovery. A cooperative can depend on one provider's records, while a private owner can finance useful tests. Claims on income and control over learning belong in the same analysis without becoming the same question.

Similarly, income distribution and entrepreneurial opportunity can diverge. Holding a stake in an existing firm does not ensure the ability to start another business, obtain inputs, reach customers, or challenge an inherited evaluation.

The future of work includes the future of assets built through work: who can retain them, redirect them, and use them to attempt something new. Affordable tools and usable routes to demonstrate judgment can change together or move in opposite directions.

These choices depend on institutions as well as contracts. Investment can coexist with closed access, while publicly financed access can become another gate. The next chapter examines the rules governing the practical ability to compete.

CHAPTER 10

Competition for future evidence

In January 2023, Britain's Competition and Markets Authority announced that six major banking providers had completed the remaining items in its Open Banking roadmap. This was an achievement with a qualification. Common standards had to remain usable. Performance and availability still needed monitoring. Completing implementation did not complete the obligation.[138]

Open Banking let customers share account information securely with authorized third parties and initiate payments through new services. Banks had to support common interfaces through which other businesses could actually use permitted information. An administrative right had to become a working connection.

Receiving your records in a form another service can interpret is more useful than an unexplained file. Being able to use that service reliably is another achievement again. A box marked data access can conceal a working route to an alternative or an interface that fails when needed.

The banks themselves had been required to create and fund the organization maintaining the standards. Completion left those funding duties in place. These were concrete obligations in a dated decision; their existence alone does not establish Open Banking's total economic benefit.[138]

Open Banking illustrates a broader question for competition policy: can a feasible access arrangement improve what participants are able to discover, while preserving productive benefits and protecting sensitive information?

Established systems can be valuable and expensive to replace. The economic case for a remedy depends on the information it supplies,

the decisions that information can change, and the costs of making the route usable.

The comparison budget identifies a feature of an allocation process. Assessing a remedy requires a further account of the value of its evidence and its consequences for production.

What would the evidence change?

Evidence has economic value through the choices it improves. A precise ranking of alternatives can leave the best available action unchanged; a modest observation about a costly failure can matter greatly. Decision relevance precedes a demand for more data.

David Blackwell supplied a foundational way to compare experiments. Under the classical conditions, if one experiment's observations can be produced from another's by adding a suitable kind of noise, the richer experiment can do at least as well across decision problems. Someone with the richer evidence can imitate the poorer procedure.[139]

Imagine a documented record of every eligible trial request: suggestion, checked outcome, and time required. A second report contains only a summary computed from those records. Someone with the interpretable full record can produce the summary; someone with the summary cannot generally recover discarded detail.

Collection costs, privacy, and implementation enter after the gross comparison of experiments. Different experiments can also answer different questions, such as operating accuracy and integration costs, without fitting one universal ranking.

Evaluation should specify feasible actions, the period under study, relevant outcomes, and how errors and costs enter the decision. A measure that rewards one dimension while ignoring another can encourage behavior that improves the score without improving the service.

Average outcomes can conceal different burdens across users. The appraisal must make consequential differences visible and compare actions an institution can actually take.

The baseline must also be feasible. Removing safeguards from the assumed status quo exaggerates a remedy's value; comparison with a perfectly informed decision maker sets an unattainable benchmark.

An experiment can earn its keep while leaving the exact contributions of programmers, models, training sources, and earlier customers unresolved. Choosing a contract is less demanding than explaining every supplier's economic merit. Taking the inference limit seriously should direct institutions toward useful, attainable learning. It need not keep them experimenting until every question about contribution disappears.

Opening the existing record

An audit of existing evidence can precede another experiment. Private tests, independent evaluations, and operational records may already resolve the question. All informative measurements belong in the analysis; disregarding internal experimentation would manufacture a deficit.

Protocols matter alongside results: eligibility, task selection, withheld cases, version changes, and whether evaluators knew whose output they were checking. Detailed records can still answer the wrong question.

Google's December 2025 search judgment makes this practical concern visible. Its specified data-sharing provisions required enough information for qualified competitors to understand the datasets, including descriptions of contents, sampling, and privacy-related transformations. It did not simply order delivery of an unexplained pile of records.[140]

This dated provision does not establish that implementation has succeeded. It highlights why interpretation can be part of access. A

million query records and an explanation of how they were selected are different assets whose usefulness can depend on receiving both.

A record limited to accepted outputs describes that selection, rather than performance on rejected outputs or tasks never encountered. An audit can identify the gap or establish that existing evidence is adequate. The analysis must be capable of supporting no further intervention.

Zahra Meghani's work on epistemic monopoly examines dependence on developers for evaluation evidence and control of research conditions.[141] Accurate reporting can coexist with unanswered questions when the producer chooses which investigations occur.

Scrutiny requires expertise, secure handling, and time. An authorized evaluator can inspect protected records and report methods, uncertainty, and conclusions. Accountability concerns what can be checked and challenged without necessarily disclosing personal data.

Disclosure can also change competition. The 2023 Merger Guidelines discuss how easier observation of rivals' terms can increase susceptibility to coordination. Giving every competitor every business record need not improve the strategic environment. An independent evaluator and a rival sales department have different uses for confidential information.

Open Banking's completion decision distinguished service-quality indicators and underlying data from current-account transaction records supporting authorized access and payments. It specified services and continuing obligations rather than publication of every internal file.[142] That specificity matters to remedies intended to create usable comparison.

A file that can travel

Portability can lower switching costs when useful records would otherwise have to be rebuilt. Open Banking's standards, conformance checks, and performance obligations provide a documented reference: compatibility was treated as a service requiring maintenance.[142]

Ipsos's February 2026 publication supplies another kind of banking evidence. Its independent service-quality survey, conducted during 2025 under a regulatory requirement, asked customers about recommending providers and particular services, including online banking and branches.[143] These experiences matter, but customers were not randomly assigned to banks. Their needs can differ.

Customer experience can inform a decision without isolating the causal effect of each input into that experience. Survey evidence and randomized assignment answer different questions; neither should acquire the other's interpretation.

Copying a record to more organizations does not create independent histories of the process that produced it. Operational gains from portability and new evidence from experimentation should be assessed separately.

Interpretation matters too. Imagine a field marked positive means successfully resolved in one record version, but problem requiring action in another. Exported files omit the version. A reader could reverse the results while processing every row correctly. Under one interpretation, following the signal helps; under the other, it hurts.

Knowing that someone with the correct interpretation could use the records does not supply the actual buyer with a usable procedure. If the records cannot distinguish the meanings, more equally ambiguous rows may not help. A verified definition can let the buyer use one procedure that reads the version and interprets the result.

Definitions alone reveal no outcomes. Outcomes with unknown meanings may be unusable. Together they can support a decision; their joint value need not equal the sum of their separate values. Freely available, understood evidence need not harm a buyer who can

ignore it. Cost and lost interpretation create the problem. Retaining the trusted record preserves the earlier procedure.

What survives migration should therefore be specified: category definitions, corrections, and the link between a result and its service version. Counting exported fields is an inadequate test of usefulness.

Interoperability concerns whether services can connect to the systems and users needed for a relevant test. A bounded interface may suffice. Security, maintenance, capacity, and changes to integration costs determine what scope is feasible and who should bear the burden.

Compatibility can bring security, capacity, and maintenance costs and alter investment incentives. The appraisal must retain these consequences and the benefits of integration. An entrant's need does not establish that someone else should bear every cost it requests.

Making the missing comparison

An audit cannot recover an outcome never observed. If that outcome matters sufficiently, a designed experiment is needed, with comparable tasks, documented assignments, checked results, and records of missing observations.

Assignments can interact through shared resources and participants' responses. A design may need to compare periods or groups, or model those interactions explicitly. Balanced assignment alone does not establish an interpretable operational comparison.

On June 3, 2019, a field experiment began calling rice farmers in India. Precision Agriculture for Development wanted to choose how to recruit farmers to its telephone service. Maximilian Kasy and Anja Sautmann helped compare six combinations: morning or evening automated calls, with no text alert, an alert an hour ahead, or one a day ahead. From a government partner's list, the service set aside ten thousand valid numbers outside the do-not-disturb register. Seventeen waves of calls followed, with assignment changing as results arrived.[144]

The morning call preceded by an hour's warning received nearly four thousand assignments. The evening call without a warning received fewer than four hundred. Every option was tried, but the assignments were unequal. At the end, the authors put the probability that the morning call with that warning was best at just over three quarters, under their statistical model. Uncertainty remained.[144]

Success meant answering five recruitment questions, rather than improved harvests or income. The evidence could guide the next recruitment campaign without establishing the agricultural service's total value. The outcome must fit the decision.

It could adapt assignments as results arrive, giving a promising service more requests while preserving informative alternatives. Serving today's requests and learning what to buy next year need not recommend identical allocations.

The comparison budget prescribes neither equal shares nor a universal quota. A small persistent chance can accumulate indefinitely; large nominal opportunities can be wasted on tasks that fail to expose the relevant difference.

A declared decision rule helps prevent changing the criterion after results arrive. The test must be capable of changing an action on grounds specified before its outcome is known.

Transportability also matters. A result obtained under unusually clear tasks, expert staff, or an obsolete version may not travel to ordinary operation. Broader testing, calibration, a narrower application, or rejection of the proposed change can each be appropriate.

Such assurance concerns expected performance across the retained accounts. It cannot promise every resident a better outcome or every realized contract a gain. Investigators should examine credible evidence and justified assumptions as well as what subsequently happened. Fortune alone cannot validate a defective method, and misfortune need not condemn a reasonable choice.

A feasible policy must use a procedure available with the information actually observed. Imagining a decision maker who is told which explanation is true does not supply such a procedure.

When every road leads to the same judge

Multiple purchasing organizations can inherit one ranking or one defective test. Separate encounters then reproduce a common judgment. Independence matters when it changes the relevant selection or measurement process.

This is the historical possibility running from Smith's towns through Bell's network to platform work. Wider markets can support more specialization, better service, and more customers. They can also extend one institution's reach over the decisions determining who may develop. A market can grow while the number of independent places to try again shrinks. That is a proposition about institutions to investigate, not a measured law of history.

Consider an explicit hypothetical. Suppose OpenAI became the only foundational AI supplier. Add a further condition: it also controlled indispensable routes to customers, the evaluations deciding admission, and access needed to develop competing systems. These are assumptions about a possible arrangement, not a description of OpenAI today or a forecast. The question concerns the structure, whichever company might occupy it.

Even a sole model supplier would not automatically possess total epistemic monopoly. Independent laboratories could retain ground truth, buyers could test alternatives, and other technologies could answer the question at issue. The stronger scenario requires control over the production and checking of relevant comparisons. Its permission becomes necessary to test the alternatives to its authority, with no workable independent route remaining.

The service might be excellent and cheap. Its owners might have invested wisely and lawfully. Those facts belong in the appraisal. But a satisfied present customer cannot report the performance of a rival that never received a feasible development route. The concern reaches

beyond a price increase: which alternative capabilities will be permitted to become observable?

Saying that markets clear does not answer this. Clearing concerns offers that can be made under existing conditions. It does not guarantee that an undeveloped supplier acquires the inputs, customers, or evidence needed to make a competitive offer. Nor do a million agents necessarily provide a million independent judgments if access to all of them passes through the same gate.

A concentrated future should be evaluated through identifiable mechanisms, reversibility, and feasible safeguards. If independent evaluation could become prohibitively costly, waiting until lost skills and institutions must be rebuilt can itself be consequential. That possibility requires evidence rather than dismissal or assumption.

Hayek treated competition as a way to discover facts that could not be known in advance.[145] The question now is how a rival obtains the computing, customers, and tests through which something unexpected can be discovered.

Bell's experience shows why the route has stages. Opening patents supported follow-on innovation outside telecommunications, while the study found a different result within the sector. Permission to use knowledge was not the entire route to a customer.[146] Publishing an incumbent's records and allowing an entrant to generate new records can therefore accomplish different things. A thousand copies of one history remain copies of one history.

Scale also supplies research, reliability, and lower costs. One large supplier can conduct informative internal tests; several firms can repeat a shared error. Private testing counts. A public trial can displace it. The aim is to preserve consequential independent learning where its benefits justify its full costs, rather than keep every proposed competitor alive.

Precaution without closing discovery

Competition policy already reaches beyond prices. The United States' 2023 Merger Guidelines discuss products, features, innovation, wages, benefits, and research competition. The proposed information rationale adds a question about future evidence; it does not discover nonprice antitrust or dispense with statutory requirements.[147]

Here is the policy position I favor. When a gatekeeper controls an indispensable route to useful independent evaluation, and there is credible evidence that closing it would be hard to reverse, the presumption should favor keeping a workable route open. Waiting for the last effective challenger to disappear can destroy the evidence needed to appraise the decision. The burden should then fall on the proposed restriction to demonstrate why a narrower arrangement cannot protect legitimate interests.

Keeping an option to learn has a cost, as does losing it. Arrow and Anthony Fisher examined how waiting for more information can affect the choice to make an irreversible investment. Their analysis treats that future information as part of a cost-benefit appraisal.[148]

The rule could require documented interfaces, workable interoperability, voluntary data portability, or access to specified knowledge needed for a credible test. Requirements should follow the obstruction. Search data access cannot supply a missing electricity connection; model access cannot repair a dishonest outcome measure. A regulator should identify what the entrant could actually do under the proposed terms.

The EU's Digital Markets Act gives this boundary a concrete form. For designated messaging services, gatekeepers must connect specified functions they already offer their own users: individual messages and attachments first, group messaging within two years of designation, voice and video calls within four. Access is upon request and free of charge. The law also requires preservation of the security provided to the service's own users, including end-to-end encryption where applicable.[149]

That is an obligation to build and maintain connections between privately owned services. A separate provision concerns access to specified search data on fair, reasonable and nondiscriminatory terms. The two duties have different designs. They demonstrate that ownership can coexist with participation rights, without establishing that every implementation succeeds or that an AI company automatically owes the same duty.[149]

For AI, an obligation might concern an interface or a secure evaluation route rather than publication of every model weight or training record. Fees may be justified; privacy, security, research incentives, and genuine technical constraints matter. Claims of unavoidable harm should receive independent scrutiny. Letting the gatekeeper alone decide whether a competing test is safe would make the safeguard depend on the permission it is intended to examine.

The public authority also needs limits. It should specify the affected decision, plausible lost comparison, legal power, expected benefit, costs, and less burdensome substitutes. A secure audit may suffice where existing evidence answers the question. Prospective trials matter when no file can contain the missing contrast. My proposed time-limited evaluation power would require legislation wherever existing authority is insufficient, along with challenge and review.[150]

I favor property rights that reward invention while leaving competitors a workable route to test and develop their own ideas. Access duties should be justified by the comparisons they enable, the decisions they improve, and their costs.

Government can also become the single judge. A mandatory benchmark, favored consultant, or official platform could synchronize mistakes across an economy. Independent appeal, alternative methods, and revision must constrain the evaluator as well as the evaluated. Precaution should keep discovery possible on both sides of the public-private boundary.

Who can keep the next entrant possible?

The rise of agent production makes access more consequential. Cheaper software work can still depend on computing, usable records, distribution, and credible evaluation. Ownership of generated output does not automatically provide those complements.

European computing access, an American research initiative, and China's industrial policy provide three documented institutional approaches. Their stated terms identify what qualifying participants may use, retain, and test. Announced objectives and formal entitlements still need to be distinguished from outcomes in practice.

EuroHPC's Playground call, opened in April 2025, offers an entry route for European startups and smaller businesses with limited computing needs. Applications arrive in order, with eligibility and technical checks rather than a competitive ranking. Allocations last one, two, or three months on a single system. The published call also warns that demand can leave particular capacity unavailable. The door has a procedure and a queue.[151]

Its Industrial Innovation modes offer eligible AI small businesses free computing for innovation. Crucially, the March 2026 terms say generated data and models remain the user's property and can be commercially exploited. They also require publication of results outside confidential portions and a final report. Public provision and private accumulation coexist in these stated terms.[151]

These terms allow publicly supported computing and private ownership of resulting assets to coexist. They do not establish that a product reaches customers, uses appropriate data, or succeeds in practice. Those are subsequent stages in development and evaluation.

An access program's value also depends on whether work can continue after an allocation ends. Reproducible tests, transferable records, waiting time, support costs, and displaced private purchases help determine whether it enables additional useful experimentation.

Property protection leaves choices about how resources may be used. Article 17 of the EU Charter protects possessions and

intellectual property while allowing regulation of use under specified conditions.[152] The practical choices concern who can use a resource, on what terms, and how investment and entry can continue. Common institutions and member states must still make the different parts of an arrangement work.

America provides a useful challenge to the claim that commitment to private enterprise leaves a state unable to act. The July 2025 AI Action Plan advocates deregulation, but also encourages open models and wider access to computing. It identifies long commitments to large cloud suppliers as an obstacle for many startups and academics, and recommends developing computing markets and the National AI Research Resource, or NAIRR. These are policy recommendations, not proof that the obstacle has disappeared.[153]

In September 2026, the National Science Foundation announced an operations center for NAIRR, the public-private research initiative. The center would coordinate providers, integrate computing and other resources, operate a portal, and support users. A researcher could potentially reach resources through an organized public route rather than negotiate every component alone. The announcement establishes an institutional undertaking; it does not measure how many otherwise impossible competitors it creates.[154]

Research access and commercial entry remain different routes. Eligibility, permitted use, and portability of results determine how far a research initiative can support later independent development. A portal can simplify access without removing dependence on its participating providers.

American authorities have also examined the private arrangements. In January 2025, FTC staff reported on partnerships between major cloud suppliers and AI developers. The report identified potential concerns involving input access, contractual and technical switching costs, and privileged information. It explicitly made no determination of illegality. Its historical scope matters: these were findings from a dated inquiry, not an audit of every contract now available.[155]

Such partnerships can finance valuable capabilities. Their terms can also shape switching and future development. Evaluating those consequences requires the agreement and feasible substitutes. The documented American initiatives show instruments available to the state; their adequacy depends on institutions in operation.

Deregulation and access protection can affect different constraints. Removing an unnecessary permit can support entry, while leaving an indispensable interface at one provider's discretion can preserve a bottleneck. A count of regulations removed does not measure the ability to conduct an independent test.

China's August 2025 AI Plus policy starts from a more explicit program of coordination. It calls for shared and interconnected computing, agent services, open-source development, and patient capital. It addresses new forms of work and employment risks as well as industrial adoption. These are announced objectives. They do not establish that an unaffiliated firm actually receives resources or that the policy improves long-run adaptation.[156]

Patient capital and shared computing could support redesign before commercialization. Their usefulness still depends on admission rules and the ability to try unfamiliar approaches. Coordinated resources can coexist with a common gate that rejects departures from an established method.

There is a further boundary. China's 2023 interim generative-AI measures encourage innovation and shared infrastructure, while imposing political content requirements on covered services offered to the mainland public. Their scope expressly excludes development and applications that do not provide such public services. That distinction prevents us from treating every private research activity as governed by the same provision. It also prevents us from treating support for computing as equivalent to unrestricted public inquiry.[157]

An economy could make it easier to try competing industrial designs while restricting which politically unwelcome conclusions a public service may generate. A test earns little independence if its

commercial or public sponsor can suppress an inconvenient result. Researchers need to be able to report that the favored system failed.

An excluded applicant's rights and the ability to report failure therefore belong in the comparison alongside computing and finance. Europe and China cannot be grouped into one alternative to America simply because both may intervene. Nor does a larger public budget settle the question. Follow the attempted product, its tests, and the grounds on which somebody can refuse the next step.

Limiting exclusion at a bottleneck can sometimes enlarge the practical scope for private enterprise. A bounded, secure interface can leave firms privately owned while changing the terms on which they compete. Whether it does so is a question about the particular design and its costs.

Public intervention can create a gate of its own. Certification costs may exclude smaller firms, or an evaluator may suppress inconvenient failures. Contestability requires limits on official control as well as private control, and reliable expectations about the assets successful experiments create.

To choose between arrangements, return to the same sequence. Can a qualified outsider obtain resources? Can it reach customers and receive informative work? Can someone check the outcomes independently? Can useful records travel? Can an exclusion or a failed official test be challenged? The answers must survive ordinary use. A published entitlement that nobody can exercise is the institutional equivalent of the exported file nobody can read.

The society with the first leading model need not be the society best able to recover from a mistake in it. Conversely, a generous access program might absorb scarce engineers and money while producing little useful knowledge. The long-run hypothesis is about preserving workable alternatives at acceptable cost. We will return to how to test it in the final chapter. These dated policies open that inquiry; they do not settle which society will prevail.

The price of finding out

Audits, portable records, interfaces, experiments, and independent evaluation address different gaps. There is no permanent ranking among them. The appropriate remedy depends on what evidence already exists, what is missing, and which feasible route can supply it.

Appraisal connects the possible mistake prevented to its probability and consequences over a stated period, then counts testing, migration, supervision, privacy protection, and lost production. A cheaper adequate substitute belongs in the comparison.[158]

Judgments about rare serious errors, residents' different burdens, and future benefits must be visible. More data cannot decide an undeclared value judgment. Costs also need honest bookkeeping. Lost operating output should not be counted twice under different headings. A payment between participants is not automatically a social resource loss; a burden outside the purchasing department is not automatically irrelevant.

Testing can improve current service and future evidence at once. Its benefits should be counted without duplication, and its continuation should depend on whether remaining uncertainty can still change a consequential choice at an acceptable cost.

Adequate choice among feasible actions can coexist with a limit on universally reliable contribution recovery. Institutions do not need an exact division of all economic merit before acting.

Arrow's analysis of information use and appropriation supplies a further question: do those deciding whether to investigate capture its wider benefits?[159] Evidence benefiting other participants can create a gap between private and social incentives. Concentration alone cannot establish that gap.

Both incumbents and entrants have interests in the design of a remedy. A trial can subsidize sales, and an audit can be too narrow to challenge an established position. The appraisal must explain the improvement sought, who benefits, and who bears the cost.

Open Banking's continuing obligations provide the concluding reference: a route maintained after the formal opening. Competition policy concerned with discovery must assess whether usable records and workable access allow meaningful comparisons over time.

The next chapter asks whether taxation could address conduct that removes useful comparison and leaves costs outside the actor's account. Before discussing a rate, we need to identify the behavior, the affected decision, and evidence for the claimed social cost.

Taxing lost comparison

Sweden introduced its nitrogen-oxide charge in 1992. Covered energy-production facilities paid for emissions, while receipts were returned in relation to useful energy output. The design separated a harmful by-product from the productive activity that generated it.[160]

A business decision can improve current production while closing a route to discovering alternatives. A case for taxation would need to show that losing that route imposes a cost on others, and that a tax is a useful way to address it.

There are two tax questions here. One asks how confidently income can be divided between productive contribution and historical position. The other asks whether particular conduct imposes an external cost by preventing valuable comparison. Explaining a fortune and changing future incentives are different tasks.

Income and its explanation

Payroll records, completed work, and measured output establish meaningful facts. They do not necessarily identify how much an input changed the probability of receiving a reward. A causal question about revenue also differs from one about a product's benefit to users. Each requires its own target, observations, and comparison.

Consider the fixed-input version of the allocation model. An admissible adjustment to hidden position can offset a change in the productive-input effect, leaving every observable allocation probability unchanged. Two candidate explanations can then assign the same law to all recorded histories while giving different answers about what changing the input would do. A rule based solely on that record has no observation on which to distinguish the exact contribution divisions.

This is an observational-equivalence argument. The underlying effects differ between candidate explanations; a shadow replay instead preserves those effects and varies the realized history. It is also separate from the finite-lifetime-comparison result. Input variation that the hidden position cannot absorb, or additional informative outcomes, may break the equivalence. The limitation concerns the specified experiment, rather than every possible form of economic evidence.

Taxation can proceed without an exact retrospective causal audit of every reward. Thomas Piketty, Emmanuel Saez, and Stefanie Stantcheva distinguish responses involving productive activity, avoidance, and compensation bargaining. The different behaviors have different implications for tax policy.[161]

Casey Rothschild and Florian Scheuer analyze taxation when income combines productive and rent-seeking activities that the authority cannot separately observe for each person. Changing a tax can alter relative returns across activities, strengthening or weakening a direct correction. Averaging the hidden components does not automatically produce the right rate.[162]

These approaches assess consequences using assumptions about behavior and welfare. They leave room for observable bases and plausible ranges without requiring exact knowledge of individual merit. The comparison-budget argument challenges a stronger confidence in what a reward record must eventually reveal; it does not make public finance impossible.

Moral desert and contractual entitlement require further judgments. A valid promise of payment can matter while its recipient's complete causal contribution remains uncertain. Security against income loss, rewards for valuable investment, and practical routes into new work pursue different aims. Evidence about alternative histories does not turn them into one policy or dictate equal incomes.

What would a charge measure?

The Swedish charge depends on a verification chain. The environmental agency describes instruments, declarations, and audits. Facility equipment must be compared at least annually with independent measurements from accredited laboratories, and the agency coordinates checks of the laboratories themselves.[160]

Lost comparison has no corresponding physical meter. A prevented assignment could have revealed a serious failure, repeated something already known, or produced an unreliable measurement. Its value depends on the decision it could change and the evidence available elsewhere. Counting missing assignments cannot establish those differences.

The comparison budget accumulates modeled allocation chance outside the current favorite. To move from that statistical object to money requires an account of how conduct changes opportunities, what observations those opportunities produce, and what their absence costs. A finite budget supplies no monetary damage estimate by itself.

The causal link must be established. Removing a restriction may still leave an experiment infeasible because participants lack resources, technical access, or a valid outcome measure. Alternatively, another source may already supply the relevant evidence. The effect of the conduct has to be evaluated against those feasible alternatives.

Scale can make replacement harder. Losing one local route may leave independent institutions elsewhere. Losing access through a gate shared across a large network can remove many practical routes together. That is a hypothesis about institutional dependence to investigate, alongside the productive gains the shared system supplies.

Beneficiaries matter too. A firm already has an incentive to buy information whose benefits it captures. Other users can benefit from an investigation without paying its producer. Kenneth Arrow examined the difficulty of appropriating the returns to information,

whether generated deliberately or as a by-product of ordinary activity.[163] This established information problem supplies a foundation for asking what useful evidence an exclusionary action prevents.

The proposed correction must count both sides. Productive integration can reduce operating costs, while secure evaluation may preserve a useful test without unrestricted disclosure. Privacy, security, disruption, and the cost of maintaining access belong in the comparison. The economic objective is a better attainable arrangement, with all its resource requirements included.

Production and private testing

A firm's response can occur on more than one margin. It can change a restrictive activity and change how much evaluation it purchases. Private testing is real evidence. If it supplies the same usable information to the same decision makers in time, treating the original loss as wholly unreplaced overstates that loss.

Equivalence needs to be demonstrated. An internal AI benchmark can use synthetic tasks while a field study observes real users; their errors may overlap, and their results may reach different people. Internal findings unavailable to an affected outside decision maker cannot automatically replace that person's missing evidence.

In their model of information acquisition, Luca Colombo, Gianluca Femminis, and Alessandro Pavan show that privately chosen information can differ from the socially efficient amount. Depending on the setting, the discrepancy can run in either direction.[164] Underinvestment therefore needs a specific account of benefits that private incentives insufficiently protect.

Different instruments affect those incentives differently. A charge on a restrictive term changes the return to imposing it. Payment for verified evaluation changes the return to producing evidence. Opening access may reduce the need for purchased tests; subsidizing tests may finance work that would have occurred anyway. Those interactions determine what a correction achieves.

Convenient tax bases can combine unlike activities. Computing produces ordinary outputs and also supports experiments investigating failures. A charge on all computing reaches both. Electricity, emissions, revenue, and distribution can justify separate tax questions, but an information-based correction needs evidence connecting its chosen base to the relevant loss.

Profit is no more direct a meter. Useful engineering, scarcity, historical advantage, and restrictive terms can all contribute to earnings. A corrective proposal must identify conduct and consequences, rather than use the size of a return as a substitute for them.

When public information displaces private effort

Colombo, Femminis, and Pavan also establish that greater public information precision reduces privately acquired precision in their specified model.[165] The result concerns an endogenous response, rather than a universal one-for-one cancellation of tests. Evaluating public provision requires counting what private participants change afterward.

A public result equivalent to one a firm planned to purchase can alter who pays without adding evidence. It may still save resources if public provision is cheaper. Different public measurements can instead expose failures private testing missed, reach other recipients, or supply work beyond private capacity. Announced test counts cannot distinguish those outcomes.

California's Electric Program Investment Charge provides a documented way to finance shared investigation. Utility customers fund research administered by the Energy Commission and participating utilities, supporting electricity-system research, development, and demonstration.[166]

During the December 20, 2022 Ferndale earthquake, the Redwood Coast Airport Microgrid supplied nearly fifteen hours of backup

during a regional outage lasting nearly sixteen hours, according to the project team's January 2026 report. Customers included an airport and a Coast Guard air station.[167]

The installation also had a persistent fault. Converters connecting the large solar array to the batteries repeatedly triggered protective trips. The report identified an unwanted current path to ground, particularly in humid conditions. Replacement equipment was eventually installed and tested. Another part of the system allowed backup operation despite the solar-converter problem.[167]

A demonstration can therefore provide service, expose a costly failure, and potentially inform later projects. These benefits have different recipients. The team found that readily monetizable benefits did not cover assessed costs, while the value of resilience depended on the community and services involved.[167] Establishing a worthwhile subsidy requires assessing wider benefits against the full costs, including what private activity it replaces.

Knowledge transmission is another step. A published diagnosis can help others only if it reaches them in a usable form and applies sufficiently to their problem. AI evaluation faces the same requirement: findings must be interpretable, relevant to subsequent tasks, and capable of changing a consequential decision. A large collection of scores can be less useful than a small test that resolves an important uncertainty.

William Kerr, Ramana Nanda, and Matthew Rhodes-Kropf describe entrepreneurship as experimentation, with staged commitments allowing learning before full investment.[168] A negative result can save resources by supporting abandonment. Its value comes from what was credibly learned, rather than failure itself.

Replacement and the feasible alternative

David Blackwell compared experiments through the decisions their information permits. If one can reproduce another's observations through a known conversion rule independent of the hidden state, a decision maker can imitate the second experiment's choices.[169] This supplies a discipline for evaluating whether evidence is an adequate substitute; acquisition and implementation costs require an additional calculation.

A lost field comparison may be replaceable by a cheaper independent evaluation. That route can reduce the sacrifice while preserving useful information. It must reach the relevant recipients in time, fit the decision, and account for capacity and delivery costs. A laboratory measurement that omits the development and learning enabled by real participation may not supply the same opportunity.

Replacement also changes the institution. A shadow replay preserves starting conditions and response rules; a new laboratory or access program changes them. The evidence such an intervention produces can be valuable without being equivalent to the lost history.

Secure infrastructure, qualified staff, independent checking, and documentation create fixed costs before the first result appears. Existing evaluators or shared facilities can reduce duplication, while common errors and centralized control can create new dependencies. A limited commission may establish feasibility before permanent provision is justified.

These considerations favor comparing whole arrangements: preserving access, buying private evaluations, supplying shared tests, or combining narrower measures. Useful institutional invention can retain the benefits of scale while reopening a consequential comparison. Preserving every old channel indefinitely is no substitute for that appraisal.

Financing and incentives

Britain's 2017 retail-banking order required providers to support implementation arrangements and fund independent service-quality surveys, with contributions linked to active accounts in the relevant survey market.[170] The account rule allocates a bill for specified duties. It does not measure each provider's responsibility for lost evidence.

Financing useful evaluation and correcting harmful conduct are different achievements. General revenue, supplier contributions, and charges tied to a demonstrated restrictive action produce different incentives. Raising enough money for a laboratory does not establish that the funding instrument targets the cause of the information loss.

The payment is also different from the resources consumed. A transfer can finance staff time, computing, and supervision; counting both the transfer and those inputs as destroyed output would double-count resource cost. Who pays still matters for distribution and behavior, including changes in prices, wages, investment, and entry.

Public provision has an opportunity cost as well. Money and expertise used for evaluation may displace another useful activity. Even a good laboratory can be financed through a charge that discourages valuable production or testing. Its funding and corrective effects require separate examination.

Verification is part of that expense. Repetitive easy tasks can inflate a test count, and selective cases can flatter an entrant. Relevant task selection, checked outcomes, version records, and independent scrutiny help establish what evidence was actually produced. A nominally cheap substitute can be costly once those safeguards are included.

Before choosing a rate

A defensible charge needs a causal chain: identified conduct, changed comparative evidence, affected decisions and beneficiaries, and an assessed loss after private responses and feasible replacement. It must then improve the attainable outcome after productive benefits, financing, enforcement, and entry are counted.

The time horizon must match on both sides. Software changes can make some findings obsolete while leaving enduring failure modes relevant. Longer effects on learning and entry require stated assumptions and evidence. A temporary information problem should not become a permanent fiscal category after its basis has changed.

Participation can change too. Qualification costs may deter useful entrants; predictable obligations may support investment. Warnings about reduced innovation need investigation, rather than automatic acceptance or dismissal. These are responses affecting the case for a particular instrument.

The evidence may favor access duties, payment for verified testing, public provision, or existing private arrangements. Where plausible accounts disagree about the direction of the welfare effect, a precise positive rate has not been established. A narrower intervention or a limited study may be the useful next step.

The contribution of shadow futures here is to make the production of comparison an economic concern. The comparison budget identifies a statistical constraint; public finance adds the causal and welfare analysis required for policy. Uncertainty about income's origins, financing shared knowledge, and correcting an external cost remain distinct arguments.

The next chapter asks how economics can study those institutions more directly. A successful history records real achievement. It can also change which alternatives remain discoverable. Understanding that interaction requires evidence about both the outcome and the arrangements producing opportunities to learn.

CHAPTER 12

An economics of possible worlds

The winter wheat planted at Broadbalk in autumn 1843 was harvested in 1844. The records of that harvest were the beginning of a much longer undertaking. Fertilizers, manure, and untreated ground supplied contrasts that successive generations of researchers continued to study. The field produced grain. Keeping differences alive helped it produce evidence too.[171]

An ordinary commercial field might sensibly receive whatever treatment its owner expects to work best. Broadbalk had another purpose. Some ground continued receiving treatments whose value could be assessed against other ground. Giving every strip the apparently best treatment would have changed what the field could teach.

That does not make the experiment an argument against productive farming. It makes its second product visible. A harvest is an outcome. A maintained comparison helps investigate an explanation. The labor, land, records, and organization required to preserve that explanation have costs and possible benefits of their own.

The long record also makes a difficulty visible. An experiment cannot remain useful simply by refusing to change. Weeds, soil acidity, new varieties, and changing farming practices can alter the question a comparison answers. Rothamsted's account of its experiments describes considered modifications alongside preserved treatment differences.[172]

By the late 1940s, some plots had become so acidic that comparing two nitrogen fertilizers no longer answered the intended question. In the separate Agdell experiment, acidity favored a disease that

depressed turnip yields; the experiment required extensive modification in 1951. Researchers applied chalk to correct acidity. Broadbalk later received shorter-strawed wheat varieties and a revised cropping arrangement in 1968. Preserving a useful experiment required changing what happened on the ground.[172]

The guide describes another consequence of persistence. Repeated applications of manure and fertilizer altered the soils themselves, leaving different stores of nutrients and organic matter. Later investigations could study those reserves and their effects. Yesterday's treatment had become part of today's condition. This is a productive reason to keep the history, rather than assume every harvest starts with identical ground. It also limits a careless reading of the result: a difference after many years can reflect the accumulated consequences of earlier management as well as the immediate application. The relevant comparison depends on whether the question concerns this year's input, a sustained practice, or a change from the inherited state.

A field trial compares treatments with accumulated histories. It cannot erase their earlier development. Likewise, evaluating available technologies and estimating how different early opportunities would have changed their development are different questions, with different evidence requirements.

There is no timeless instruction here to hold everything fixed. A researcher must know which difference is being maintained, what else has changed, and whether the resulting observation still bears on the intended question. A familiar treatment name cannot guarantee an unchanged experiment.

AI evaluation faces a similar continuity problem. An unchanged benchmark can become a precise answer to an obsolete question. Revising it only when a favored system performs badly can undermine comparability. Documented changes are part of the evidence.

Broadbalk's plan distinguishes cropping and management histories across treatment strips. The name alone does not identify the ground, treatment, or period producing an observation. The same discipline applies to software: a service label can survive changes in models,

interfaces, and review procedures. Version history helps separate those changes from differences in assigned work.

The fields yield crops and measurements: harvests, soil conditions, and changes across treated plots. Maintaining those comparisons is part of Rothamsted's productive work. Keeping an experiment going can make questions answerable that an ordinary harvest would leave unresolved.

The future of economics needs more attention to that supply. We already study how resources produce goods and services, how incentives influence behavior, and how uncertainty affects choices. We should also investigate which arrangements let consequential alternatives develop interpretable records, and what is lost when those arrangements disappear.

A second use for the harvest

Rothamsted's sample archive gives this argument an unexpected extension. Material retained from earlier harvests can be examined with methods unavailable when it was collected. The preserved record includes physical samples, not only numbers in a table. A later researcher can sometimes ask the material a new question.[173]

Sarah Bearchell and colleagues took small portions of old leaves and stems from archive containers, powdered them using liquid nitrogen, and extracted DNA. They looked for short DNA sequences to accommodate damage in the old material. Their 2005 study examined two wheat pathogens in samples spanning 1844–2003. Gaps in the archive required combining material from two plots, after checks of comparability. Changes in the pathogens' relative abundance were associated with sulfur-dioxide emissions. The researchers considered other correlated factors; they had not randomized atmospheric pollution.[174]

The useful surprise is the question made possible. Harvest material retained for one scientific purpose could contain evidence relevant to another. The archive connected a modern laboratory method with a

sequence of older observations. It did not manufacture those earlier samples after discovering that they would be useful.

This gives preservation a potential value that cannot always be specified completely in advance. Someone designing a record for today's decision may not know every future question. Keeping suitable material and documentation can make later investigation possible. The archive supplies a real example, rather than a promise that every retained dataset will eventually justify its storage bill.

The distinction between storage and comparison remains essential. A sample can reveal another measurement of a harvest that occurred. It cannot supply the harvest from a treatment that was never applied to that ground. Preserving material expands the observations available from a history; preserving treatment contrasts helps create a different part of the evidence.

Digital records likewise differ in what they preserve. Requests, versions, outputs, corrections, and checked outcomes can support more questions than a success total. They still cannot invent an unobserved alternative's response to a task.

Neither statement is a reason to collect everything. Residents' records can contain sensitive information. Storage, documentation, security, and authorized access consume resources. The institution must choose what to preserve for justified purposes and how to protect it. A comparison worth maintaining may need a carefully limited record rather than unrestricted retention.

Preservation has an option value and a cost. Retention policies should specify permitted uses, provenance, exclusions, and limits on later claims while protecting the people whose data make investigation possible. The future researcher is one beneficiary among others.

The institutional task is to make those tradeoffs reviewable. A preservation decision should identify the anticipated use, likely costs, permissions, review date, and consequences of deletion. Some judgments will prove mistaken as technology changes. A documented reason for the original choice gives later decision makers a basis for

revision. At Rothamsted, the connection between retained material and recorded management allows new questions to reach backward. In software procurement, a comparable ambition is a protected record that remains intelligible after an interface, model, or contractor changes. Neither ambition requires promising that every future question will be answerable.

The wheat archive also cautions against treating old records as self-explanatory. Researchers had to use methods suited to aged material and check what their measurements meant. An economist recovering old procurement logs would likewise need to understand missing entries, changing definitions, and the process that created the archive. Historical depth is valuable when it remains interpretable.

A final success label can discard the original input and the correction needed to diagnose a failure. Retaining those records may enable re-evaluation, subject to reliable checks, relevance, privacy, and reproducible software versions. A useful archive preserves enough context for a justified question.

Archives, natural variation, additional measured outcomes, and designed experiments can all supply evidence about an alternative. Each route depends on a connection between the observations available and the question being asked.

The arrangement makes a difference

Ronald Fisher's 1926 paper on field experiments began with a familiar agricultural difficulty: apparently similar ground can yield differently. A treatment's success cannot be assessed solely by weighing the crop accurately. The arrangement of the comparison affects what the difference means.[175]

His discussion moved from replication to the way plots were allocated. A collection of treatment labels arranged systematically could produce a misleading estimate of error. Random arrangements, including appropriate restrictions that the analysis respects, could support a different inferential claim. The method depended on how

the experiment was generated, rather than the appearance of its final diagram alone.

The original Broadbalk experiment should not be retroactively described as a modern randomized trial. Its lasting treatment contrasts, subsequent changes, and accumulated observations require their own interpretation. Fisher's paper supplies a separate methodological point: design is part of the evidence, and uncertainty calculations must correspond to that design.

The same design issues recur in service evaluation. Different task difficulty, inconsistent judgment, or a shared faulty reference can distort apparently balanced comparisons. Repetition alone does not repair a common measurement error.

The remedy is more specific than asking for more data. Which requests entered? How were they allocated? Which outcomes were independently checked? Which service version produced them? What could the system retain or learn between requests? The answers describe the experiment that the analyst actually has.

Fisher's question about apparently similar ground remains useful for economics: what would justify attributing an observed difference to a treatment rather than the allocation process? Designed experiments, credible existing variation, and structural restrictions offer different routes. Counting an outcome accurately does not identify its cause.

Experimental design, causal inference, structural modeling, sequential learning, and partial identification already provide ways to study counterfactual questions. The comparison budget focuses attention on one part of the evidence supply: the accumulated chances that an allocation goes outside the current favorite.

Under the model's conditions explained in Chapter 3, those chances limit how precisely productive contribution can be recovered from the allocation history. A researcher can then ask whether the real market supplies other observations that overcome that limit.

An application should start with the available records. Does the analyst know the assignment probabilities? Are customer outcomes,

independent tests, or other measurements available? Those observations determine whether the model describes the actual information problem.

An empirical study need not begin by claiming that an industry's lifetime budget is finite. A finite record cannot establish that conclusion merely because comparison appears to be slowing. It can investigate a defined horizon, assess uncertainty about the allocation mechanism, and compare feasible ways of producing evidence. That would already be useful progress.

Simulation offers a narrower test. Build an artificial market whose rules are known, run the proposed estimator, and check whether it recovers what was put in. Tim Morris, Ian White, and Michael Crowther explain why a simulation study must state its aims, data-generating mechanisms, and measures of performance. A method can work in that artificial world while failing in a real market whose feedback rules are different.[176]

A credible account of an allocation process records eligibility, assignment, internal testing, rejected cases, and checked outcomes. Known probabilities should have a documented basis; estimated probabilities need uncertainty. Completed-work shares cannot substitute for them.

An experiment observing paired outcomes contains more information than an allocation-winner record. Its number of cases is not automatically a comparison budget. Applying an allocation-only limit while ignoring checked responses misstates the observation scheme. Designing additional informative outcomes can change what is learnable even when production remains concentrated.

What an empirical test would need

Kasy and Sautmann's agricultural recruitment application supplies a documented example of matching experimentation to policy choice. Six recruitment options were tested through an adaptive calling design with 10,000 numbers over seventeen waves. The outcome was completion of five recruitment questions, rather than later agricultural income.[177]

The call experiment measured recruitment. To assess the agricultural service's effects on yields or income, researchers would need to follow those outcomes as well. What gets recorded determines which decisions the evidence can support.

A useful study would distinguish assignment chances, observed outcomes, and the effect being estimated. It would ask what can be learned from allocations alone, how checked outcomes change the answer, and how sensitive the conclusion is to plausible feedback rules.

Dependence belongs in that account. Repeated records from the same underlying event, common shocks, and shared evaluation errors cannot be treated as fresh independent evidence without justification. Recording versions, exclusions, and failures makes those judgments inspectable.

Feasibility includes power to detect a consequential difference, the cost of supervision, privacy protection, and what action the result can change. A spending limit controls expenditure; it does not by itself establish adequate information or net social value.

The resulting report should connect methods, uncertainty, costs, and decisions. Negative and missing results need to remain visible. A successful evaluation can favor an existing arrangement, recommend a narrower change, or show that further investigation is not worth its cost.

The permissions behind the experiment

The design still needs an institution capable of carrying it out. Requests may be controlled by the buyer, service interfaces by suppliers, and verification resources by another organization. Having each ingredient somewhere does not mean the usable experiment has been assembled.

Kenneth Arrow's 1963 study of medical care supplies an earlier institutional argument. The patient, in his account, cannot inspect the service fully before consuming it and depends on knowledge the physician possesses. Expectations that the doctor should put the patient's welfare first help organize this unequal relationship. Arrow interpreted medical institutions partly as responses to uncertainty about illness, treatment, and information. He also warned that an attempted remedy can introduce other obstacles. Institutions deserve investigation as arrangements for managing uncertainty, with consequences that require appraisal.[178]

James Buchanan and Yong Yoon analyze an anticommons: multiple rights to exclude can leave a resource underused. Their parking example requires complementary permits from separate holders.[179] Information can face a related institutional problem when useful records, interpretation, and permission to evaluate are controlled separately.

Contracting can sometimes assemble those complements. Commitment, verifiable delivery, compensable costs, and financing determine whether a feasible agreement exists. A nominal right of access does not provide the remaining inputs for free.

This should make us wary of explaining every refusal as a plan to suppress competition. A holder may face genuine privacy, security, staff, or productive costs. The useful question is whether those costs exceed the attainable benefit, whether the beneficiary can compensate the provider, and whether the arrangement can be enforced. A refusal can be privately rational without settling the wider economic case.

It should also make us wary of assuming that a mandate supplies every missing input. A rule can require access while leaving nobody responsible for defining the outcomes or paying the evaluator. Another can fund a laboratory that lacks permission to observe the relevant tasks. A useful institutional design assembles the complements needed for the stated comparison.

Rothamsted's maintained fields and archive offer a historical reference for this less glamorous work. Treatments, measurements, material, and documentation must remain connected across time. An isolated bag of grain and a beautifully organized treatment register are different things. Their joint usefulness depends on retaining the connection between them.

Institutional alternatives include existing qualified evaluators, secure contracts, shared infrastructure, and independently verified private tests. Their costs and possible common failures should be compared before establishing a new public body.

An experiment must permit a result favoring the incumbent. An institution whose survival requires the challenger to win has compromised its evidentiary purpose.

Credibility also requires stable criteria, recorded revisions, and relevant rights of challenge. Independent review can test whether the evidence supports a conclusion; the institution acting on it remains responsible for its decision.

The same principle applies to the researcher. A study of comparison opportunities should preserve evidence that a market already generates useful tests or that no practical alternative meets the requirements. Negative findings help distinguish a mechanism from a slogan. They can show where the comparison-budget perspective adds little to an existing explanation.

When the leading arrangement fails

The country comparison proposes a question about robustness: can an economy discover and adopt workable alternatives after a technological, commercial, or policy failure, at acceptable cost? A large computing fleet alone cannot answer it.

A failure can be isolated by independent tests or reproduced across many services sharing a common checklist and restricted evaluation route. The number of brands says little about which situation applies. Robustness concerns the practical ability to expose the failure and develop a workable alternative.

Track the route from discovering the problem to putting a better arrangement into service: permissions, time, expense, mistaken decisions, and productive capacity lost. Ask whether a dormant supplier's retained records helped, whether existing private tests supplied the answer, and whether a cheaper substitute was available. Then investigate whether public access, an enforceable interface, or transferable records improved recovery. Differences between countries would also require attention to their resources and other institutions; a faster recovery by itself would not identify the cause. Record unsuccessful applications too; successful recipients alone cannot describe the access system.

The hypothesis must be able to lose. Subsidized computing might produce unusable results. Official checks might suppress failures or spread a common error. Compliance burdens might deter the investment an alternative needs. Count displaced private experiments and the benefits of faster initial development as well as recovery. The relevant time horizon and consequences must be stated.

Announced American, European, and Chinese policies open this inquiry; they cannot settle it. Begin with comparable decisions and observable routes through failure. Ask who could challenge the accepted explanation, what evidence they obtained, and whether acting on it improved the outcome. Those questions let the institutional argument risk an answer against itself.

What an economist should report

An economic account of a widely used service should establish its costs, accuracy, and reach, then explain what its operating arrangement permits participants to learn about alternatives. Productive improvement and comparative learning can move together or apart.

An informative report would state its target. Is it estimating the value of a supplier's recommendation, the effect of access on entry, the effect of a training design on workers, or the effect of an input on winning rewards? These questions can be connected, but a success on one is not an answer to all the others.

It would identify the observation process. The routing rule, the outcomes retained, internal experiments, and external tests can all matter. A conclusion drawn from allocation winners alone should say so. If more information exists, the analyst should explain what changes when it is included.

It would report the relevant uncertainty about the mechanism. Two histories with the same market shares may have different assignment processes. Two models can forecast the observed process similarly while disagreeing about how an intervention would work. Precise estimates under one maintained explanation should not conceal uncertainty about that explanation.

Blackwell and Dubins show how conditional forecasts can merge under specified assumptions. In the allocation model, that agreement can coexist with unresolved contribution. Accurate prediction of the next winner does not automatically identify the input effect behind the process.[180]

Forecasts of workload and inference about a different technology answer different questions. Each can be valuable when its evidence and assumptions fit the intended use.

It would also distinguish statistical information from decision value. A measured contrast could be precise but economically trivial. Another could remain uncertain while affecting a consequential

choice. Blackwell's decision perspective supplies a foundation; collection costs, interpretation, incentives, and legal feasibility still determine whether obtaining the evidence is worthwhile.[181]

A precise but tiny contrast may leave an action unchanged; uncertainty about a costly failure may justify a targeted test. The report must identify the consequential decision rather than substitute a larger comparison budget for a value argument.

Finally, the report would state which counterfactual it is evaluating. A shadow replay holds underlying effects, initial conditions, and response rules fixed while allowing the realized history to differ. A new access rule or training policy changes the institution. Estimating the consequences requires assumptions about behavior under that intervention. The two operations should not slide into each other halfway through the argument.

A useful procurement report can be brief: what question did the buyer ask, which services were compared, what happened, what remains uncertain, and what decision follows? The methods and records should be available for anyone authorized to check them.

A research program with room to fail

The comparison budget is not a ready-made ranking of economies. Application requires a specified allocation process, justified conditional probabilities, relevant alternatives, and a target the available observations can distinguish.

A first useful project could reconstruct a procurement process with known assignment rules and checked outcomes. It would ask how much the allocation record alone reveals, what the additional outcomes contribute, and how conclusions change under plausible feedback mechanisms. An estimate that changes sharply when a reasonable mechanism is altered should be reported as sensitive, rather than polished into false certainty.

A second could compare feasible evaluation designs within that process. Extend one history, create a fresh task cohort, retain a

previous measurement, or buy another source of evidence. The aim would be to discover which option improves a defined inference or purchasing decision at acceptable cost. Calling a cohort fresh would not establish independence if it retains the same suppliers' learned state and shared shocks.

A third could investigate the institution supplying access. Which permissions, costs, and incentives prevent a useful test? Can a contract solve the problem? Do public tests replace private ones? Does a narrow interface preserve scale benefits while making a missing outcome observable? These are questions for evidence, and their answers could weaken as well as strengthen a case for intervention.

A useful next study could begin with one purchase. Record which alternatives were eligible, how they were selected, which outcomes reached the buyer, and what additional test changed the decision. That would not measure an entire economy’s lost possibilities. It could establish something more modest and actionable: whether a specific institution could learn more, at a cost worth paying.

Hayek's discovery argument remains an important predecessor. Competition can reveal facts that were not known or used beforehand. A serious extension should ask where the discovery process works, which questions it answers, and where other arrangements could improve it. It should not treat every departure from a particular model as a reason to dismiss decentralized learning.[182]

Likewise, acknowledging the costs of duplicated production should not end the inquiry before considering duplicated evaluation. Several firms need not each build a complete frontier training operation to contribute useful tests of deployed services. Some comparisons can occur within an integrated organization; others may need independent access. The capability, cost, and information questions must each be examined.

The physical AI economy makes this separation consequential. Computing capacity, power connections, skilled staff, and working systems are productive constraints. Preserving a hypothetical

competitor with no feasible route to meeting them is not an experiment. A bounded test of an available alternative might be. The researcher must specify the alternative well enough for somebody to attempt it.

The prospect of failure is part of the program's value.

A convincing first application should therefore be small enough to inspect. An observation inventory, an auditable allocation rule, a set of checked outcomes, and a plainly stated decision would be more valuable than an industry ranking built from convenient public shares. Other researchers could then examine the same target under a different mechanism, add an omitted measurement, or run a feasible replication. If replacing a lost comparison costs little, a dramatic damage story weakens. If a proposed intervention destroys more productive value than its evidence can recover, the recommendation should change.

There is a corresponding possibility on the other side. A seemingly efficient arrangement may omit the only feasible route through which an alternative can demonstrate a consequential capability. Discovering that fact could justify a well-targeted institutional change. The policy case becomes stronger when the missing comparison and the attainable improvement are shown, rather than merely imagined.

The record we choose to leave

At Rothamsted, someone had to preserve the samples, label them, and keep the treatment records intelligible. Later researchers could put new questions to that material. An archive that had been discarded could not have answered them.

Markets leave records too. Some come from ordinary sales; others from deliberate tests. A customer database, a trial's assignment log, and an independent evaluation can help answer different questions. Economics should investigate what each costs to produce and which decisions it can improve.

MusicLab gave the same recordings more than one opportunity to develop a following. Lockdown finished first in one world and fortieth in another.[183] Seeing both charts changes the explanation we can give. An account built around the winning result has to accommodate the other outcome too.

To judge an institution, follow a decision. A buyer considering a new supplier should be able to use a trial's result. A training program should be able to change when a different method helps beginners. Keeping records, arranging access, and checking outcomes matter when they give someone a workable reason to act.

The value of another comparison lies in the choice it can improve. Economics should help us find where that value is greatest, and how to obtain it.

The next opportunity

At Rothamsted in England, wheat grows in an experiment begun in 1843. Across its plots, researchers have compared mineral fertilizers, manure, and no fertilizer. They have kept records and saved samples of crops and soil. Generations of researchers have inherited both the harvests and the differences that help explain them.[184]

Give every plot the same treatment, and the field might produce more wheat. It would also stop producing that continuing comparison. An untreated plot can look like wasted ground when this year's crop is all that counts. Its value changes when we count what it helps people learn.

Someone has to keep making room for it. Soil must be tended, samples preserved, records understood. The experiment has changed as agricultural methods and scientific questions have changed. The knowledge available to later researchers depends on work that earlier generations chose to do.

A market leaves a different kind of field behind it. Orders, wages, investments, and download counts record where activity went. They can tell us a great deal about the businesses and people who received those opportunities. The rejected application usually leaves a thinner file.

For a beginner, an assignment can bring pay, practice, a reference, and something to show the next employer. Turn someone down for lack of experience, and the decision can help perpetuate the very lack it cites. Another year of work by the experienced applicant will not reveal what the beginner would have learned. To find that out, someone has to offer a beginning.

The same difficulty can arise when software does the work. A company might send thousands of tasks to AI agents using one model and build an impressive record of completed jobs. To discover

whether another system would perform better, it needs suitable tasks, access to that system, and a way to compare results. Another million completions by the incumbent do not automatically provide that comparison.

A rival listed on a website has permission to exist. Getting its work into an evaluation, its product in front of customers, or its findings into a decision is a further step. When a gatekeeper controls those routes, it can help decide which alternatives acquire evidence and which remain promises. That is where epistemic monopoly reaches into everyday economic life.

We can spend heavily on producing answers while spending little on discovering whether we are asking the right system. We can reward experience while closing the routes through which someone else could acquire it. A busy economy can keep repeating a choice without getting much better at explaining it.

The practical response begins with something tangible. A buyer can put a rival through a useful test. A shared computing facility can let a researcher attempt a different design. An interface can let customers try another service without abandoning everything they have already built. Each proposal needs a reason to justify its cost. But the value of finding out belongs in that calculation alongside the value of the next completed job.

A well-designed comparison can confirm that the established supplier is still the best choice. It can also expose a better method, a neglected skill, or a mistake shared by everyone using the same system. Its purpose is to let what happens in the test change what happens afterward.

At MusicLab, that possibility was built into the experiment. The same forty-eight recordings entered separate worlds. Listeners made choices, those choices became visible to later listeners, and the charts developed differently. Lockdown finished first in one world and fortieth in another.[185]

Imagine having only the first chart. We could play the winning recording, examine its qualities, and write a persuasive account of

why it deserved to lead. With only the other chart, we could do the same for its failure. The second history makes both explanations harder to deliver with such confidence.

That discomfort is useful. It gives us room to admire an achievement without treating its reward as the final word on everyone else's ability. It asks us to look again at the applicant, the business, or the idea whose thin record may partly reflect the opportunities it never received.

Most shadow futures will leave no second chart. We cannot recover every missed opportunity or run every possible experiment. We can, however, make room for comparisons that might change an important choice. What seems inevitable after success can still be open before the next decision.

Lockdown's other chart existed because someone made room for another history. The recording was ready. The opportunity had to be built.

The chart tells us what won. What gets a chance next is still up to us.

Notes

The other life of a song

1. Matthew J. Salganik, *Bit by Bit: Social Research in the Digital Age*, online open review edition, section 4.6.1, "Create zero variable cost data," especially figures 4.20–4.21 and the paragraph reporting Lockdown's rankings. Source for first and fortieth place, separate worlds with different participants, and the grid and ranked-list presentations. Author's account.

2. Matthew J. Salganik, Peter Sheridan Dodds, and Duncan J. Watts, "Experimental Study of Inequality and Unpredictability in an Artificial Cultural Market," *Science* 311, no. 5762 (2006): 854–856, doi:10.1126/science.1121066. Design, participant count, and zero initial downloads: p. 854. Inequality, variation across worlds, and the continuing role of independently assessed appeal: pp. 854–855, figures 1–3. The later thousand-listener extension is an explicitly imagined illustration, not another experimental finding. Original article and supporting materials.

3. Arnout van de Rijt, Soong Moon Kang, Michael Restivo, and Akshay Patil, "Field Experiments of Success-Breeds-Success Dynamics," *Proceedings of the National Academy of Sciences* 111, no. 19 (2014): 6934–6939, doi:10.1073/pnas.1316836111. Kickstarter sample and random assignment: pp. 6934–6935; 39 versus 70 percent receiving subsequent third-party funding: p. 6935, figure 1. Diminishing marginal returns to larger initial treatments: pp. 6936–6937, figure 3. The 70 percent is not the proportion achieving their funding target. The initial experiment selected projects with funding goals up to $1,000; methods and the later treatment comparison are on p. 6937. Full paper hosted by the European University Institute.

4. Robert K. Merton, "The Matthew Effect in Science," *Science* 159, no. 3810 (1968): 56–63, especially pp. 56–57, doi:10.1126/science.159.3810.56. Merton's account draws in part on Harriet Zuckerman's interviews with Nobel laureates; p. 57 connects recognition with resources for further research. Original article. Armen A. Alchian, "Uncertainty, Evolution, and Economic Theory," *Journal of Political Economy* 58, no. 3 (1950): 211–221, especially pp. 211–213, doi:10.1086/256940, treats profit-based selection without requiring optimal individual foresight. Fuller credit for reinforcement, estimation failure, and predictive merging belongs in Chapters 2–3.

5. The name's inspiration is Steven Johnson's description of Stuart A. Kauffman's adjacent possible. See Johnson, "The Genius of the Tinkerer," *The Wall Street Journal*, September 25, 2010, adapted from *Where Good Ideas Come From: The Natural History of Innovation*, archived print view, PDF p. 2, paragraphs introducing Kauffman and the

expanding-house analogy. The introduction explains the connection and the narrower economic use developed here.

6. Martin Watzinger, Thomas A. Fackler, Markus Nagler, and Monika Schnitzer, "How Antitrust Enforcement Can Spur Innovation: Bell Labs and the 1956 Consent Decree," *American Economic Journal: Economic Policy* 12, no. 4 (2020): 328–359, doi:10.1257/pol.20190086, abstract; June 9, 2017 author working paper, pp. 1–4 and 6–8.

7. Erik Brynjolfsson, Danielle Li, and Lindsey Raymond, "Generative AI at Work," *The Quarterly Journal of Economics* 140, no. 2 (2025): 889–942, doi:10.1093/qje/qjae044. See the abstract and Section I for the 5,172-agent sample and average 15 percent gain; Sections III and VI.A for agents' control over suggestions; Sections V.B and VI.B for heterogeneity and learning. The study follows a staggered rollout at one firm; it does not establish economy-wide employment or wage effects. See also the corresponding November 2024 working paper, pp. 1–3. Published article; author working paper. Earlier working-paper figures of 5,179 agents and 14 percent are not used here.

8. Joel Becker, Nate Rush, Beth Barnes, and David Rein, "Measuring the Impact of Early-2025 AI on Experienced Open-Source Developer Productivity", 2025, abstract and pp. 1–5; Joel Becker, Nate Rush, Tom Cunningham, David Rein, and Khalid Mahamud, "We Are Changing Our Developer Productivity Experiment Design", METR, February 24, 2026, sections "Wider adoption of AI has made it more difficult to measure task-level productivity" and "Other means of measuring productivity."

9. Rothamsted Research, "Why Do We Make Changes to the Long-Term Experiments at Rothamsted?", January 17, 2024, opening paragraphs and the Broadbalk discussion. Source for experiments begun from 1843 by John Lawes and Henry Gilbert, retained soil and plant samples, continuing treatments, and changes in methods. The institution also cautions that small-plot findings do not automatically generalize to farming at landscape scale. Institutional account. The harvest-versus-comparison discussion is an interpretation of experimental design, not a measured claim about forgone yields.

The evidence success leaves behind

10. Thomas Blake, Chris Nosko, and Steven Tadelis, "Consumer Heterogeneity and Paid Search Effectiveness: A Large-Scale Field Experiment," *Econometrica* 83, no. 1 (2015): 155–174, doi:10.3982/ECTA12423. See the authors' May 2014 NBER working paper 20171, printed pp. 1–5 and the brand-keyword experiment in Section 3. The findings concern the studied company, campaigns, populations, and short-term outcomes. Full working paper. The opening customer journey is an illustration of the documented attribution problem, not a reconstructed individual customer.

11. Martin Erlic, "Shadow Futures: Contribution Uncertainty and the Self-Reinforcing Market," SSRN 6003994. The original formulation of the comparison budget and

contribution-uncertainty argument; the model and its conditions are explained in Chapters 2–3.

12. Salganik, Dodds, and Watts, "Experimental Study of Inequality and Unpredictability in an Artificial Cultural Market," *Science* 311 (2006): 854–856, especially pp. 854–855, original article. Lockdown's specific ranks are reported in Salganik's *Bit by Bit*, open review edition, section 4.6.1.

13. George A. Akerlof, "The Market for 'Lemons': Quality Uncertainty and the Market Mechanism," *The Quarterly Journal of Economics* 84, no. 3 (1970): 488–500, especially pp. 488–490. The used-car example is an explicitly simplifying model in the original paper. Original paper. It concerns information asymmetry and selection; the observed-input attribution question here is a distinct problem that can coexist with it.

14. Arnout van de Rijt, Soong Moon Kang, Michael Restivo, and Akshay Patil, "Field Experiments of Success-Breeds-Success Dynamics," *PNAS* 111 (2014): 6934–6939, especially 6934–6937. doi:10.1073/pnas.1316836111.

15. Steven Johnson, "The Genius of the Tinkerer," *The Wall Street Journal*, September 25, 2010, adapted from *Where Good Ideas Come From: The Natural History of Innovation* (Riverhead Books, 2010), archived print view, PDF p. 2; Stuart A. Kauffman, *At Home in the Universe: The Search for the Laws of Self-Organization and Complexity* (Oxford University Press, 1995); Kauffman, "The Adjacent Possible", *Edge*, November 9, 2003.

16. Robert K. Merton, "The Matthew Effect in Science," *Science* 159 (1968): 56–63, especially pp. 56–57, original paper; Armen A. Alchian, "Uncertainty, Evolution, and Economic Theory," *Journal of Political Economy* 58 (1950): 211–221, especially pp. 211–213, doi:10.1086/256940. Detailed attribution of increasing returns, reinforcement, estimation failure, and predictive merging follows in Chapters 2–4.

17. Thijs Bol, Mathijs de Vaan, and Arnout van de Rijt, "The Matthew Effect in Science Funding," *PNAS* 115, no. 19 (2018): 4887–4890, especially pp. 4888–4889, figures 1 and 3, doi:10.1073/pnas.1719557115. Full paper.

18. Erik Brynjolfsson, Danielle Li, and Lindsey Raymond, "Generative AI at Work," *The Quarterly Journal of Economics* 140 (2025): 889–942, abstract and Sections I, V.B, and VI, doi:10.1093/qje/qjae044. This is the published 5,172-agent study used in the prologue. Its measured productivity and learning results do not settle economy-wide employment, wage allocation, or alternative long-term training arrangements.

19. Joel Becker, Nate Rush, Tom Cunningham, David Rein, and Khalid Mahamud, "We Are Changing Our Developer Productivity Experiment Design," METR, February 24, 2026, sections on wider adoption and alternative measurement designs, research team's account.

20. Martin Watzinger, Thomas A. Fackler, Markus Nagler, and Monika Schnitzer, "How Antitrust Enforcement Can Spur Innovation: Bell Labs and the 1956 Consent Decree,"

American Economic Journal: Economic Policy 12 (2020): 328–359, published record. For the historical and foreclosure analysis, see the June 2017 author working paper, pp. 1–4 and 7–8, full text. The study's innovation estimates do not measure a comparison budget.

21. Rothamsted Research, "Why Do We Make Changes to the Long-Term Experiments at Rothamsted?", January 17, 2024, opening and Broadbalk discussion, institutional account. Maintained treatments coexist with changes in cultivars and methods; plot-scale findings have limits when applied to farming at wider scales.

The winner and the explanation

22. Matthew J. Salganik, Peter Sheridan Dodds, and Duncan J. Watts, "Experimental Study of Inequality and Unpredictability in an Artificial Cultural Market," *Science* 311 (2006): 854–856, especially pp. 854–855 and figures 1–3, full article. Lockdown's ranks are reported in Salganik's *Bit by Bit*, open review edition, section 4.6.1. The retrospective critic and missing-chart exercise are illustrations, not reported individual reactions.

23. Arnout van de Rijt, Soong Moon Kang, Michael Restivo, and Akshay Patil, "Field Experiments of Success-Breeds-Success Dynamics," *PNAS* 111 (2014): 6934–6939, doi:10.1073/pnas.1316836111. Intervention design and four settings: pp. 6934–6935; marginal returns: pp. 6936–6937 and figure 3; matching, sampling, and procedures: p. 6937. Full text. The study establishes intervention effects; discussion of specific donor reasoning is a possible mechanism, not an observed private motive.

24. Lev Muchnik, Sinan Aral, and Sean J. Taylor, "Social Influence Bias: A Randomized Experiment," *Science* 341, no. 6146 (2013): 647–651, doi:10.1126/science.1240466. Design, site ordering, and sample: pp. 647–648; first-viewer effects and final scores: pp. 648–649, figure 1; negative-treatment correction: p. 649. Full paper.

25. Robert K. Merton, "The Matthew Effect in Science," *Science* 159 (1968): 56–63, doi:10.1126/science.159.3810.56. Forty-first chair and named examples: p. 56; recognition as access to resources: p. 57; credit for Harriet Zuckerman's interviews: opening. Original article. The fixed number of honors and the subsequent feedback from an honor are analytically distinct.

26. Thijs Bol, Mathijs de Vaan, and Arnout van de Rijt, "The Matthew Effect in Science Funding," *PNAS* 115 (2018): 4887–4890, doi:10.1073/pnas.1719557115. Threshold design: p. 4888; participation, output measures, and further funding excluding the initial grant: p. 4889, figures 1–3; institutional discussion: p. 4890. Full paper.

27. Yang Wang, Benjamin F. Jones, and Dashun Wang, "Early-Career Setback and Future Career Impact," *Nature Communications* 10 (2019), article 4331, doi:10.1038/s41467-019-12189-3. See "Future career impact," analyses of screening, and the Discussion. The higher-impact comparison is conditional on remaining active in the

NIH system. The authors investigate and argue against screening as a complete explanation; the text does not present screening as their settled conclusion. Full article.

28. Vincent Traag, Emer Brady, Philippe Vincent-Lamarre, Flavie Bidel, Cindy Lopes-Bento, Jens Peter Andersen, and Carter Bloch, "The Matthew Effect and Early-Career Setbacks in Research Funding—a Replication Study," *eLife*, reviewed preprint version 1, October 13, 2025, doi:10.7554/eLife.109042.1. Abstract, Sections 2–4, and Appendix A describe scope, robustness, reapplication, and selection. Generalization beyond near-threshold comparisons uses additional modeling assumptions. Full reviewed preprint.

29. Armen A. Alchian, "Uncertainty, Evolution, and Economic Theory," *Journal of Political Economy* 58 (1950): 211–221, doi:10.1086/256940, pp. 213–214.

30. George A. Akerlof, "The Market for 'Lemons': Quality Uncertainty and the Market Mechanism," *The Quarterly Journal of Economics* 84 (1970): 488–500, especially pp. 488–490, full paper. Used cars are a simplifying model. The inspection, dealership, and salesperson extensions here are labeled thought experiments rather than empirical claims made by Akerlof.

31. Thomas Blake, Chris Nosko, and Steven Tadelis, "Consumer Heterogeneity and Paid Search Effectiveness: A Large-Scale Field Experiment," *Econometrica* 83 (2015): 155–174, doi:10.3982/ECTA12423. See the May 2014 NBER working paper, pp. 1–5, Section 3 and Section 4.1, printed p. 11, full working paper. The geographic test used 68 areas without nonbrand ads, 65 matched controls and 77 remaining controls. Random selection of candidate areas was followed by historical-sales matching; this was not pure individual randomization. The sixty-day duration is stated on p. 4. See the introduction for the distinction between brand and nonbrand results.

32. Martin Watzinger, Thomas A. Fackler, Markus Nagler, and Monika Schnitzer, "How Antitrust Enforcement Can Spur Innovation: Bell Labs and the 1956 Consent Decree," *American Economic Journal: Economic Policy* 12 (2020): 328–359, published record. For the historical and foreclosure analysis, see the June 2017 author working paper, pp. 1–4 and 7–8, full working paper. No actual supplier's comparison budget is estimated here.

33. Matthew J. Salganik and Duncan J. Watts, "Leading the Herd Astray: An Experimental Study of Self-Fulfilling Prophecies in an Artificial Cultural Market," *Social Psychology Quarterly* 71 (2008): 338–355, doi:10.1177/019027250807100404. Inversion design and the two named songs: pp. 341–343, table 2; appeal, recovery, and downloads: pp. 349–351, figures 8–9. Original article. The deliberate inversion is an intervention, distinct from independently evolving histories under unchanged rules.

The comparison budget

34. Ashok Chandrashekar, Fernando Amat, Justin Basilico, and Tony Jebara, "Artwork Personalization at Netflix", Netflix TechBlog, December 7, 2017, sections "Challenges," "Contextual bandits approach," and "Performance evaluation."

35. The information bound uses a conditional-logit allocation model with bounded differences in input profiles. The lifetime limit additionally requires common predictable design, local equivalence, comparison-dominated separation, and an almost-surely finite comparison budget. All informative observed outcomes must be included. An infinite budget alone need not identify contribution, and almost-sure finiteness does not imply finite expectation. Chapter 3 explains these distinctions. The supplier probabilities and halving schedule are hypothetical.

36. Lihong Li, Wei Chu, John Langford, and Robert E. Schapire, "A Contextual-Bandit Approach to Personalized News Article Recommendation," *Proceedings of WWW 2010*: 661–670, doi:10.1145/1772690.1772758. See arXiv version 2 (2012), Sections 2.1, 4, and 5.2.1, full paper. Offline replay assumes the stated encounter distribution and randomized logging; the experiment records click outcomes as well as selection. It is distinct from replaying a reinforced economic history under unchanged starting conditions and response rules.

37. Karthik Rajkumar, Guillaume Saint-Jacques, Iavor Bojinov, Erik Brynjolfsson, and Sinan Aral, "A Causal Test of the Strength of Weak Ties," *Science* 377 (2022): 1304–1310, doi:10.1126/science.abl4476. See design and main results, pp. 1304–1309; tie-strength definitions, pp. 1305–1306; and interference discussion, pp. 1309–1310, full paper hosted by MIT. The different hypothetical career record is an illustration. Observed jobs in the experimental population are not all jobs caused by the experiment.

38. Robin Pemantle, "A Survey of Random Processes with Reinforcement," *Probability Surveys* 4 (2007): 1–79, doi:10.1214/07-PS094. See Section 2.1, especially pp. 3–5, for the classical urn and its variants, full survey. The elementary urn is a credited mathematical thought experiment, not an empirical description of a procurement market.

39. Roberto Imbuzeiro Oliveira, "The Onset of Dominance in Balls-in-Bins Processes with Feedback," *Random Structures & Algorithms* 34 (2009): 454–477, doi:10.1002/rsa.20261. The author preprint's introduction and Theorem 1, pp. 1–2, credit earlier monopoly results, full preprint. Thomas Gottfried and Stefan Grosskinsky, "Asymptotics of Generalized Pólya Urns with Non-Linear Feedback," *Electronic Journal of Probability* 29 (2024): 1–56, doi:10.1214/24-EJP1157; see arXiv version 3, Sections 2.1–2.3, especially Theorem 2.2, full preprint. Strong and weak monopoly are distinct. The contribution result in Chapter 3 requires further information assumptions.

40. Maximilian Kasy and Anja Sautmann, "Adaptive Treatment Assignment in Experiments for Policy Choice," *Econometrica* 89 (2021): 113–132,

doi:10.3982/ECTA17527, pp. 113–114 and Section 6, pp. 122–123, full published paper; authors' November 10, 2021 correction, especially corrected Theorem 1.

41. Yaneer Bar-Yam, "From Big Data to Important Information," *Complexity* 21, supplement 2 (2016): 73–98, doi:10.1002/cplx.21785, abstract and Sections I–II; author preprint.

42. Ron Kohavi, Alex Deng, Brian Frasca, Toby Walker, Ya Xu, and Nils Pohlmann, "Online Controlled Experiments at Large Scale," *Proceedings of KDD 2013*: 1168–1176, doi:10.1145/2487575.2488217; author version, introduction and Section 1.1, pp. 1–2.

Shadow futures

43. Matthew J. Salganik, Peter Sheridan Dodds, and Duncan J. Watts, "Experimental Study of Inequality and Unpredictability in an Artificial Cultural Market," *Science* 311 (2006): 854–856, full paper, pp. 854–855 and figures 1–3. Lockdown's specific rankings are reported in Salganik's *Bit by Bit*, Section 4.6.1. Separate randomized listener groups supplied observed parallel markets, not literal replications of identical individuals or a test of the lifetime information limit.

44. The comparison budget sums conditional allocation probability outside the current favorite. The lifetime result requires a common predictable design, local equivalence, comparison-dominated separation in both directions, and an almost-surely finite lifetime budget at every parameter. Additional informative observations must be included. Replays preserve the parameter, initial state, and response rules. Intervention effects require those rules to remain valid under the specified change. The observed-input setting differs from Akerlof's hidden-quality model, discussed in Chapter 1; the reinforcement results of Pemantle (2007), Oliveira (2009), and Gottfried and Grosskinsky (2024) are discussed in Chapter 2.

45. Erik Brynjolfsson, Danielle Li, and Lindsey R. Raymond, "Generative AI at Work," *Quarterly Journal of Economics* 140 (2025): 889–942, published article, abstract and Sections I and IV. This is the published 5,172-agent study, rather than an earlier working-paper sample. The staggered introduction informs workplace productivity in its studied setting; it is not a decomposition of individual desert or a general employment forecast.

46. Line Chloé Le Goff and Philippe Soulier, "Parameter Estimation of a Two-Colored Urn Model Class," *International Journal of Biostatistics* 13 (2017), article 20160029, doi:10.1515/ijb-2016-0029; author preprint, version 2, October 4, 2016, Sections 3.2–3.3, pp. 14–15, and Sections 4.1–4.2, pp. 19–20.

47. Lihong Li, Wei Chu, John Langford, and Robert E. Schapire, "A Contextual-Bandit Approach to Personalized News Article Recommendation," *WWW 2010*, 661–670, author version 2, Sections 2.1, 4, and 5.2.1. Randomized logging, recorded click

responses, and the assumptions of offline evaluation matter. Its replay method is distinct from observing complete counterfactual economic histories.

48. Shizuo Kakutani, "On Equivalence of Infinite Product Measures," *Annals of Mathematics* 49 (1948): 214–224, doi:10.2307/1969123, treats product measures. Yu. M. Kabanov, R. Sh. Liptser, and A. N. Shiryaev, "On the Question of Absolute Continuity and Singularity of Probability Measures," *Mathematics of the USSR-Sbornik* 33 (1977): 203–221, doi, extends the analysis to general sequences. Jean Jacod and Albert N. Shiryaev, *Limit Theorems for Stochastic Processes*, second edition (2003), Chapter IV, and S. S. Gabriyelyan (2011), author version 4, Theorem 1, supply filtered-space continuity criteria. Lucien Le Cam and Grace Lo Yang, *Asymptotics in Statistics*, second edition (2000), Chapter 2, provide experiment-comparison foundations for precision limits. These credits distinguish the proposed economic bound from its inherited machinery.

49. John Snow, *On the Mode of Communication of Cholera*, second edition (London: John Churchill, 1855), Part 3, especially the discussions of Lambeth's new supply, intermingled customers, and the inquiry into household supplies; full section transcribed by UCLA. The historical comparison was not investigator-randomized. The discussion here concerns its inferential design; it does not identify two companies with a measured comparison budget.

50. David Blackwell and Lester Dubins, "Merging of Opinions with Increasing Information," *Annals of Mathematical Statistics* 33 (1962): 882–886, doi:10.1214/aoms/1177704456, main theorem, p. 883. Merging concerns conditional future distributions under its hypotheses. The application here combines that result with the equivalence of history laws at different fixed contribution parameters. Predictive merging by itself does not rule out parameter learning.

51. Hao Zhang, "Inconsistent Estimation and Asymptotically Equal Interpolations in Model-Based Geostatistics," *Journal of the American Statistical Association* 99 (2004): 250–261, full published paper, introduction, Section 3, Theorem 2, Corollaries 1–2, and Theorem 3, pp. 250–253. Fixed-domain sampling, model conditions, and the distinction between individual parameters and an estimable combination are essential. The soil field is an illustration, not a reported dataset.

52. Thomas Blake, Chris Nosko, and Steven Tadelis, "Consumer Heterogeneity and Paid Search Effectiveness: A Large-Scale Field Experiment," *Econometrica* 83 (2015): 155–174, May 2014 author working paper, introduction and Section 3. Brand-search substitution and nonbrand heterogeneity should not be generalized to all advertising or long-run brand investment.

53. Line Chloé Le Goff and Philippe Soulier, "Parameter Estimation of a Two-Colored Urn Model Class," *International Journal of Biostatistics* 13(1) (2017), article 20160029, Sections 2 and 3.2, doi:10.1515/ijb-2016-0029. Their repeated-experiment estimation framework distinguishes independent repetitions from extending one reinforced path.

Consistency requires identification and the relevant regularity conditions; separate organizational units do not automatically supply independent experiments.

54. Martin Watzinger, Thomas A. Fackler, Markus Nagler, and Monika Schnitzer, "How Antitrust Enforcement Can Spur Innovation: Bell Labs and the 1956 Consent Decree," *American Economic Journal: Economic Policy* 12 (2020): 328–359, published record; June 2017 author paper, introduction and institutional background. The decree and subsequent innovation evidence are developed in Chapter 4; this is the 1956 licensing intervention.

The histories that became infrastructure

55. Adam Smith, *An Inquiry into the Nature and Causes of the Wealth of Nations* (1776), Book I, Chapters I and III (pins, specialization, market extent and water carriage), and Chapter X, Part II (corporation and settlement restrictions). Original text, Project Gutenberg. The carpenter and two towns are thought experiments. Geographic separation does not establish independent selection or freer historical markets.

56. *United States v. United States Steel Corp.*, 251 U.S. 417 (1920), opinion, pp. 438–439.

57. P. W. Anderson, "More Is Different," *Science* 177, no. 4047 (1972): 393–396, opening page and Bell Telephone Laboratories affiliation. Published article. The economic use is an analogy. The proposed centuries-long economic progression remains a historical hypothesis; Anderson supplies a physical analogy.

58. John Bardeen, "Semiconductor Research Leading to the Point Contact Transistor," Nobel lecture, December 11, 1956, full lecture, pp. 318–321: research program, earlier theory, materials, complementary contributions and the failed field-effect proposal directing surface research. For the December 1947 device, see the Computer History Museum's collection account, with original logbook and publication references. This documents an achievement; it does not estimate the causal benefit of Bell's organizational structure.

59. U.S. Department of Justice, *United States Attorneys' Bulletin*, February 17, 1956, vol. 4, no. 4, full issue, Antitrust Division, p. 119. The contemporary summary records licensing, technical-information, patent-list, and other requirements. Licensing included exceptions and reciprocal arrangements; this was not a general abolition of patents. Reported portfolio counts differ by source and definition, so the narrative uses "thousands." The 1956 settlement is distinct from the later Bell breakup.

60. Computer History Museum, "1952: Bell Labs Licenses Transistor Technology", symposium and manufacturing-visit account, with proceedings and collection references. The stated $25,000 fee and nine-day event predate the 1956 decree. This distinction prevents attributing the beginning of outside transistor development to the later settlement.

61. *Hush-A-Phone Corp. v. United States*, 238 F.2d 266 (D.C. Cir. 1956), November 8, 1956, opinion, pp. 267–269 and notes 8–12.

62. Martin Watzinger, Thomas A. Fackler, Markus Nagler, and Monika Schnitzer, "How Antitrust Enforcement Can Spur Innovation: Bell Labs and the 1956 Consent Decree," *American Economic Journal: Economic Policy* 12 (2020): 328–359, published record. See the June 9, 2017 author paper, introduction, institutional background, and empirical design/results. The published abstract confirms the matching design and contrasting sector findings. No effect size is taken from earlier working-paper versions. Patent-based follow-on innovation is not total welfare or an estimated comparison budget.

63. W. Brian Arthur, "Competing Technologies, Increasing Returns, and Lock-In by Historical Events," *Economic Journal* 99 (1989): 116–131, full published paper, pp. 116–118 and 128. The adoption mechanism and possible historical selection are prior work. Increasing returns alone does not establish a finite lifetime comparison budget.

64. Paul A. David, "Clio and the Economics of QWERTY," *American Economic Review* 75 (1985): 332–337, full paper, pp. 332 and 335–336. S. J. Liebowitz and Stephen E. Margolis, "The Fable of the Keys," *Journal of Law and Economics* 33 (1990): 1–25, author text, introduction and discussion of typing studies.

65. F. A. Hayek, "Competition as a Discovery Procedure," translated by Marcellus S. Snow, *Quarterly Journal of Austrian Economics* 5 (2002): 9–23, full translation, pp. 9–11. This is a 2002 translation of the 1968 Kiel lecture.

66. Sony Group History, Volume 1, Chapter 3, "Developing the Transistor Radio", Part 2. Corporate retrospective; reports, equipment and TR-52/TR-55 chronology are attributed. No reconstructed dialogue, motives, or precise launch month is used; these events predate the 1956 decree.

67. Robert K. Merton, "The Matthew Effect in Science," *Science* 159 (1968): 56–63, especially 56–57. Article.

68. Arnout van de Rijt, Soong Moon Kang, Michael Restivo, and Akshay Patil, "Field Experiments of Success-Breeds-Success Dynamics," *PNAS* 111 (2014): 6934–6939, full paper, pp. 6934–6937. Subsequent third-party support differs from meeting a Kickstarter target; larger treatment effects were not unbounded or proportional.

69. Armen A. Alchian, "Uncertainty, Evolution, and Economic Theory," *Journal of Political Economy* 58 (1950): 211–221, pp. 213–214.

70. AT&T, *Annual Report 1955*, original report scan, printed pp. 13 and 16, PDF pp. 15 and 18. Americus equipment and solar-power trials are described on p. 13; the company's account of integration, decree terms, and expected business consequences is on p. 16. These are attributed corporate descriptions and predictions, not independently established policy effects.

71. FCC, *Use of the Carterfone Device in Message Toll Telephone Service*, 13 FCC 2d 420 (1968), adopted June 26, 1968, decision, pp. 420–425, especially p. 424.

Who gets to be tested

72. Amit P. Mehta, *United States v. Google*, memorandum opinion, August 5, 2024, full court opinion, printed pp. 1–4. Quality findings, default arrangements, 2021 payments, and liability are dated findings, not descriptions of current shares or an appellate disposition. The search case is distinct from the separate ad-tech litigation.

73. Mehta, August 2024 opinion, pp. 115–117, especially paragraphs 337–339. The Firefox–Yahoo switch, query-share changes, advertising pressure, and reversal are the court's account. This observed intervention is not an investigator-randomized experiment or an estimate of an invariant default effect.

74. Mehta, August 2024 opinion, pp. 34–39, especially paragraphs 86–93.

75. Mehta, August 2024 opinion, p. 49, paragraphs 136–137. Human-rated comparisons and the Bing latency comparison supplied internal learning; Project Folly concerns the particular speed-improvement effort described in the record, not every aspect of search quality.

76. Zahra Meghani, "Regulations Matter: Epistemic Monopoly, Domination, Patents, and the Public Interest," *Philosophy & Technology* 34 (2021): 1449–1474, published text, introduction and Section 6.

77. *National Family Farm Coalition v. EPA*, June 3, 2020, No. 19-70115, opinion, pp. 5–7 and 27–32.

78. *United States v. Google*, final judgment, December 5, 2025, Sections III–V and IX.V, pp. 2–10 and 32–33; Mehta, remedies opinion, September 2, 2025, introduction and remedy-specific conclusions. The judgment specifies conditional contracts, datasets, syndication, eligibility, safeguards, and tapering. It does not require divestiture of Chrome or ban all search payments.

79. Mehta, August 2024 opinion, p. 116, paragraph 335. The payment and budget share concern 2021. They are not current Mozilla financial estimates or a complete causal measure of the payments' welfare effect.

80. Laura Chambers, "Defending an Open Web: What the Google Search Ruling Means for the Future", September 8, 2025. Mozilla's funding, flexibility, implementation, privacy, and concentration arguments are attributed stakeholder positions.

81. Mehta, September 2025 remedies opinion, payment-ban discussion, printed pp. 119–128. Requested relief, anticipated consequences, and ordered relief are distinct. The plaintiffs subsequently challenged rejection of the payment ban on cross-appeal.

82. F. A. Hayek, "Competition as a Discovery Procedure", 2002 English translation of the 1968 lecture, pp. 9–11.

83. Mehta, September 2025 remedies opinion, user-side data discussion, especially pp. 163–164; December final judgment, Sections IV.B–C.

84. Jesper Akesson, Michael Luca, Gemma Petrie, and Kush Amlani, *Can Browser Choice Screens Be Effective?* (2023), full report, acknowledgments, executive summary, and experimental design, PDF pp. 2–5, 22–25 and 38–40, especially Table 2 and the T3/T4 comparison. Mozilla commissioned the compensated research. Rounded shares: 11% and 19%; reported difference: 8.2 percentage points. Timing was randomized; the click mechanism is interpretation. Long-run use was not observed.

85. Plaintiffs, second compliance and enforcement status report, September 17, 2026, pp. 2–3 and 10–12; public case index, accessed October 16, 2026.

86. Lee-Anne Mulholland, Google's appeal statement, January 16, 2026; plaintiffs, response and opening cross-appeal brief, July 28, 2026, introduction, issues, and cross-appeal. These establish opposing arguments at those dates, not an appellate ruling.

87. Matthew 2:1–18, King James Version.

88. Colleen Cunningham, Florian Ederer, and Song Ma, "Killer Acquisitions," *Journal of Political Economy* 129, no. 3 (2021): 649–702, especially pp. 649–653, published article, doi:10.1086/712506.

Who deserves the winnings

89. Matthew J. Salganik, Peter Sheridan Dodds, and Duncan J. Watts, "Experimental Study of Inequality and Unpredictability in an Artificial Cultural Market," *Science* 311 (2006), 854–856, full paper, pp. 854–855 and Figures 1–3. The particular first-place and fortieth-place rankings come from Salganik's *Bit by Bit*, Section 4.6.1. The study compares markets with different listeners, not cloned people. These ranks do not document musician income, working hours, or personal reactions.

90. Shuji Nakamura, "Background Story of the Invention of Efficient Blue InGaN Light Emitting Diodes", Nobel lecture, December 8, 2014, printed pp. 70 and 72–75, especially pp. 73–74.

91. Nichia Corporation, "Nichia Settled with Former Employee over 'Reasonable Remuneration'", January 11, 2005, opening, settlement scope and payment paragraphs, and company statements 2–4.

92. Replays preserve parameters, starting conditions, and input and state response rules; interventions change a specified input or policy. A contribution claim therefore requires a defined intervention and horizon, together with assumptions about how the system responds. See Chapter 3.

93. DARPA, "Grand Challenge", institutional timeline, first three paragraphs; "The DARPA Grand Challenge: Ten Years Later", March 13, 2014, opening through the 2007 Urban Challenge.

94. William R. Kerr, Ramana Nanda, and Matthew Rhodes-Kropf, "Entrepreneurship as Experimentation," *Journal of Economic Perspectives* 28, no. 3 (2014): 25–48, especially 27–32 and 43–44. doi:10.1257/jep.28.3.25.

95. Amit P. Mehta, *United States v. Google*, liability opinion, August 5, 2024, p. 49, paragraphs 136–137.

96. Thomas Piketty, Emmanuel Saez, and Stefanie Stantcheva, "Optimal Taxation of Top Labor Incomes: A Tale of Three Elasticities," *American Economic Journal: Economic Policy* 6, no. 1 (2014): 230–271, especially 230–235. doi:10.1257/pol.6.1.230.

97. Casey Rothschild and Florian Scheuer, "Optimal Taxation with Rent-Seeking," *Review of Economic Studies* 83(3) (2016): 1225–1262, doi:10.1093/restud/rdw017; December 2015 author version, abstract and introduction, printed pp. 1–4.

98. Arnout van de Rijt, Soong Moon Kang, Michael Restivo, and Akshay Patil, "Field Experiments of Success-Breeds-Success Dynamics," *PNAS* 111 (2014): 6934–6939, especially 6934–6935 and Figure 1. Article.

99. Robert K. Merton, "The Matthew Effect in Science," *Science* 159 (1968): 56–63, full paper, pp. 56–57.

The intelligence factory

100. IEA, *Key Questions on Energy and AI* (2026), full report, published April 16, 2026; executive summary, pp. 9–11; physical bottlenecks, p. 21; demand outlook and accelerator definitions, pp. 24–26. The summary rounds 2030 consumption to 950 TWh. These are total data-center estimates/projections, not AI-only consumption or observed future demand. Wait times vary across jurisdictions.

101. Matt Lakin, ORNL, "Super Construction at Super Scale: How ORNL Built a New Home for Frontier", November 14, 2023, construction account from cooling plant and piping through foundations and the pressure test.

102. ORNL, "Computer Engineers at ORNL Pioneer Approaches to Energy Efficient Supercomputing", section on David Grant's cooling-system work and the transition from Titan to Frontier.

103. LUMI consortium, "Sustainable Future", opening paragraphs on waste-heat use, accessed October 16, 2026.

104. Llama Team, AI at Meta, *The Llama 3 Herd of Models*, arXiv version 3, November 23, 2024, Section 3.3.4, pp. 13–14 and Table 5. The reported fifty-four-day snapshot contains 466 interruptions: 47 planned and 419 unexpected. More than 90 percent

effective training time concerns the reported operation; it is not a measurement of economic efficiency or a claim that the whole training run lasted fifty-four days. The same section reports single-GPU restart risk, slow stragglers affecting thousands of GPUs, and significant manual intervention only three times in the snapshot. Developer self-report, not an independent operational audit.

105. Argonne Leadership Computing Facility, "INCITE Program Awards Supercomputing Time to 75 High-Impact Projects", May 5, 2026, opening, selection paragraphs and listed awards; *INCITE Overview and Policies*, version 13, December 2024, distributed for the 2026 cycle, Section 1.2.1, printed pp. 1–2. Awards describe allocated opportunity, not completed research or a causal evaluation of public computing. The source's dated policy version is preserved.

106. Llama Team, *The Llama 3 Herd of Models*, introduction, p. 1; Section 3, training infrastructure; Section 4, pre-training and post-training, p. 15.

107. DeepSeek-AI, *DeepSeek-V3 Technical Report*, arXiv version 2, February 18, 2025, p. 5, Table 1 and accompanying paragraphs. The calculation values 2.788 million H800 GPU hours at an assumed USD2 per hour, yielding USD5.576 million across the specified official training stages. The report explicitly excludes prior research and ablations. It is an attributed developer calculation, not a total company spending figure, audited invoice, or cost-equivalent comparison with another model or firm. The original report appeared in December 2024.

108. Stanford HAI, *Artificial Intelligence Index Report 2026*, full report, Section 1.1, printed pp. 16–18, particularly p. 18 and Figure 1.1.4. Industry's reported share is 91.2 percent. The Epoch AI dataset is manually curated for notability; the report records an April 22, 2026 snapshot. This is neither all-model market share nor a causal estimate of exclusion, and a model's sector differs from an application's ownership or its users' options.

109. Jordan Hoffmann and colleagues, "Training Compute-Optimal Large Language Models," *NeurIPS* 2022, published paper, abstract, pp. 1–2, Section 4 and Section 4.1, p. 7. The same-compute comparison is Chinchilla versus Gopher, not a universal constant across architectures, dates, or objectives. Section 4.1 records additional optimizer, tokenizer and data-mix changes; no isolated randomized model-size effect is claimed. The authors' April 12, 2022 account corroborates the qualitative result.

110. William R. Kerr, Ramana Nanda, and Matthew Rhodes-Kropf, "Entrepreneurship as Experimentation," *JEP* 28(3) (2014), 25–48, published paper, pp. 27–30 and 43–44. Chapter 6 develops its role and boundaries. A staged experiment's value and stopping option are prior ideas; the comparison-budget bound is a different proposed contribution.

111. Kenneth J. Arrow, "Economic Welfare and the Allocation of Resources for Invention," in *The Rate and Direction of Inventive Activity* (1962), 609–626, NBER chapter, pp. 609 and 614–617, especially pp. 615 and 617. Arrow supplies the established

uncertainty, use and appropriation problem. No particular AI access or intellectual-property rule follows automatically from it.

112. IEA, *Powering Artificial Intelligence: Strategies for Integrating Data Centres into Electricity Systems*, October 14, 2026, full PDF, pp. 7–10 and 52–57, especially the AEP Ohio entry on p. 57.

A billion answers

113. Wei-Lin Chiang and colleagues, "Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference," *ICML*, PMLR 235 (2024): 8359–8388, published paper, PDF pp. 1–3, Figure 1 and Section 3.1.

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117. Lihong Li, Wei Chu, John Langford, and Xuanhui Wang, "Unbiased Offline Evaluation of Contextual-bandit-based News Article Recommendation Algorithms," *WSDM* 2011, 297–306, DOI 10.1145/1935826.1935878; author version 2, March 1, 2012, Sections 3–4, especially Theorem 1 and the Yahoo study. The theorem assumes independent identically distributed events and uniformly randomized logging; modifications require corresponding conditions. A finite-stream ratio and concentration for arbitrary adaptive algorithms need additional care. Logged policy evaluation differs from a structural replay of economic development.

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125. Stanford HAI, *AI Index Report 2026*, Section 1.4, inference subsection, printed p. 36.

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126. Erik Brynjolfsson, Danielle Li, and Lindsey Raymond, "Generative AI at Work", *The Quarterly Journal of Economics* 140(2) (2025): 889–942, Section VI.B and Figure VII.

127. Brynjolfsson, Li, and Raymond, abstract, Sections III and V. Use the published 5,172-agent sample and 15 percent average gain in issues resolved per hour; older working-paper headlines used different figures. The setting is text-based support for business software. Effects vary by prior performance and experience. Suggestions were optional, and the human agent remained responsible for the interaction.

128. Same paper, Section VI.B, Figure VII. Chat-level outage analysis measures handling duration, not resolution outcomes unavailable at that granularity. Outages varied in duration and coverage. The authors discuss noisy estimates and possible differences in chat composition. Retained improvements are evidence consistent with learning, rather than proof that all assistance gains persist or that outages randomly assigned a training intervention.

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130. Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakcı, and Rei Mariman, "Generative AI without Guardrails Can Harm Learning", *PNAS* 122(26), 2025,

e2422633122. Relevant full author version: education_llm.pdf, pp. 4–11 for design/results and pp. 13–18, Section 4, for message/error analyses. The authors interpret asking-for-answer and engagement patterns as consistent with reliance on supplied solutions; these analyses do not separately randomize each safeguard or establish a unique mediator. Classroom allocation uses the school's random student assignment plus constrained classroom assignment. Individual laptop use was not independently randomized. The outcome is short-run assessment performance. GPT Tutor's unassisted advantage was not statistically established. The August 2025 correction, DOI 10.1073/pnas.2518204122, concerns an affiliation production error, not the results.

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133. David H. Autor, "Why Are There Still So Many Jobs? The History and Future of Workplace Automation", *Journal of Economic Perspectives* 29(3), 2015, 3–30, especially pp. 5–7. See also James Bessen, "Toil and Technology", *Finance & Development* 52(1), March 2015, 16–19, ATM discussion. The figures concern average US urban branches in the stated historical window. Autor also identifies deregulation. These accounts do not isolate an ATM treatment effect or describe current bank employment.

134. Daron Acemoglu and Pascual Restrepo, "Robots and Jobs: Evidence from US Labor Markets", *Journal of Political Economy* 128(6), 2020, 2188–2244; author-hosted journal-format proof, PDF pp. 1–4. Historical local exposure design, including European adoption as an instrument, differs from random robot assignment. No local coefficient is presented as an unconditional national or generative-AI forecast.

135. International Labour Organization, *World Employment and Social Outlook 2021: The Role of Digital Labour Platforms in Transforming the World of Work* (2021), Chapter 4 conclusion, printed p. 192 (PDF p. 194), and deactivation discussion/footnote 7, printed p. 182 (PDF p. 184). Full report. Reported conditions and surveyed workers, not a current universal sole-platform diagnosis. Footnote 7 explicitly records alternative-platform use after permanent deactivation. The moving driver and shared-ranking scenario are hypothetical.

136. Directive (EU) 2024/2831, Article 9(6) and Article 29(1), official consolidated text, November 11, 2024, PDF pp. 9–10 and 19.

137. Anton Korinek and Donghyun Suh, *Scenarios for the Transition to AGI*, March 2024, sections 3 and 4.1, 22–27, and conclusion, 36–37.

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138. CMA, "Roadmap Completion Decision", January 2023, paragraphs 1.1–1.4, 2.1–2.6, 4.4–4.9. Six providers' completion, continuing funding and monitoring; dated obligations, not a welfare estimate or current implementation audit.

139. Blackwell, "Equivalent Comparisons of Experiments", 1953, original pp. 265–267, decision losses and Theorems 3, 5–6; Section 4, pp. 268–269, extends beyond finite outcomes. Earlier Sherman–Stein result credited. De Oliveira, November 2017 author paper, pp. 1–5, published 2018. Fixed decision setting, known interpretation and state-independent garbling matter; gross value excludes collection costs.

140. *United States v. Google*, Final Judgment, December 5, 2025, ECF 1462, Section IV.B–C, pp. 6–7. Qualified competitors, specified user-side data, documentation, safeguards and exclusions including algorithms, ranking signals and post-trained LLMs. Dated ordered terms; no delivery, success or appellate-disposition finding here.

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146. Watzinger et al., "How Antitrust Enforcement Can Spur Innovation", 2020; full June 2017 author version, introduction and institutional/empirical-design sections. Outside-sector finding and telecommunications contrast; patent proxies do not measure total welfare or Bell's comparison budget.

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148. Kenneth J. Arrow and Anthony C. Fisher, "Environmental Preservation, Uncertainty, and Irreversibility," 1974, pp. 316–319, doi:10.2307/1883074, article copy.

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151. EuroHPC, Playground call, April 9, 2025, Description/capacity notice; access modes, Industrial Innovation; Terms of Reference, V3, March 31, 2026, Sections 1.3–1.4, 2–3.1. Published terms, not delivered capacity or demonstrated additional entry.

152. Charter of Fundamental Rights of the European Union, Article 17(1)–(2), and accompanying explanations.

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Acknowledgments

This book depends on researchers who made their experiments, methods, working papers and findings available for others to examine, and on the people who preserve public records and scientific archives. The notes and references identify the work behind its arguments and examples. Particular debts run to the MusicLab researchers and to the long traditions of information economics, experimental design, reinforcement, discovery and public finance discussed throughout the book.

The comparison-budget argument, its interpretation, and its proposed applications are my responsibility.